Modelling Diabetes Health Indicators

MA7442 Modelling Data — Assessment 1 (Task 2)

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1. Introduction and Reflection on Task 1 Feedback

This report constitutes Task 2 of the MA7442 *Modelling Data* assessment. It builds directly on the exploratory data analysis (EDA) and research questions developed in Task 1, incorporating the feedback received and extending the analytical scope accordingly.

In Task 1, the EDA was intentionally centred on diabetes status as the primary outcome variable. This focus was adopted to maintain analytical coherence and prevent the report from becoming overly expansive in scope. Given the breadth of variables available in the dataset, conducting a fully comprehensive EDA across multiple outcomes at that stage would likely have diluted the conceptual focus and resulted in excessive length. The analysis was therefore deliberately targeted to support a diabetes-oriented modelling framework.

Feedback indicated, however, that restricting the EDA exclusively to diabetes limited the diversity of subsequent research questions and led to conceptual overlap. Task 2 therefore broadens the analytical framework and introduces dedicated non-diabetes outcome analyses, substantially expanding the empirical foundation of the study.

1.1 Reflection on Task 1 Feedback

Three areas of feedback were identified for improvement.

Scope of the EDA.
The Task 1 EDA effectively characterised the dataset but examined associations solely in relation to diabetes status. This constrained the diversity of the research questions. In Task 2, Section 3 introduces a comprehensive EDA block examining cardiovascular indicators, general health measures, and behavioural variables as primary outcomes in their own right.

Research question overlap.
Because the initial EDA centred exclusively on diabetes, the research questions were conceptually similar. In response, the structure has been revised. RQ1 is retained in refined form; the previous RQ2 and RQ3 have been consolidated into a single modelling-focused question (new RQ2); and two new research questions (RQ3 and RQ4) are derived directly from the expanded non-diabetes EDA.

Figure line type.
In Task 1, a `geom_line` layer was used to connect mean unhealthy days across age groups. Since age group is an ordered categorical variable rather than a continuous measure, this implied unjustified linear interpolation. In Task 2, age-structured visualisations have been revised to use `geom_col` or lollipop-style representations, preserving ordinal structure without implying continuity between categories.

1.2 Rationale for Expanded Exploratory Analysis

The expanded EDA presented in Task 2 is intentionally broader in scope than in Task 1. This extension responds directly to feedback requesting analysis of non-diabetes outcomes and greater utilisation of the dataset. As a result, Section 3 systematically examines cardiovascular indicators, health status variables, and behavioural correlates using multiple complementary visualisations.

Given the size and multidimensional nature of the dataset, this comprehensive exploration necessarily increases the length of the EDA section. However, the extended analysis is structured purposefully: each block serves to establish demographic gradients, socioeconomic patterning, behavioural clustering, and outcome progression. The additional length reflects analytical depth rather than redundancy, and it provides the empirical basis for the revised and diversified research questions developed in Task 2.

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2. Data Preparation

The dataset and all variable recoding steps are identical to those described in Task 1 and are reproduced in the setup chunk below (Hidden) to ensure a fully self-contained, reproducible document. No additional transformations are required for the EDA in Section 3. All 253,680 records are retained; no missing values were present.

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3. Additional Exploratory Data Analysis: Non-Diabetes Outcomes

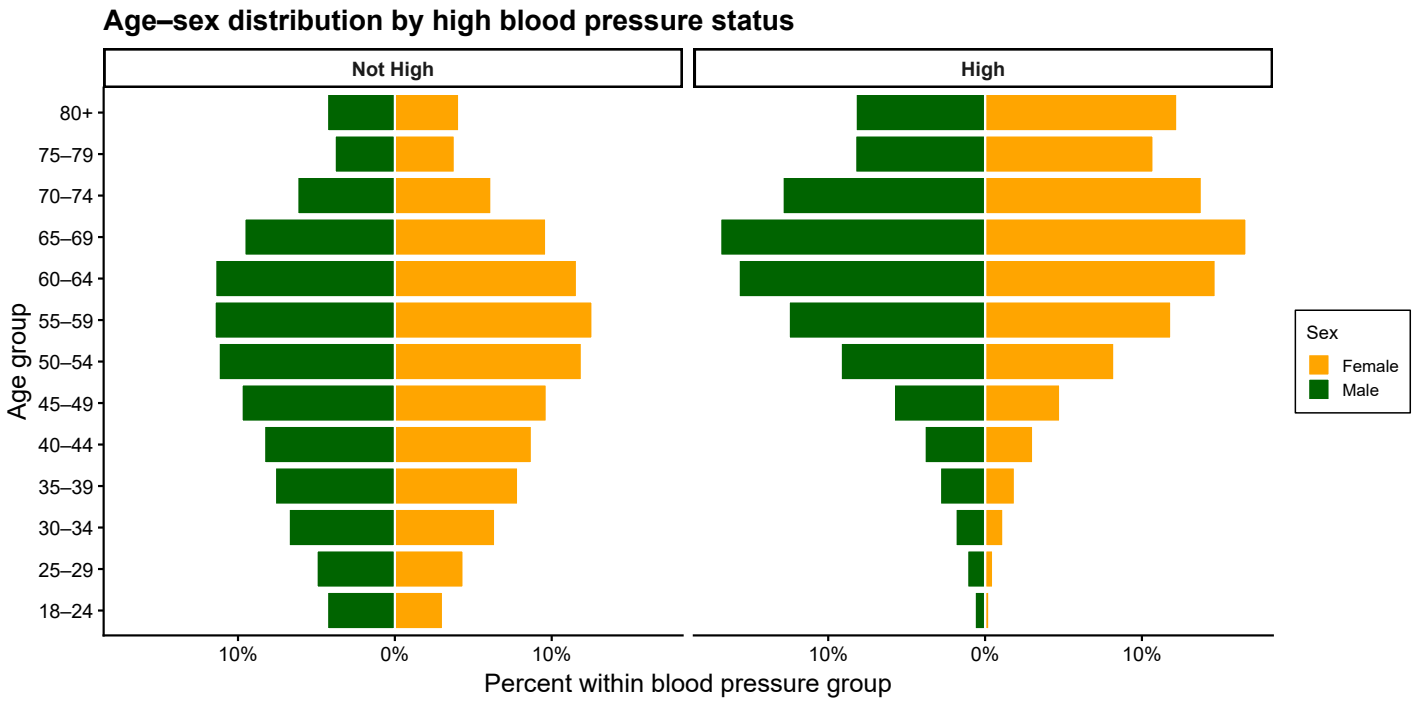
The following analyses explore relationships among health, lifestyle, and socioeconomic variables *independently of diabetes status*. The goal is to characterise the joint structure of these variables, in particular to understand what shapes self-reported general health, mental health burden, and functional limitation, and to motivate the non-diabetes research questions introduced in Section 4.

3.1 Cardiovascular Risk Indicators: Demographic, Socioeconomic and Behavioural Gradients

This section explores the population structure and correlates of key cardiovascular risk indicators, namely high blood pressure, high cholesterol, and stroke. The analysis moves beyond diabetes to examine how these outcomes are patterned across age, sex, socioeconomic position, and behavioural factors. Emphasis is placed on identifying demographic concentration, socioeconomic gradients, and clustering with related health behaviours. The objective is to establish structural regularities in the data that may inform subsequent modelling and risk interpretation.

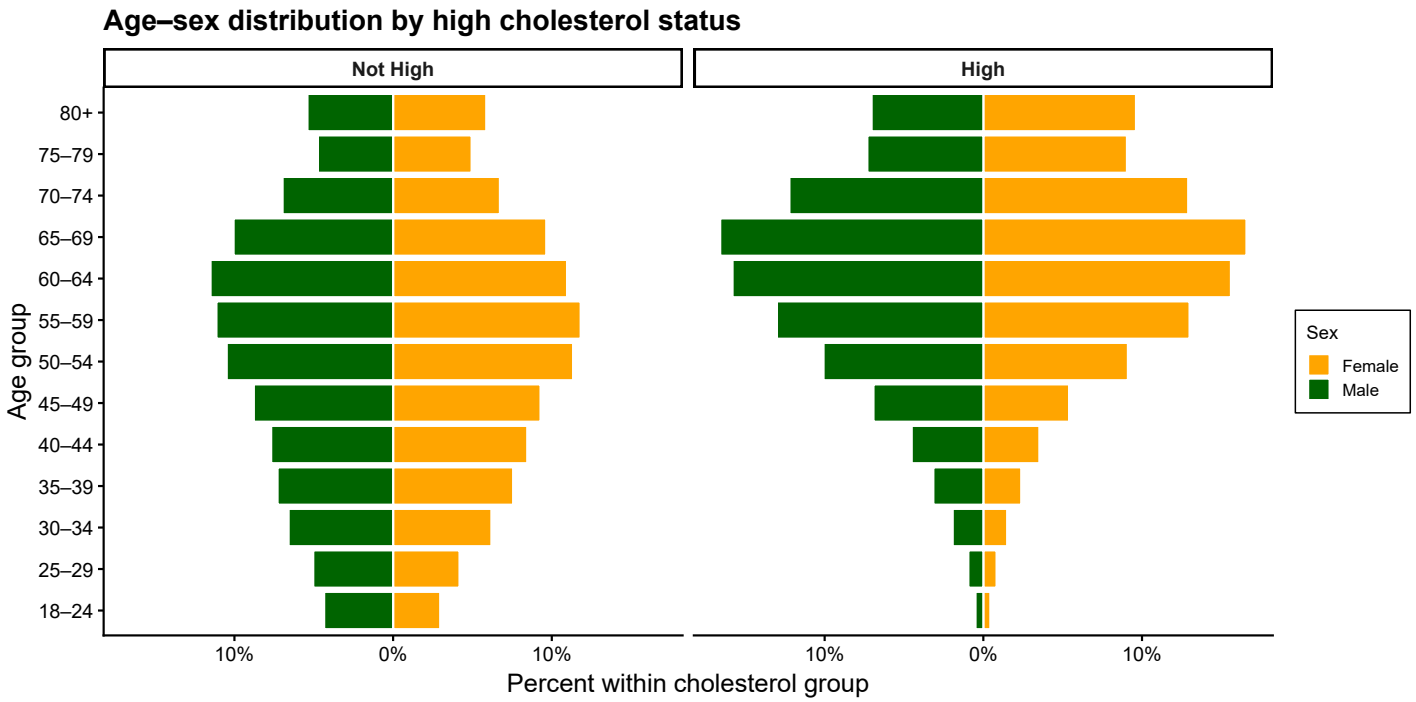
3.1.1 Age–Sex Distribution by High Blood Pressure Status

```
ggplot(bp_pyramid,
       aes(x = Age, y = percent_signed, fill = Sex)) +
  geom_col(width = 0.9, colour = "white") +
  coord_flip() +
  facet_wrap(~ HighBP, nrow = 1) +
  scale_y_continuous(
    labels = function(x) paste0(abs(x), "%")
  ) +
  scale_fill_manual(
    values = pal_sex
  ) +
  labs(
    title = "Age-sex distribution by high blood pressure status",
    x     = "Age group",
    y     = "Percent within blood pressure group",
    fill  = "Sex"
  )
```



3.1.2 Age–sex distribution by high cholesterol status

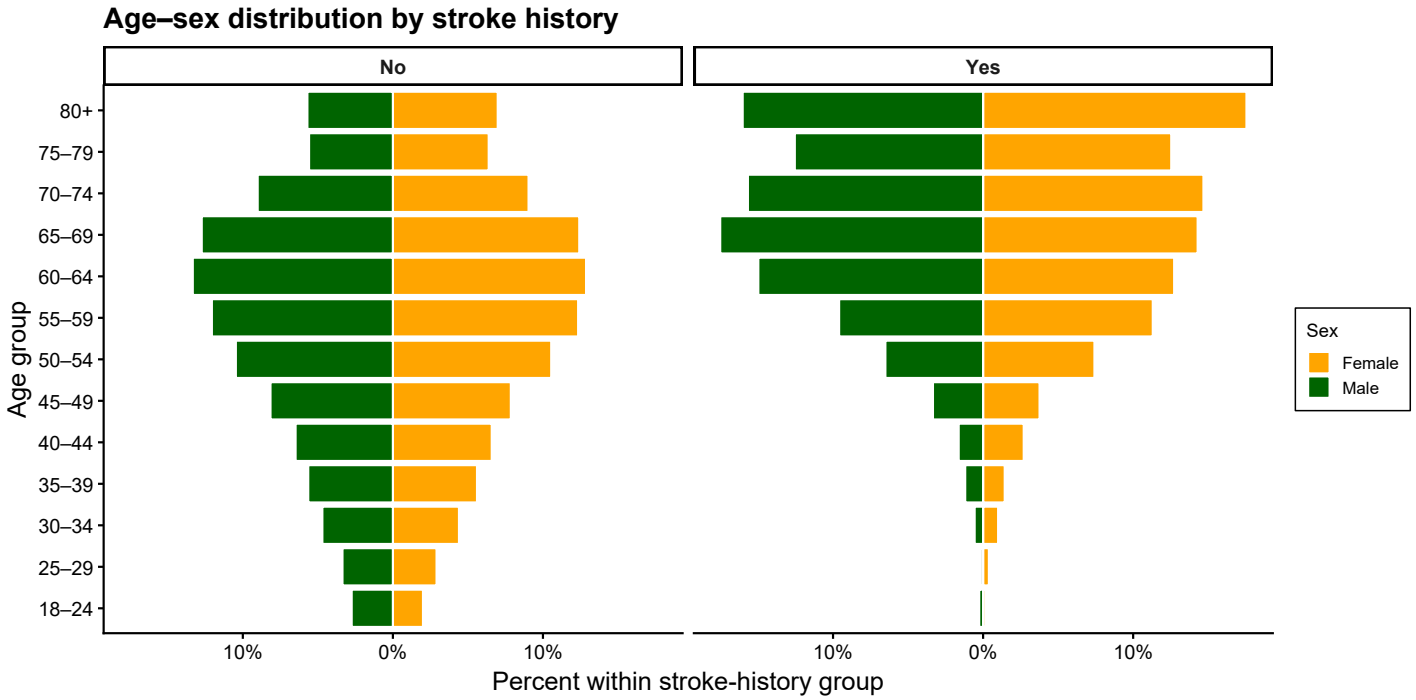
```
ggplot(chol_pyramid,
       aes(x = Age, y = percent_signed, fill = Sex)) +
  geom_col(width = 0.9, colour = "white") +
  coord_flip() +
  facet_wrap(~ HighChol, nrow = 1) +
  scale_y_continuous(
    labels = function(x) paste0(abs(x), "%")
  ) +
  scale_fill_manual(
    values = pal_sex
  ) +
  labs(
    title = "Age-sex distribution by high cholesterol status",
    x     = "Age group",
    y     = "Percent within cholesterol group",
    fill  = "Sex")
```



3.1.3 Age–sex distribution by stroke history

```
stroke_pyramid <- diab |>
  count(Stroke, Age, Sex) |>
  group_by(Stroke, Sex) |>
  mutate(
    percent = 100 * n / sum(n)
  ) |>
  ungroup() |>
  mutate(
    percent_signed = case_when(
      Sex == "Male"   ~ -percent,
      Sex == "Female" ~  percent,
      TRUE            ~ NA_real_
    )
  )

ggplot(stroke_pyramid,
       aes(x = Age, y = percent_signed, fill = Sex)) +
  geom_col(width = 0.9, colour = "white") +
  coord_flip() +
  facet_wrap(~ Stroke, nrow = 1) +
  scale_y_continuous(
    labels = function(x) paste0(abs(x), "%")
  ) +
  scale_fill_manual(
    values = pal_sex
  ) +
  labs(
    title = "Age-sex distribution by stroke history",
    x     = "Age group",
    y     = "Percent within stroke-history group",
    fill  = "Sex"
```

```
  )
```



Across all three outcomes, high blood pressure, high cholesterol, and stroke, the dominant structural feature is age concentration. Each "Yes" panel is heavily skewed toward the 55–69 and older age bands, while representation below age 40 is minimal. This confirms that cardiovascular burden in this dataset is fundamentally age-structured rather than evenly distributed across the life course.

However, the degree of age concentration differs meaningfully between outcomes. High blood pressure and high cholesterol show similar mid-to-late life clustering, with substantial presence beginning in the 50–54 band and peaking around 60–69. Stroke, by contrast, exhibits a far sharper right-skew: the "Yes" group is overwhelmingly concentrated in older age bands, with negligible representation in younger adults. This suggests stroke is not merely age-associated but strongly age-dependent relative to the other indicators.

Sex differences are comparatively modest within age strata for high blood pressure and high cholesterol. Once age is conditioned upon, male–female distributions appear broadly parallel, implying that sex operates as a secondary structuring factor relative to age. In the stroke group, however, there is a slight female predominance in the oldest age bands, which may reflect differential survival into advanced age rather than a pure sex-specific risk gradient.

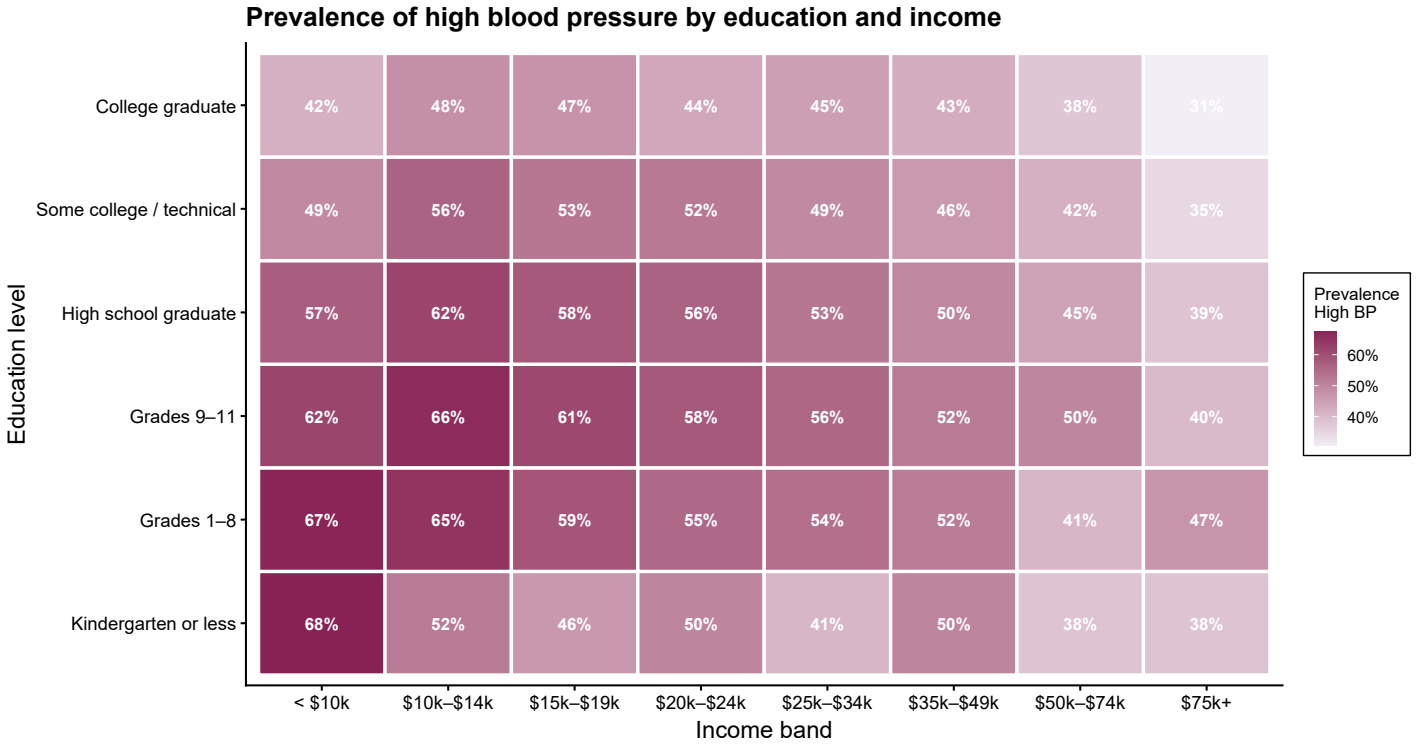
Importantly, these pyramids depict the **composition within outcome groups**, not prevalence across the population. They therefore highlight demographic concentration rather than risk probability. The consistency across high blood pressure and high cholesterol suggests shared cardiometabolic ageing processes, whereas stroke displays a more extreme demographic compression, indicating it may represent the culmination of accumulated vascular risk rather than an early-stage cardiovascular condition.

Taken together, the figures establish age as the primary axis of cardiovascular risk patterning in this sample. Any socioeconomic or behavioural gradients observed later must therefore be interpreted within the context of this strong underlying demographic structure.

3.1.4 Joint SES heatmap of High Blood Presure prevalence.

```
bp_ses |>
  ggplot(aes(x = Income, y = Education, fill = prev_bp)) +
  geom_tile(colour = "white", linewidth = 1.2) +
  geom_text(
```

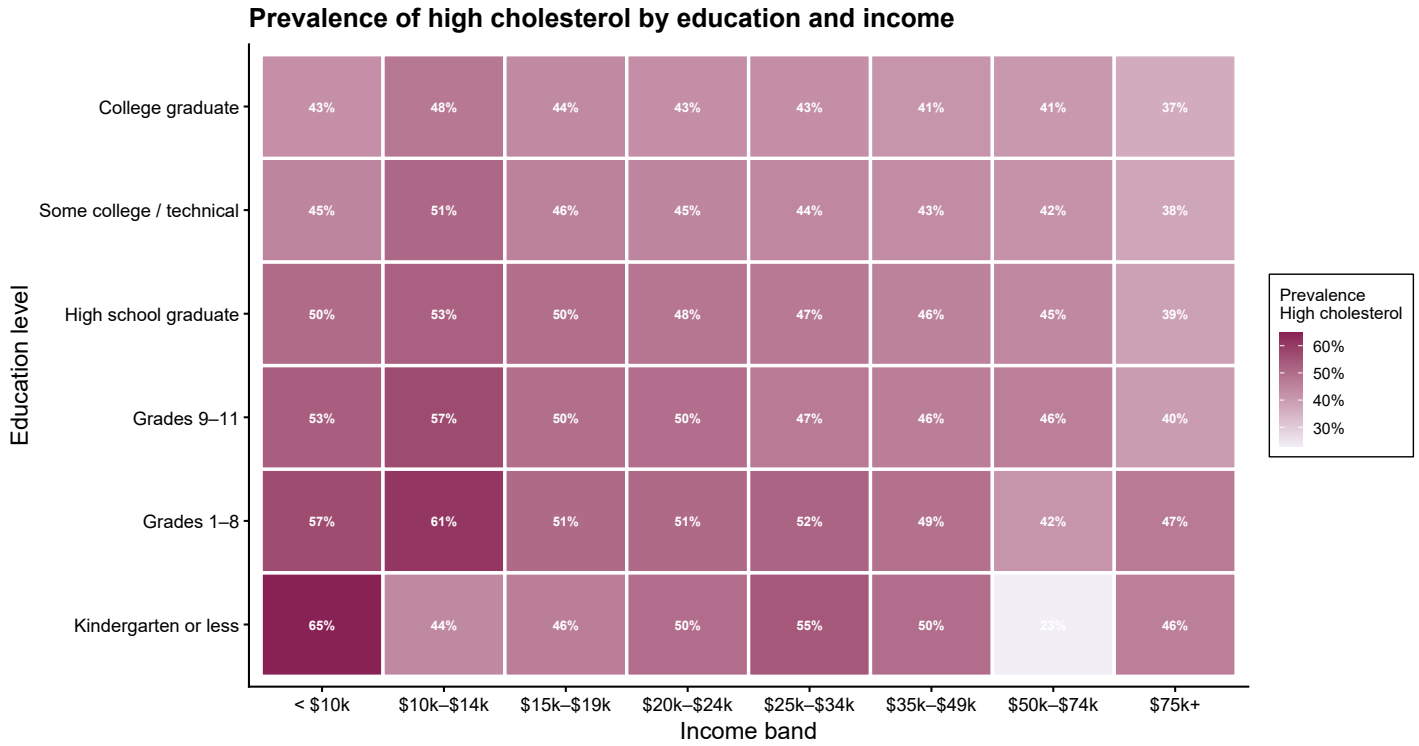
```
    aes(label = percent(prev_bp, accuracy = 1)),
    colour   = "white",
    fontface = "bold",
    size     = 4.5) +
  scale_fill_gradient(
    low  = "#F1EEF6",
    high = "#882255",
    name = "Prevalence\nHigh BP",
    labels = percent_format(accuracy = 1)) +
  labs(title = "Prevalence of high blood pressure by education and income",
    x     = "Income band",
    y     = "Education level")
```



3.1.5 Joint SES heatmap of High Cholestrol prevalence.

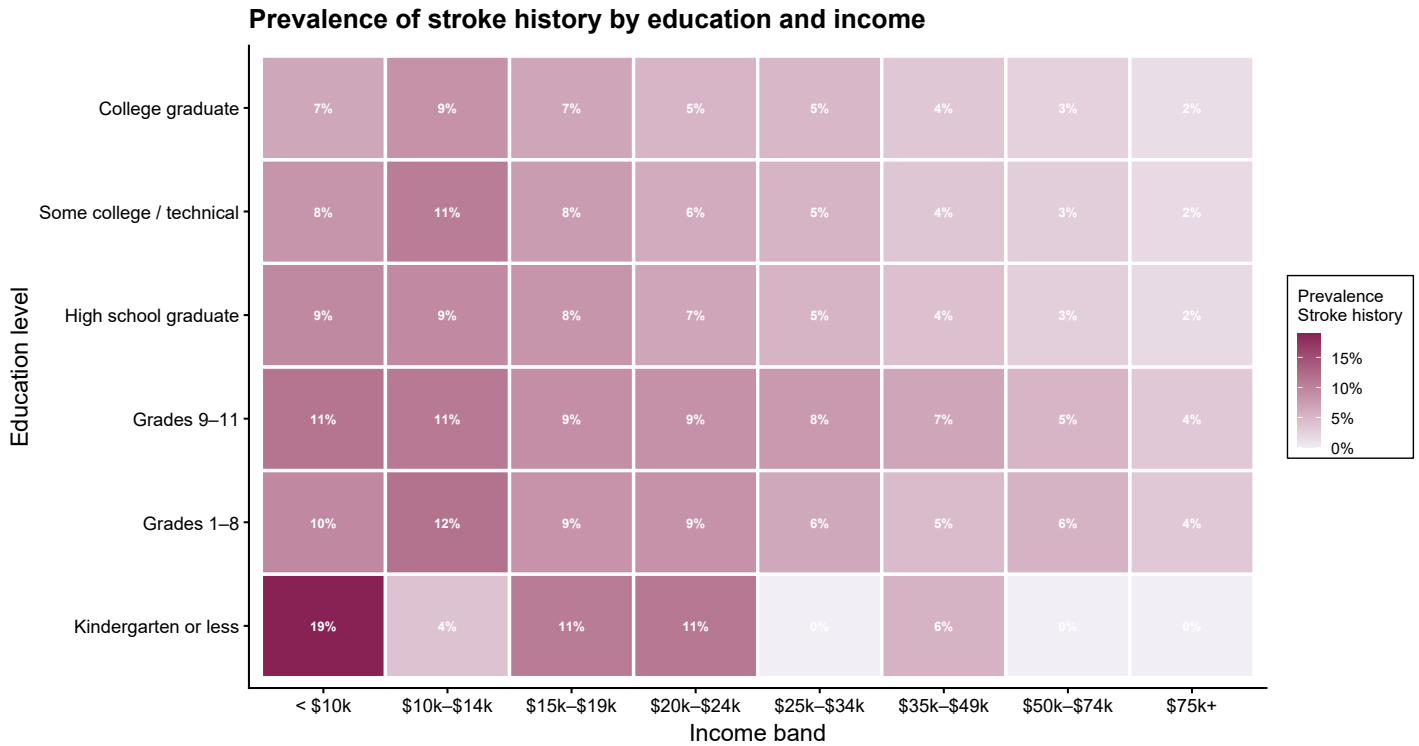
```
chol_ses |>
  ggplot(aes(x = Income, y = Education, fill = prev_chol)) +
  geom_tile(colour = "white", linewidth = 1.2) +
  geom_text(
    aes(label = percent(prev_chol, accuracy = 1)),
    colour   = "white",
    fontface = "bold",
    size     = 3.5
  ) +
  scale_fill_gradient(
    low    = "#F1EEF6",
    high   = "#882255",
    name   = "Prevalence\nHigh cholesterol",
    labels = percent_format(accuracy = 1)
  ) +
  labs(
    title = "Prevalence of high cholesterol by education and income",
```

```
    x      = "Income band",
    y      = "Education level"
  )
```



3.1.6 Joint SES heatmap of Stroke prevalence.

```
stroke_ses |>
  ggplot(aes(x = Income, y = Education, fill = prev_stroke)) +
  geom_tile(colour = "white", linewidth = 1.2) +
  geom_text(
    aes(label = percent(prev_stroke, accuracy = 1)),
    colour = "white",
    fontface = "bold",
    size = 3.5) +
  scale_fill_gradient(
    low = "#F1EEF6",
    high = "#882255",
    name = "Prevalence\nStroke history",
    labels = percent_format(accuracy = 1)) +
  labs(title = "Prevalence of stroke history by education and income",
       x = "Income band",
       y = "Education level")
```



Across all three outcomes, a clear socioeconomic gradient is evident. Prevalence is consistently highest in the lowest education and lowest income strata and declines with increasing educational attainment and income. The lower-left cells (least educated, lowest income) exhibit the greatest burden, while college graduates in the highest income bands show the lowest prevalence. This pattern indicates cumulative socioeconomic disadvantage rather than a single-dimension effect.

High blood pressure and high cholesterol display particularly strong and stable two-dimensional gradients. Within nearly every education category, prevalence decreases progressively as income rises. Likewise, within each income band, higher education corresponds to lower prevalence. The monotonic pattern across both axes suggests that both income and education independently structure cardiometabolic risk, rather than one merely proxying for the other.

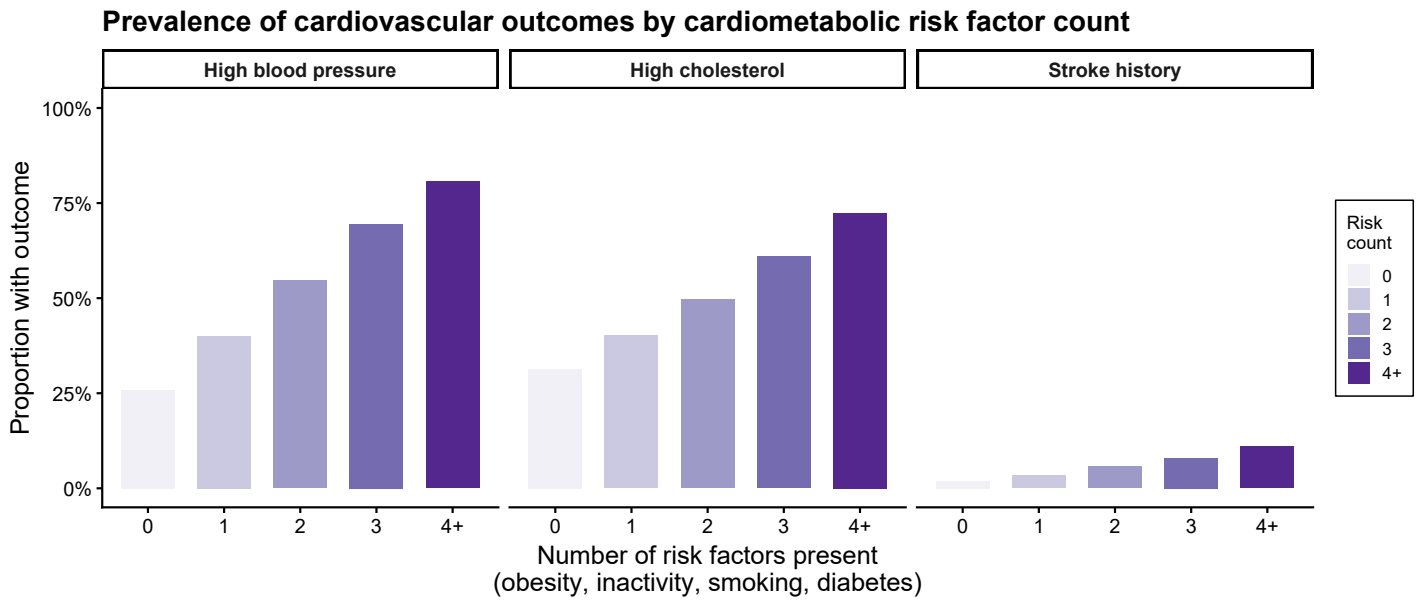
Stroke exhibits the same directional gradient but at substantially lower absolute prevalence and with more variability across cells. Some low-education, low-income cells show markedly elevated stroke prevalence (e.g., 19%), while higher SES cells approach very low levels. The noisier pattern likely reflects smaller counts and the relative rarity of stroke, but the overall concentration in socioeconomically disadvantaged groups remains clear.

Importantly, these are crude prevalence estimates. As shown in the age–sex pyramids, lower socioeconomic strata contain disproportionately older individuals, and age is a dominant determinant of all three outcomes. Therefore, part of the observed SES gradient likely reflects demographic composition. However, the persistence of a consistent gradient across both income and education dimensions suggests that socioeconomic position is not merely a proxy for age but a meaningful structural correlate of cardiovascular burden.

3.1.7 Prevalence Cardiovascular Outcomes Accumulated Risk Factors

```
ggplot(cvd_long, aes(x = risk_group, y = prop, fill = risk_group)) +
  geom_col(width = 0.7) +
  facet_wrap(~ Outcome, nrow = 1) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    limits = c(0, 1)
  ) +
  scale_fill_brewer(palette = "Purples", name = "Risk\ncount") +
```

```
  labs(
    title = "Prevalence of cardiovascular outcomes by cardiometabolic risk factor count",
    x     = "Number of risk factors present\n(obesity, inactivity, smoking, diabetes)",
    y     = "Proportion with outcome"
  )
```



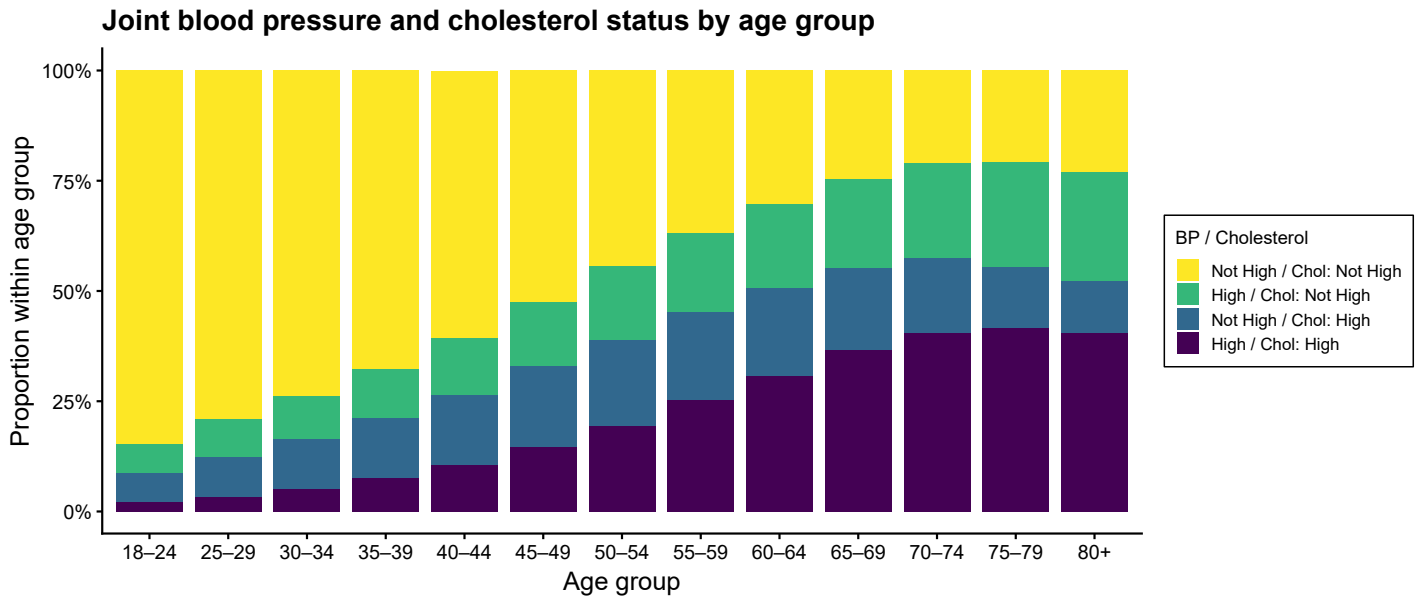
Cardiovascular outcomes show a clear **dose–response relationship with the accumulation of cardiometabolic risk factors**. For both **high blood pressure** and **high cholesterol**, prevalence increases steadily as additional risk factors are present. Individuals with no risk factors show relatively low prevalence (around one quarter to one third), while those with **four or more risk factors exhibit very high prevalence**, exceeding roughly **70–80%** for both conditions. This pattern indicates that these outcomes are strongly associated with the combined burden of obesity, physical inactivity, smoking, and diabetes.

Stroke history follows the same upward trend, but at substantially lower levels overall. Even among individuals with multiple risk factors, stroke prevalence remains below roughly **10–12%**, reflecting the relative rarity of stroke compared with hypertension and hypercholesterolaemia in the population.

Overall, the figure demonstrates that **cardiometabolic risk factors act cumulatively rather than independently**. As risk exposures accumulate, the likelihood of cardiovascular conditions, particularly hypertension and high cholesterol, increases markedly, highlighting the importance of addressing multiple lifestyle and metabolic risk factors simultaneously.

3.1.8 Joint blood pressure and cholesterol profiles across age groups

```
ggplot(bp_chol_age,
       aes(x = Age, y = prop, fill = interaction(HighBP, HighChol,
                                                 sep = " / Chol: "))) +
  geom_col(position = "fill", width = 0.85) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  scale_fill_viridis_d(option = "D", direction = -1) +
  labs(
    title = "Joint blood pressure and cholesterol status by age group",
    x     = "Age group",
    y     = "Proportion within age group",
    fill  = "BP / Cholesterol"
  )
```



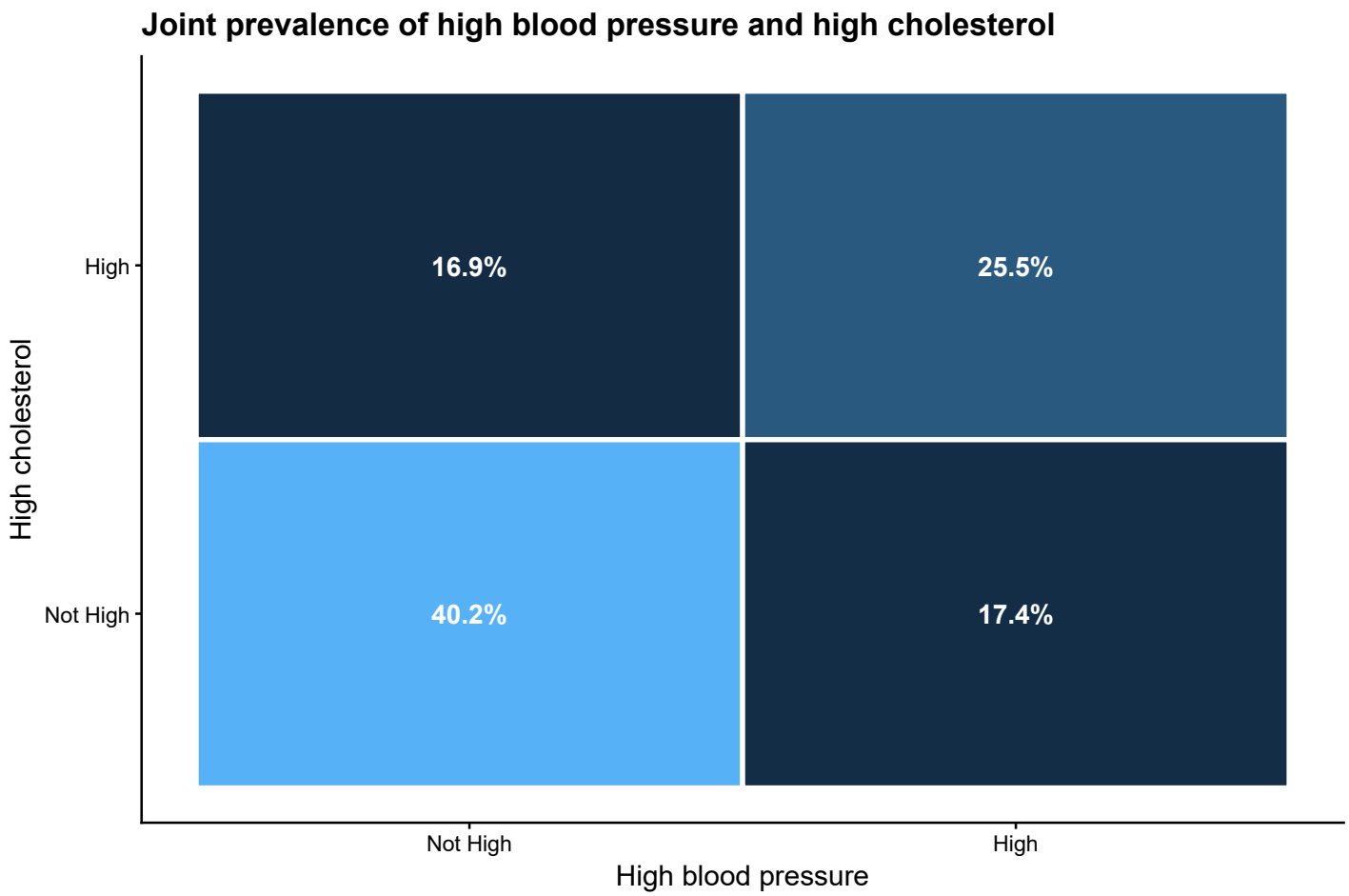
The distribution of cardiovascular profiles shifts systematically with age. Younger groups are overwhelmingly characterised by normal blood pressure and cholesterol, whereas the proportion with dual elevation (high/high) increases sharply from midlife onward and becomes dominant in older age bands. This pattern reflects progressive cardiometabolic accumulation rather than isolated risk emergence.

Importantly, the high/high category expands more rapidly than either single-risk category, indicating clustering rather than independence of these conditions. By ages 65+, dual dysregulation accounts for a substantial share of the population within each age group, suggesting convergence of vascular risk factors with ageing.

The pattern implies that hypertension and hypercholesterolaemia are not parallel but interacting processes within cardiovascular ageing, reinforcing their role as intermediate markers in the pathway toward more severe vascular outcomes.

3.1.9 Co-occurrence of Hypertension and High Cholesterol

```
diab |>
  count(HighBP, HighChol) |>
  mutate(prop = n / sum(n)) |>
  ggplot(aes(x = HighBP, y = HighChol, fill = prop)) +
  geom_tile(colour = "white", linewidth = 1.5) +
  geom_text(
    aes(label = percent(prop,accuracy = 0.1)),
    size     = 6,
    fontface = "bold",
    colour   = "white"
  ) +
  labs(
    title = "Joint prevalence of high blood pressure and high cholesterol",
    x     = "High blood pressure",
    y     = "High cholesterol"
  ) +
  theme(
    legend.position = 'none'
  )
```



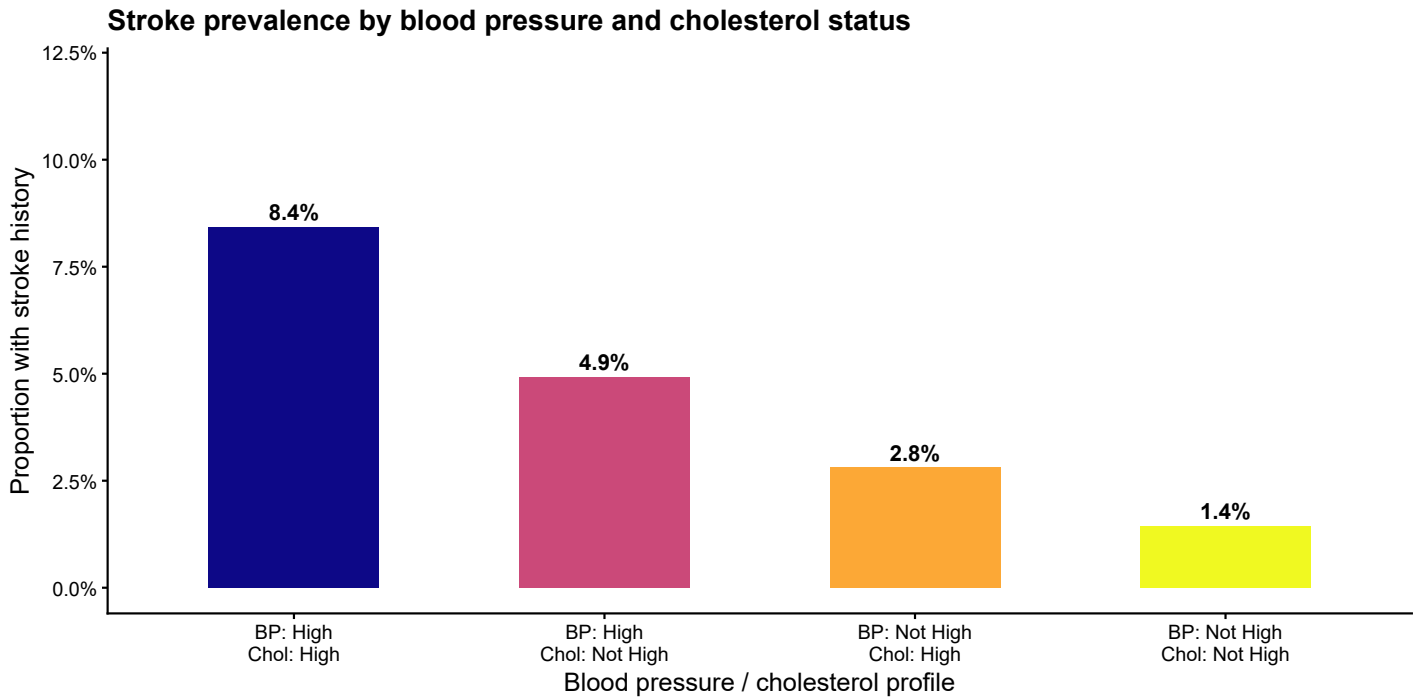
Approximately **40.2%** of the sample has neither high blood pressure nor high cholesterol. The remaining **59.8%** have at least one of the two conditions, indicating that cardiometabolic risk is common in this population.

The largest single elevated profile is **concurrent high blood pressure and high cholesterol (25.5%)**, which exceeds either isolated condition (17.4% BP only; 16.9% cholesterol only). This suggests meaningful clustering rather than independent occurrence of the two risk factors.

Overall, hypertension and hypercholesterolaemia appear strongly co-patterned, with dual elevation more common than either single-condition state, reinforcing their joint role in cardiovascular risk stratification.

3.1.10 Stroke prevalence by blood pressure and cholesterol status

```
ggplot(stroke_bpchol,
       aes(x = BP_Chol, y = prop, fill = prop)) +
  geom_col(width = 0.55) +
  geom_text(aes(label = percent(prop, accuracy = 0.1)),
            vjust = -0.4, size = 5.5, fontface = "bold") +
  scale_y_continuous(labels = percent_format(accuracy = 0.1),
                     limits = c(0, 0.12)) +
  scale_fill_viridis_c(option = "C", direction = -1, guide = "none") +
  labs(
    title = "Stroke prevalence by blood pressure and cholesterol status",
    x     = "Blood pressure / cholesterol profile",
    y     = "Proportion with stroke history"
  )
```

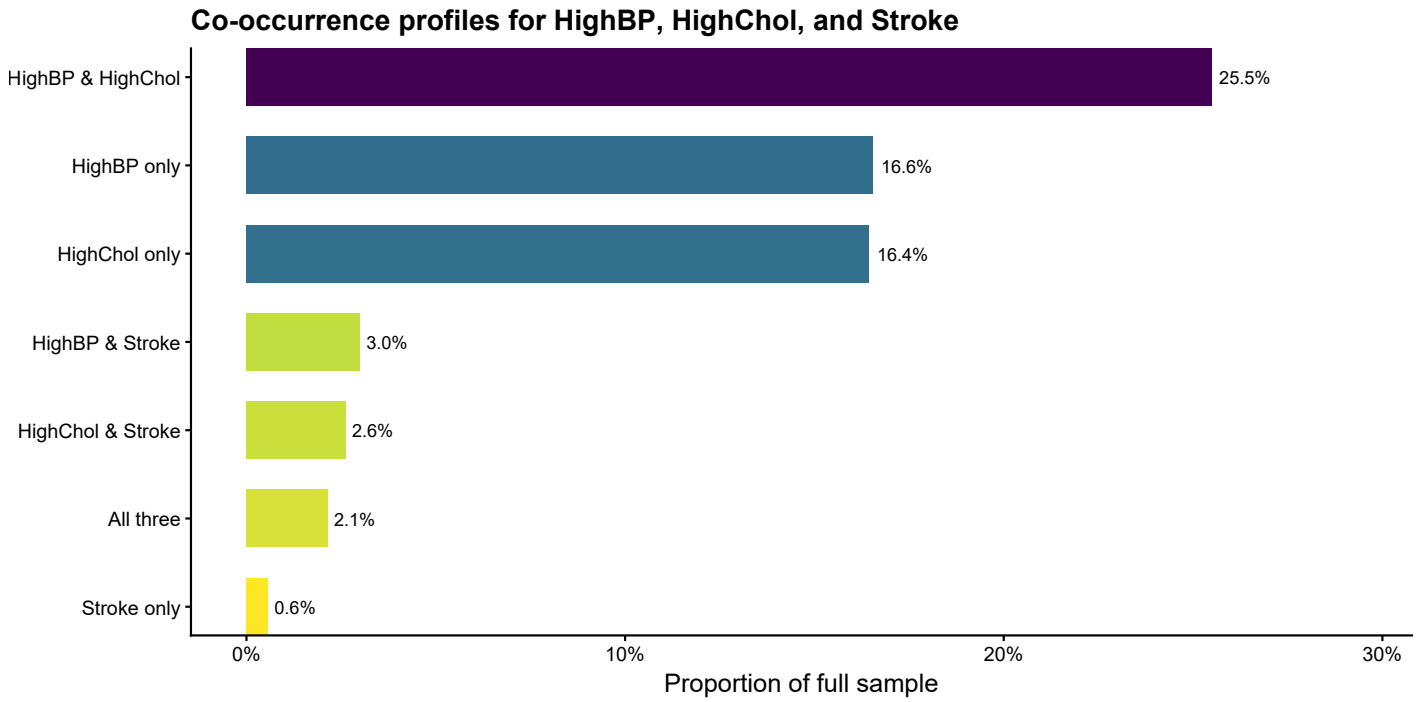


Stroke prevalence increases monotonically across cardiovascular profiles. The highest prevalence is observed among individuals with both high blood pressure and high cholesterol, followed by isolated high blood pressure, isolated high cholesterol, and neither condition.

The gradient suggests cumulative vascular burden, with blood pressure appearing more strongly associated with stroke than cholesterol alone. Stroke thus reflects downstream risk concentration rather than isolated cardiometabolic dysfunction.

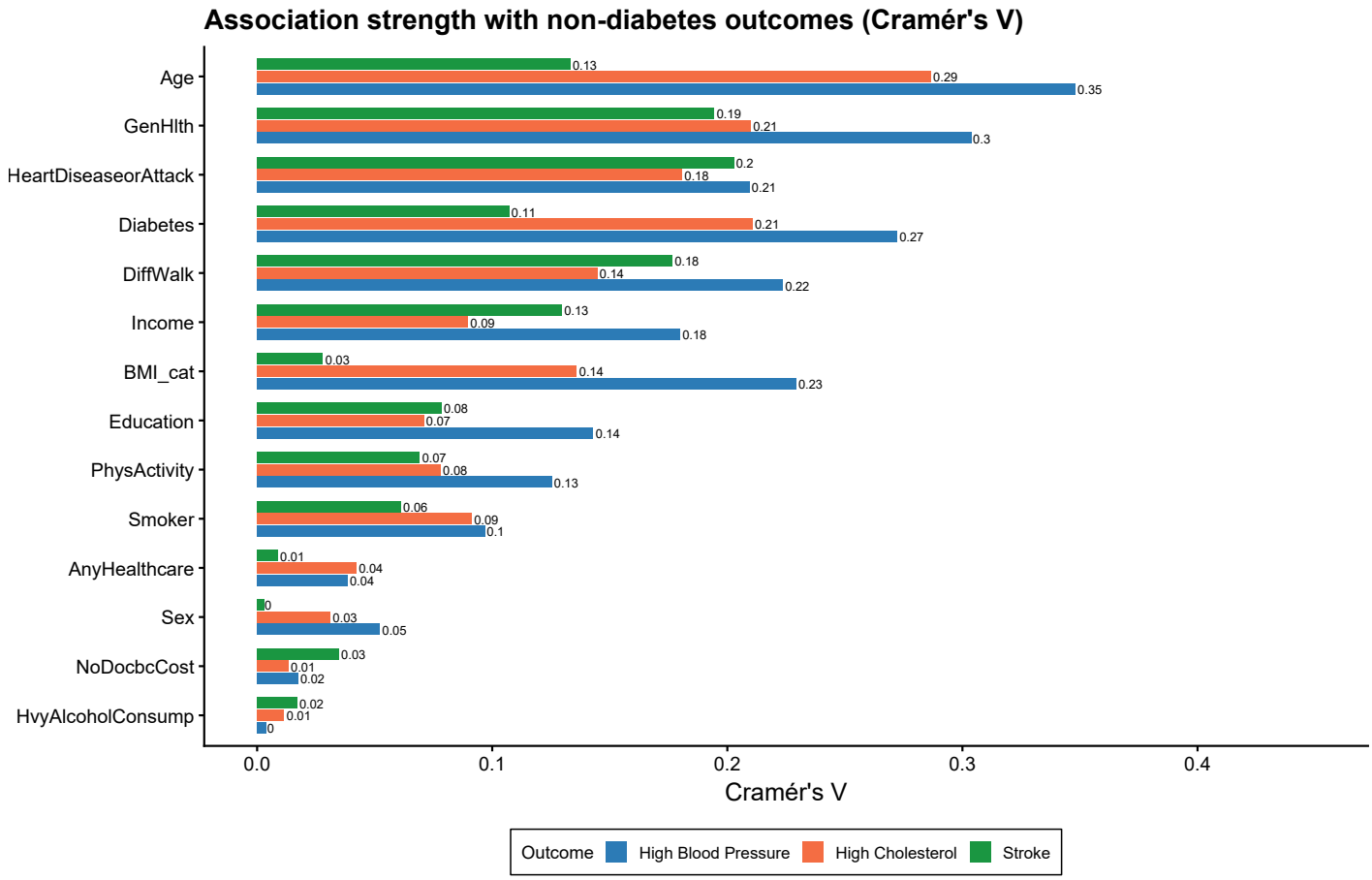
3.1.11 Joint co-occurrence profiles of high blood pressure, high cholesterol, and stroke

```
ggplot(joint, aes(x = reorder(Profile, Proportion), y = Proportion, fill = Proportion)) +
  geom_col(width = 0.65) +
  geom_text(
    aes(label = percent(Proportion, accuracy = 0.1)),
    hjust = -0.15,
    size  = 4.5
  ) +
  coord_flip() +
  scale_x_discrete(expand = expansion(mult = c(0.05, 0.05))) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    limits = c(0, max_prop * 1.15)
  ) +
  scale_fill_viridis_c(option = "D", direction = -1, guide = "none") +
  labs(
    title = "Co-occurrence profiles for HighBP, HighChol, and Stroke",
    x     = NULL,
    y     = "Proportion of full sample"
  )
```



3.1.12 Association Strength - Cramér's V

```
ggplot(cv_nondm,
       aes(x = reorder(Predictor, CramersV),
           y = CramersV,
           fill = Outcome)) +
  geom_col(position = position_dodge(width = 0.75), width = 0.7) +
  geom_text(aes(label = round(CramersV, 2)),
            position = position_dodge(width = 0.75),
            hjust = -0.1, size = 3.2) +
  coord_flip() +
  expand_limits(x = max(cv_nondm$CramersV) + 0.03) +
  scale_fill_manual(values = c(
    "High Blood Pressure" = "#2c7bb6",
    "High Cholesterol"    = "#f46d43",
    "Stroke"              = "#1a9641"
  )) +
  scale_y_continuous(limits = c(0, 0.45)) +
  labs(
    title = "Association strength with non-diabetes outcomes (Cramér's V)",
    x     = NULL,
    y     = "Cramér's V",
    fill  = "Outcome"
  ) +
  theme(legend.position = "bottom")
```



This chart highlights how different risk factors weigh against cardiovascular outcomes using **Cramér's V** (a measure of association where higher values indicate a stronger relationship).

The data reveals two distinct "profiles" of disease:

1. **Metabolic & Age-Driven (High BP & Cholesterol)**

Hypertension and High Cholesterol are primarily driven by **systemic, long-term factors**.

- **Primary Drivers: Age** ($V \approx 0.35$) and **General Health** ($V \approx 0.30$) are the strongest predictors.
- **Secondary Drivers:** Strong links to **Diabetes** and **BMI**, suggesting these are markers of general metabolic decline.
- **Takeaway:** These conditions follow a broad, population-wide risk gradient shaped by getting older and overall physical state.

2. **Clinical & Pathological (Stroke)**

Stroke follows a different logic, focusing more on **existing damage** than just age.

- **Primary Drivers: Prior Heart Disease** ($V \approx 0.20$), **General Health** ($V \approx 0.19$), and **Walking Difficulty** ($V \approx 0.18$).
- **Key Difference:** Age is significantly less dominant here than it is for Blood Pressure. BMI has almost no direct association with Stroke in this specific view.
- **Takeaway:** Stroke appears as a "clinical endpoint", it is most strongly associated with patients who already show signs of cardiovascular failure or physical impairment.

Outcome	Strongest Predictor	Second Predictor
High Blood Pressure	Age (0.35)	General Health (0.30)
High Cholesterol	Age (0.29)	General Health (0.21)

Stroke	Heart Disease (0.20)	General Health (0.19)

3.1.13 Summary of Key Findings

Section 3.1 establishes that cardiovascular risk in this dataset is structured along three primary axes: **age, socioeconomic position, and cardiometabolic clustering**.

First, **age is the dominant organising factor**. High blood pressure and high cholesterol rise steadily from midlife onward, while stroke is far more sharply concentrated in older age groups. Stroke appears less as a gradual condition and more as a late-stage vascular outcome.

Second, a **clear and consistent socioeconomic gradient** is evident across all three outcomes. Lower education and lower income strata show substantially higher crude prevalence, with monotonic declines across both dimensions. Although partly influenced by age composition, the two-dimensional gradient suggests an independent structural role for socioeconomic position.

Third, **cardiometabolic conditions cluster rather than occur independently**. Dual high blood pressure and high cholesterol (25.5%) exceed either isolated condition, and stroke prevalence increases monotonically across BP/cholesterol profiles, peaking in the dual-high group. This supports a cumulative vascular burden framework.

Fourth, **obesity strongly structures hypertension and cholesterol but not stroke**, reinforcing the interpretation of BP and cholesterol as intermediate metabolic states. Stroke instead aligns more strongly with prior heart disease, general health, and walking difficulty.

Finally, association analyses (Cramér's V) reveal two distinct disease profiles:

- **Hypertension and cholesterol** are predominantly age and general-health driven.
- **Stroke** is more closely associated with established cardiovascular pathology and functional impairment.

Overall, Section 3.1 demonstrates that cardiovascular outcomes in this dataset are not randomly distributed but follow coherent demographic, socioeconomic, and pathological structures. These findings provide the empirical foundation for the modelling strategy developed in Section 4.

3.1.14 Implications for Modelling

The exploratory analysis of cardiometabolic outcomes suggests several considerations for subsequent modelling. First, the strong accumulation pattern observed across cardiometabolic risk factors indicates that lifestyle and metabolic variables such as obesity, smoking, physical inactivity, and diabetes are likely to act as key predictors of cardiovascular outcomes. Second, the association strength analysis shows that demographic and health-status variables—particularly age and self-rated general health—have the strongest relationships with high blood pressure and high cholesterol, indicating that these variables should be prioritised in predictive models.

For stroke, the strongest associations appear with prior cardiovascular disease history, poorer general health, and functional limitation (difficulty walking), suggesting that stroke risk may be more closely linked to underlying health status and comorbidity than to behavioural factors alone. Finally, the relatively low prevalence of stroke compared with hypertension and hypercholesterolaemia suggests that models predicting stroke may face class imbalance and may require approaches suitable for rare outcomes.

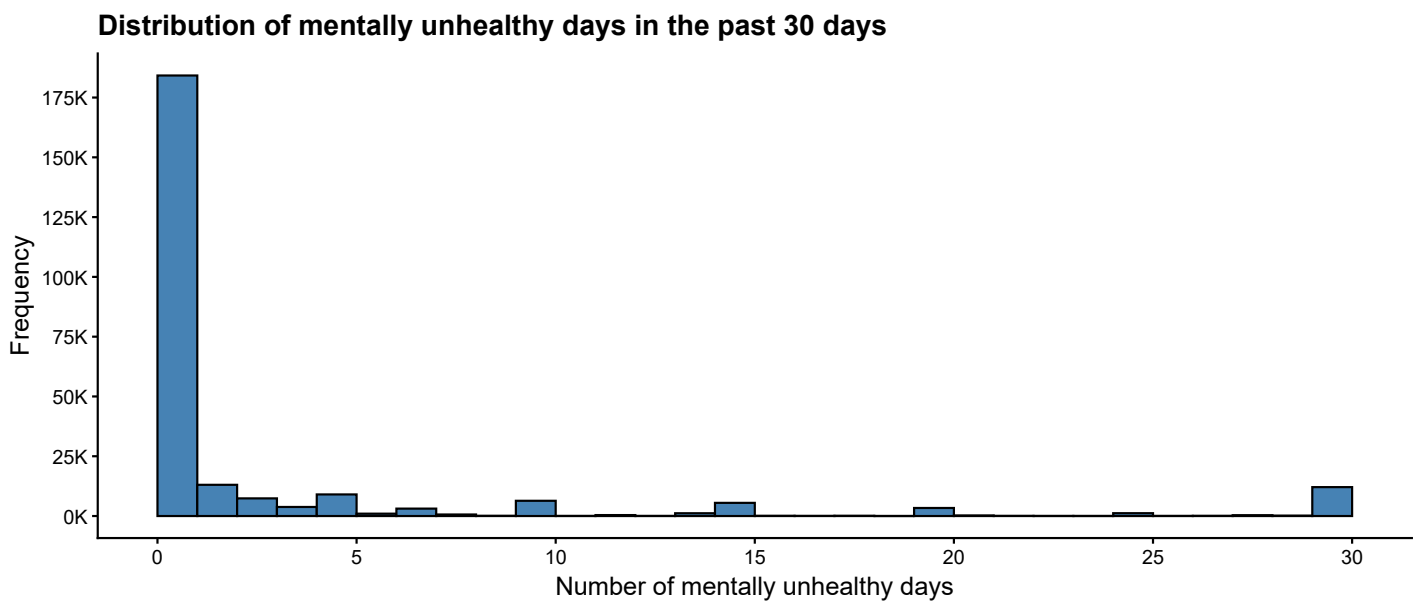
Overall, the EDA indicates that **demographic factors, general health status, and cardiometabolic risk profiles** are likely to be central components of the modelling framework used to investigate cardiovascular outcomes.

3.2 Patterns and determinants of mental health burden

This section examines the distribution and key correlates of **mental health burden**, measured as the number of mentally unhealthy days reported in the past 30 days. The analysis explores how mental health burden varies across demographic, socioeconomic, and health-related factors, including age, physical health, and lifestyle behaviours. By identifying systematic patterns in mental health outcomes, this section aims to highlight the factors most strongly associated with elevated psychological distress in the population.

3.2.1 Distribution of mentally unhealthy days in the past 30 days

```
ggplot(diab, aes(x = MentHlth)) +
  geom_histogram(
    binwidth = 1,
    boundary = 0,
    fill = "steelblue",
    colour = "black"
  ) +
  scale_x_continuous(
    breaks = seq(0, 30, by = 5),
    limits = c(0, 30)
  ) +
  scale_y_continuous(
    labels = label_number(scale = 1e-3, suffix = "K"),
    breaks = seq(0,200000,by = 25000)
  ) +
  labs(
    title = "Distribution of mentally unhealthy days in the past 30 days",
    x = "Number of mentally unhealthy days",
    y = "Frequency")
```

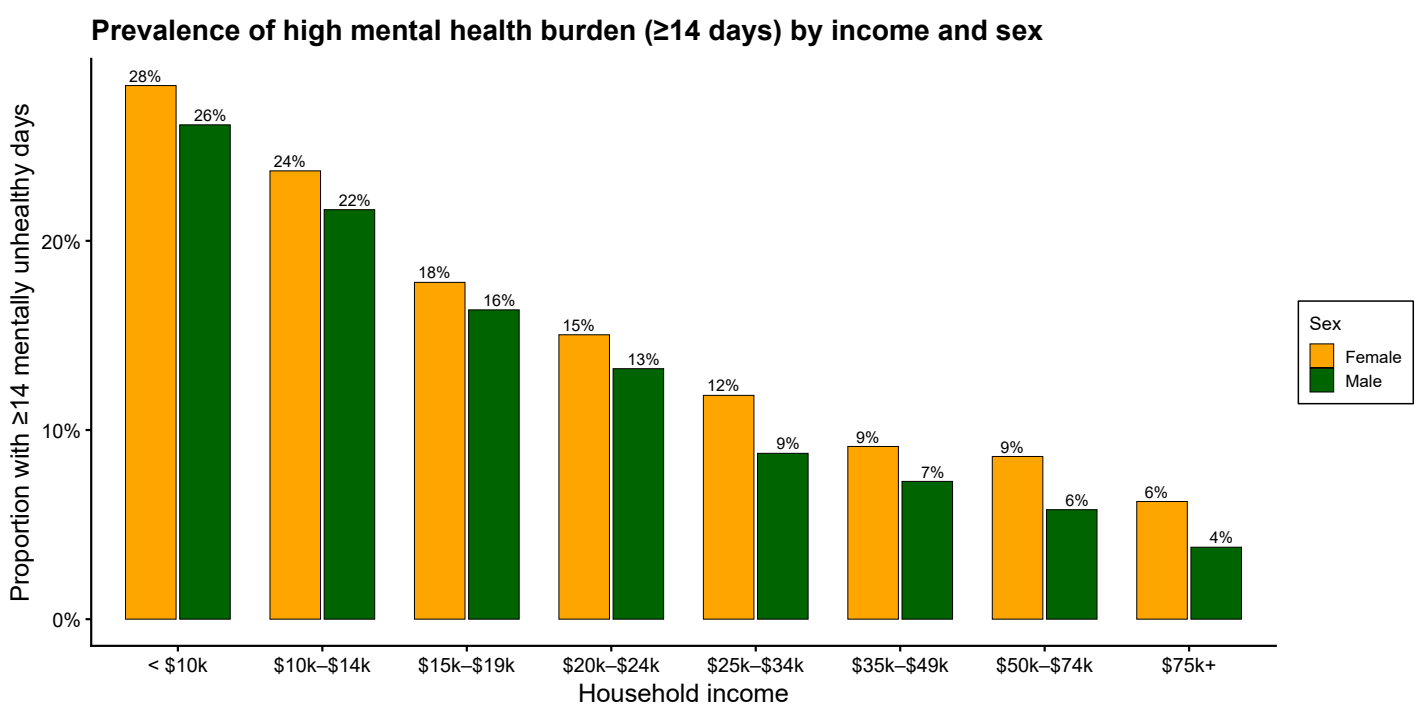


The distribution of mentally unhealthy days is **highly right-skewed**, with a very large proportion of respondents reporting **zero mentally unhealthy days in the past month**. Frequencies decline sharply as the number of unhealthy days increases, producing a long right tail that extends toward the upper bound of 30 days. This pattern indicates that while most individuals report little or no recent mental health burden, a smaller subset experiences **frequent or persistent distress**.

The distribution also shows evidence of **excess zeros and strong dispersion**, with the spike at zero far exceeding the frequencies of neighbouring values. Such a structure suggests that standard linear modelling would be inappropriate, and motivates the use of **count-based models**, potentially including **zero-inflated or overdispersed models (e.g., Poisson or zero-inflated negative binomial)** when analysing mental health burden as an outcome in subsequent modelling.

3.2.3 Mental Health Burden by Income and Sex

```r
diab |>
  mutate(HighMH = MentHlth >= 14) |>
  group_by(Income, Sex) |>
  summarise(prop_high = mean(HighMH), .groups = "drop") |>

  ggplot(aes(x = Income, y = prop_high, fill = Sex)) +
  geom_col(
    position = position_dodge(width = 0.75),
    width = 0.7,
    colour = "black",
    linewidth = 0.2
  ) +
  geom_text(
    aes(label = percent(prop_high, accuracy = 1)),
    position = position_dodge(width = 0.9),
    vjust = -0.3, size = 4) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    limits = c(0, NA)
  ) +
  scale_fill_manual(
    values = pal_sex
  ) +
  labs(
    title = "Prevalence of high mental health burden ( 14 days) by income and sex",
    x     = "Household income",
    y     = "Proportion with  14 mentally unhealthy days",
    fill  = "Sex"
  )
```



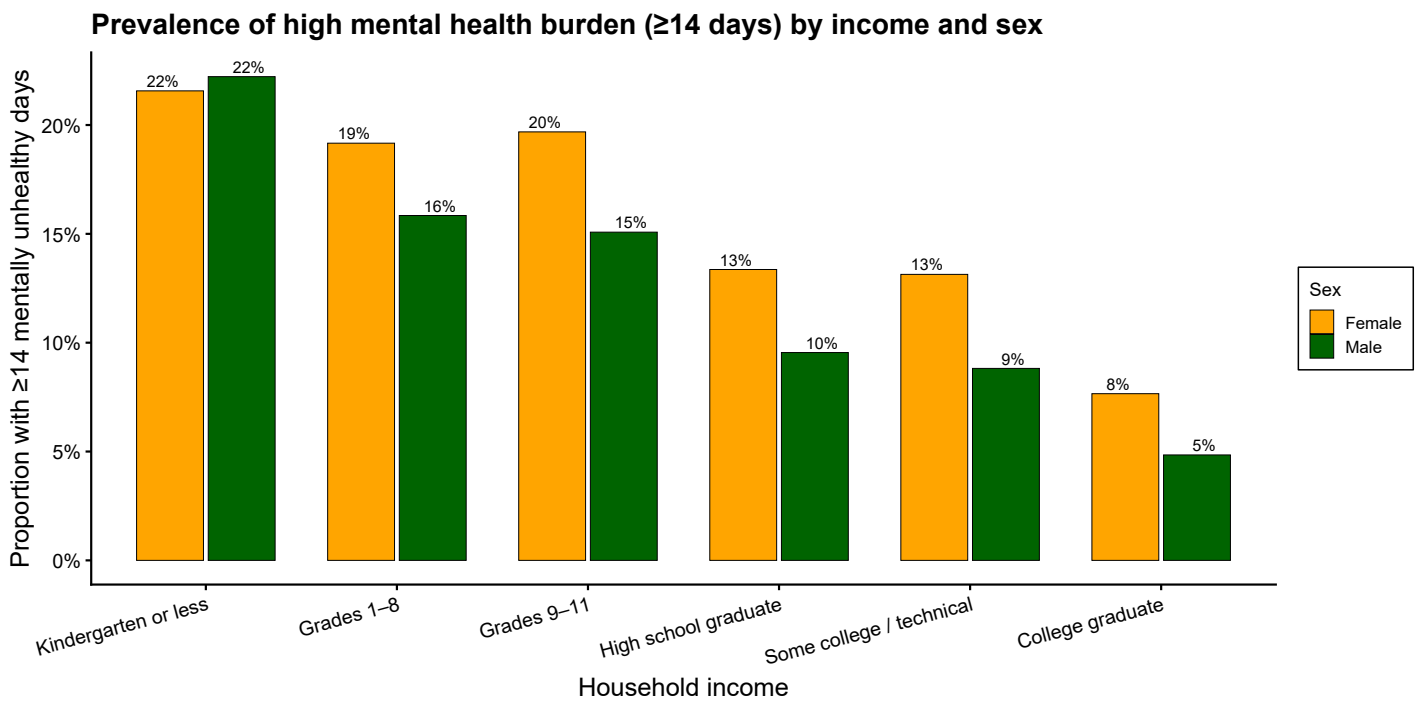
A clear socioeconomic gradient is evident in the prevalence of high mental health burden. The proportion of respondents reporting ≥14 mentally unhealthy days declines steadily as household income increases, indicating that individuals in lower-income groups experience substantially higher levels of psychological distress. In the lowest income category (<$10k), roughly 28% of females and 26% of males report high mental health burden,

compared with only 6% and 4% respectively in the highest income group ($75k+).

Across all income categories, females consistently report higher prevalence than males, although the difference between sexes is modest relative to the overall income gradient. These results suggest that socioeconomic position is a strong determinant of mental health burden, with financial disadvantage associated with markedly higher levels of frequent psychological distress. (Money doesn't buy happiness :))

3.2.3 Mental Health Burden by Educatioand Sex

```
diab |>
  mutate(HighMH = MentHlth >= 14) |>
  group_by(Education, Sex) |>
  summarise(prop_high = mean(HighMH), .groups = "drop") |>

  ggplot(aes(x = Education, y = prop_high, fill = Sex)) +
  geom_col(
    position = position_dodge(width = 0.75),
    width = 0.7,
    colour = "black",
    linewidth = 0.2
  ) +
  geom_text(
    aes(label = percent(prop_high, accuracy = 1)),
    position = position_dodge(width = 0.9),
    vjust = -0.3, size = 4) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    limits = c(0, NA)
  ) +
  scale_fill_manual(
    values = pal_sex
  ) +
  labs(
    title = "Prevalence of high mental health burden ( 14 days) by income and sex",
    x     = "Household income",
    y     = "Proportion with  14 mentally unhealthy days",
    fill  = "Sex"
  ) +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))
```



A clear educational gradient is visible in the prevalence of high mental health burden. Individuals with lower levels of educational attainment report substantially higher proportions of ≥14 mentally unhealthy days compared with those with higher education. For example, among respondents with kindergarten or less education, approximately 22% report high mental health burden, whereas this proportion declines to around 8% for females and 5% for males among college graduates.

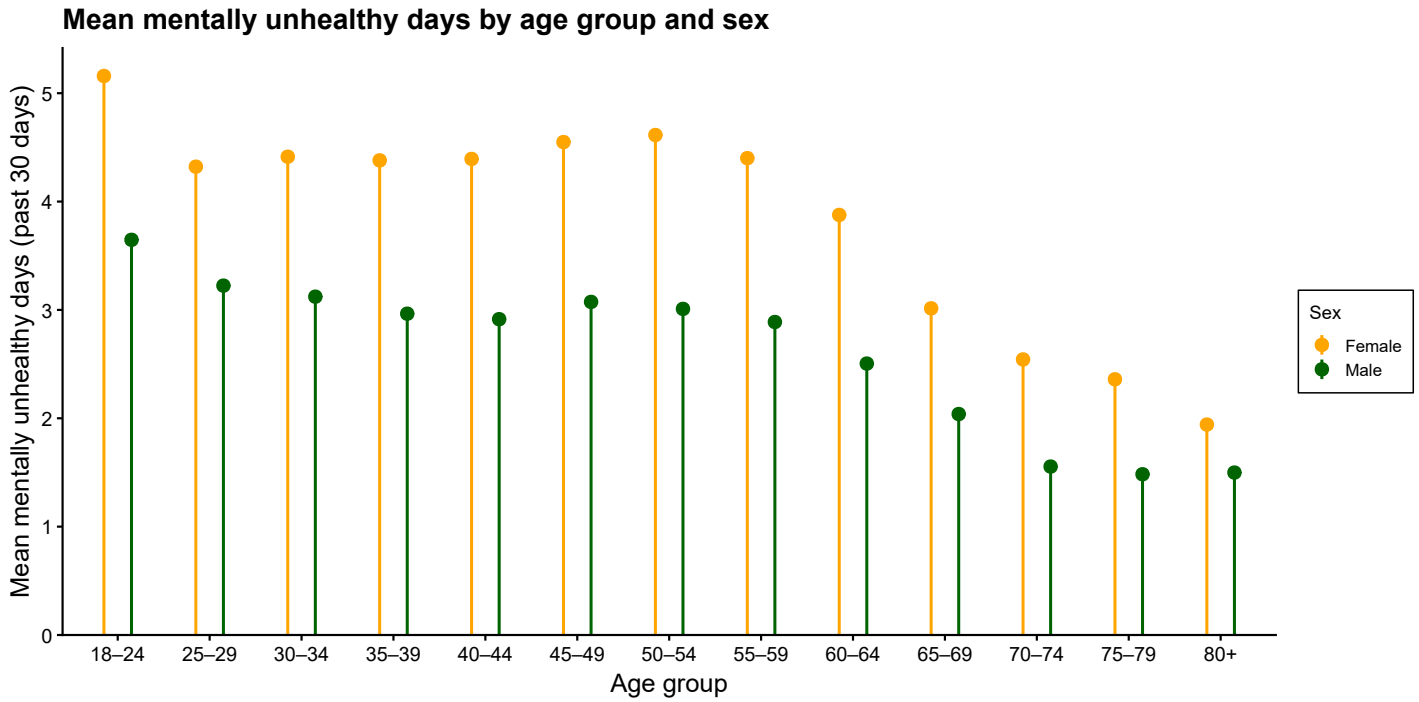
Across all education categories, females consistently report higher levels of mental health burden than males, although the difference between sexes is relatively modest compared with the overall education gradient. The steady decline in prevalence with increasing educational attainment suggests that socioeconomic resources and opportunities associated with education may play an important role in mental health outcomes.

Overall, the figure indicates that lower educational attainment is associated with substantially higher levels of frequent psychological distress, reinforcing the broader pattern of socioeconomic inequalities in mental health observed in the income analysis.

3.2.4 Mental Health Burden by Age Group and Sex

```
mh_age_sex <- diab |>
  group_by(Age, Sex) |>
  summarise(mean_mh = mean(MentHlth, na.rm = TRUE), .groups = "drop")

pos_dodge <- position_dodge(width = 0.6)

ggplot(mh_age_sex, aes(x = Age, y = mean_mh, colour = Sex)) +
  geom_linerange(
    aes(ymin = 0, ymax = mean_mh),
    position = pos_dodge,
    linewidth = 1
  ) +
  geom_point(
    position = pos_dodge,
    size = 3.8
  ) +
  scale_y_continuous(
    limits = c(0, NA),
```

```
    expand = expansion(mult = c(0, 0.05))
  ) +
  scale_colour_manual(
    values = pal_sex
  ) +
  labs(
    title  = "Mean mentally unhealthy days by age group and sex",
    x      = "Age group",
    y      = "Mean mentally unhealthy days (past 30 days)",
    colour = "Sex"
  )
```

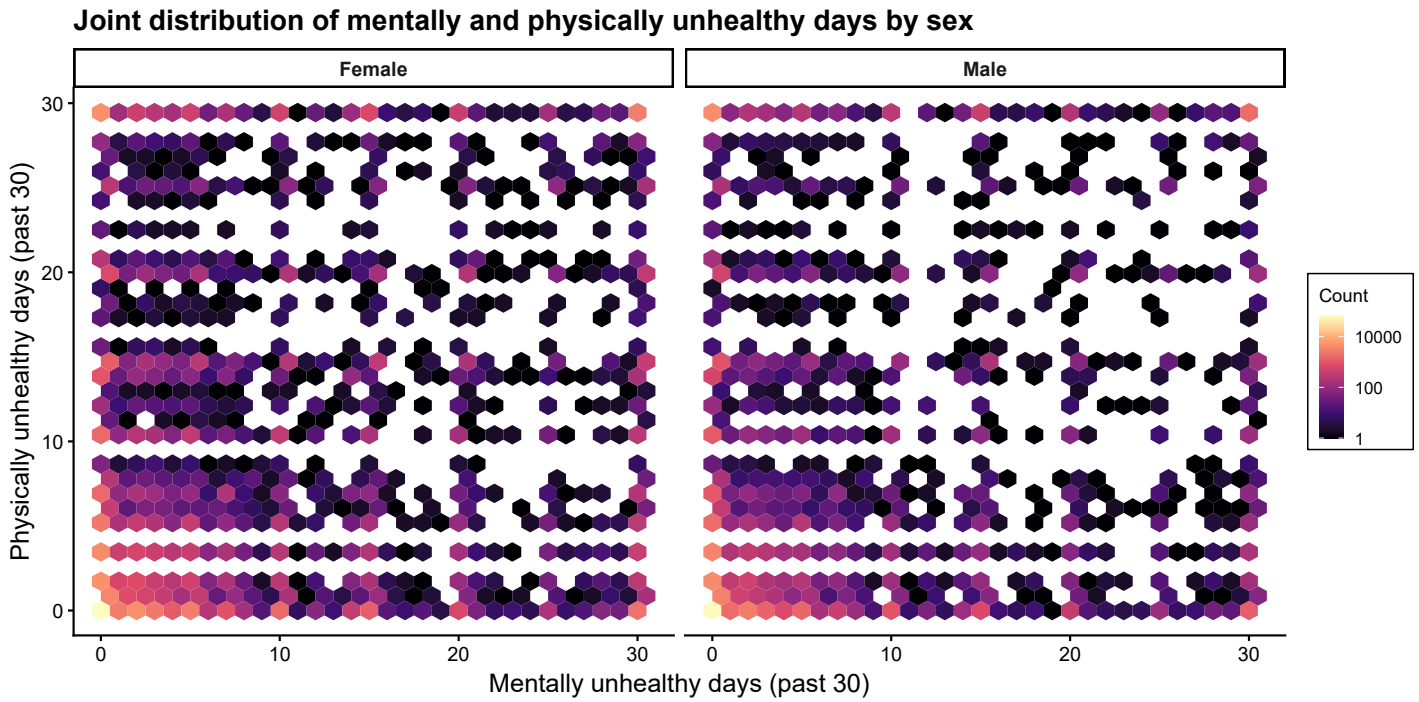


Mean mentally unhealthy days are higher among women than men in every age group. The difference is most visible in younger and middle adulthood, where women report roughly 1–2 more days on average than men.

Across age, the pattern is not monotonic. Mean days are relatively high in early adulthood, remain similar through midlife, and then decline progressively after around age 60 for both sexes. The lowest averages are observed in the oldest age groups.

3.2.5 Joint Distribution of Mental and Physical Health Burden by Sex

```
ggplot(diab, aes(x = MentHlth, y = PhysHlth)) +
  geom_hex(bins = 30) +
  scale_fill_viridis_c(option = "magma", name = "Count", trans = "log10") +
  facet_wrap(~ Sex, nrow = 1) +
  labs(
    title = "Joint distribution of mentally and physically unhealthy days by sex",
    x     = "Mentally unhealthy days (past 30)",
    y     = "Physically unhealthy days (past 30)"
  )
```



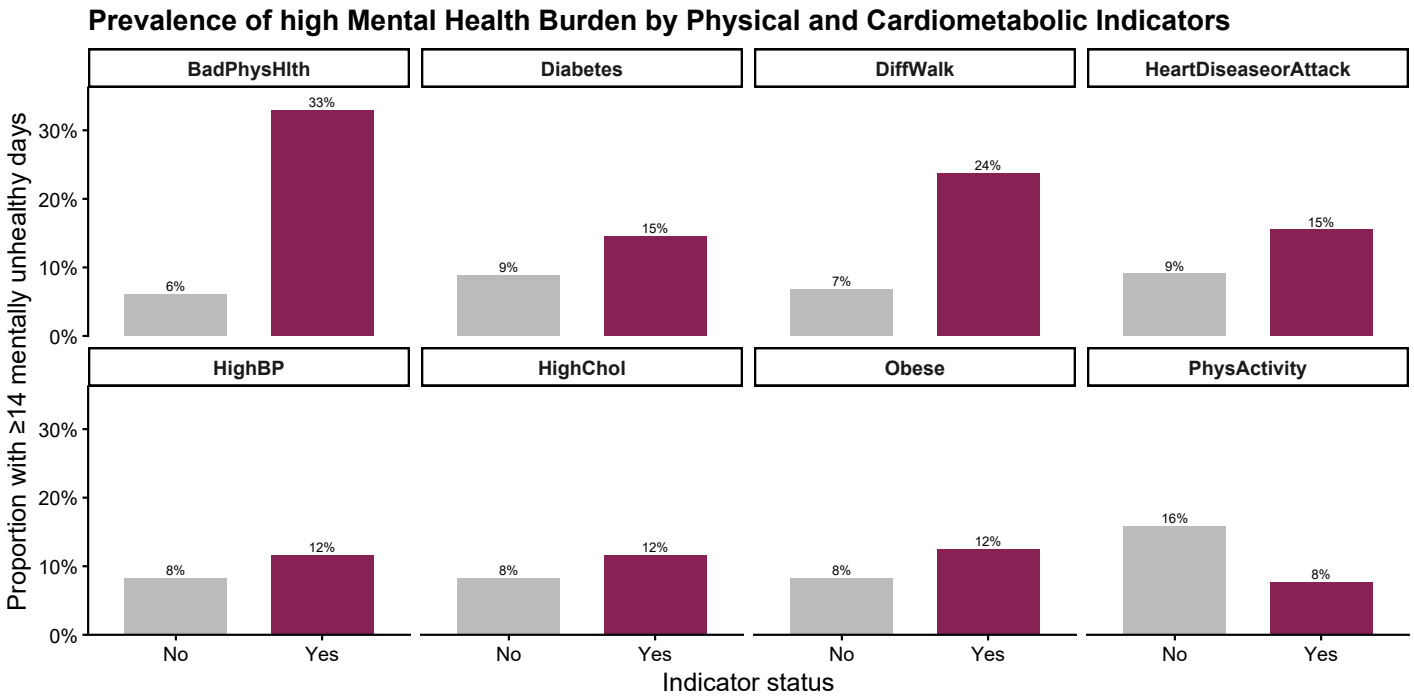
In both panels, the highest concentration of observations occurs near zero mentally and physically unhealthy days. Density decreases as values increase along either axis, and combinations of simultaneously high mental and physical burden are relatively sparse.

There is visible clustering along diagonal regions, indicating that higher numbers of mentally unhealthy days tend to coincide with higher numbers of physically unhealthy days rather than being concentrated along a single axis.

The overall distributional shape is similar for women and men, with broadly comparable density patterns across low, moderate, and higher values.

3.2.6 Prevalence of High Mental Health Burden by Physical and Cardiometabolic indicators

```
diab_mh_phys |>
  ggplot(aes(x = Status, y = prop, fill = Status)) +
  geom_col(width = 0.7, show.legend = FALSE) +
  geom_text(
    aes(label = percent(prop, accuracy = 1)),
    vjust = -0.3,
    size  = 3.3
  ) +
  facet_wrap(~ Indicator, nrow = 2, ncol = 4) +
  scale_y_continuous(
    labels = percent_format(accuracy = 1),
    limits = c(0, NA),
    expand = expansion(mult = c(0, 0.10))
  ) +
  scale_fill_manual(values = pal_binary) +
  labs(
    title = "Prevalence of high Mental Health Burden by Physical and Cardiometabolic Indicators",
    x     = "Indicator status",
    y     = "Proportion with  14 mentally unhealthy days"
  )
```



High mental health burden (≥14 mentally unhealthy days) varies substantially across physical and cardiometabolic indicators. The strongest associations are observed for indicators of functional and physical health limitations. Individuals reporting high physical health burden (≥14 physically unhealthy days) show a markedly elevated prevalence of high mental health burden (≈ 33%), compared with only about 6% among those without high physical burden. A similarly large difference appears for difficulty walking, where prevalence rises from roughly 7% to 24%.

Cardiometabolic conditions such as diabetes, heart disease or myocardial infarction, high blood pressure, and high cholesterol are also associated with higher levels of mental health burden, although the magnitude of these differences is smaller. For example, respondents with diabetes or heart disease exhibit prevalence around 15%, compared with approximately 9% among those without these conditions.

Lifestyle factors show a mixed pattern. Obesity is associated with moderately higher mental health burden (≈12% vs ≈8%), while physical activity appears protective, with prevalence roughly halved among individuals reporting recent physical activity (≈ 8%) compared with those reporting none (≈16%).

Overall, the results suggest that mental health burden is most strongly linked with physical health limitations and functional impairment, while cardiometabolic disease and lifestyle factors exhibit weaker but still noticeable associations.

3.2.7 Summary of Key Findings

Several clear structural patterns emerge in the analysis of mental health burden. First, the distribution of mentally unhealthy days is highly right-skewed, with a large concentration at zero and a long tail toward higher values. This indicates substantial heterogeneity in psychological distress across the population and motivates the use of count-based models with potential zero inflation in subsequent modelling.

Second, strong **socioeconomic gradients** are visible. The prevalence of high mental health burden declines consistently with increasing income and education, suggesting that socioeconomic disadvantage is associated with substantially higher levels of frequent psychological distress.

Third, **demographic differences** are evident. Women report higher average numbers of mentally unhealthy days than men across all age groups, while mental health burden tends to decline gradually in older age groups after midlife.

Fourth, mental and physical health burdens show **moderate co-occurrence**, with higher levels of mentally unhealthy days tending to coincide with higher levels of physically unhealthy days.

Finally, mental health burden is **most strongly associated with indicators of physical limitation**, particularly poor physical health and difficulty walking. Cardiometabolic conditions such as diabetes, heart disease, hypertension, and high cholesterol also show positive associations with mental health burden, although these relationships are weaker than those linked to functional impairment.

Overall, the results suggest that mental health burden in this population is structured primarily by **socioeconomic position, physical health status, and functional limitation**, providing a clear empirical basis for the modelling analyses developed in later sections.

3.2.8 Implications for Modelling

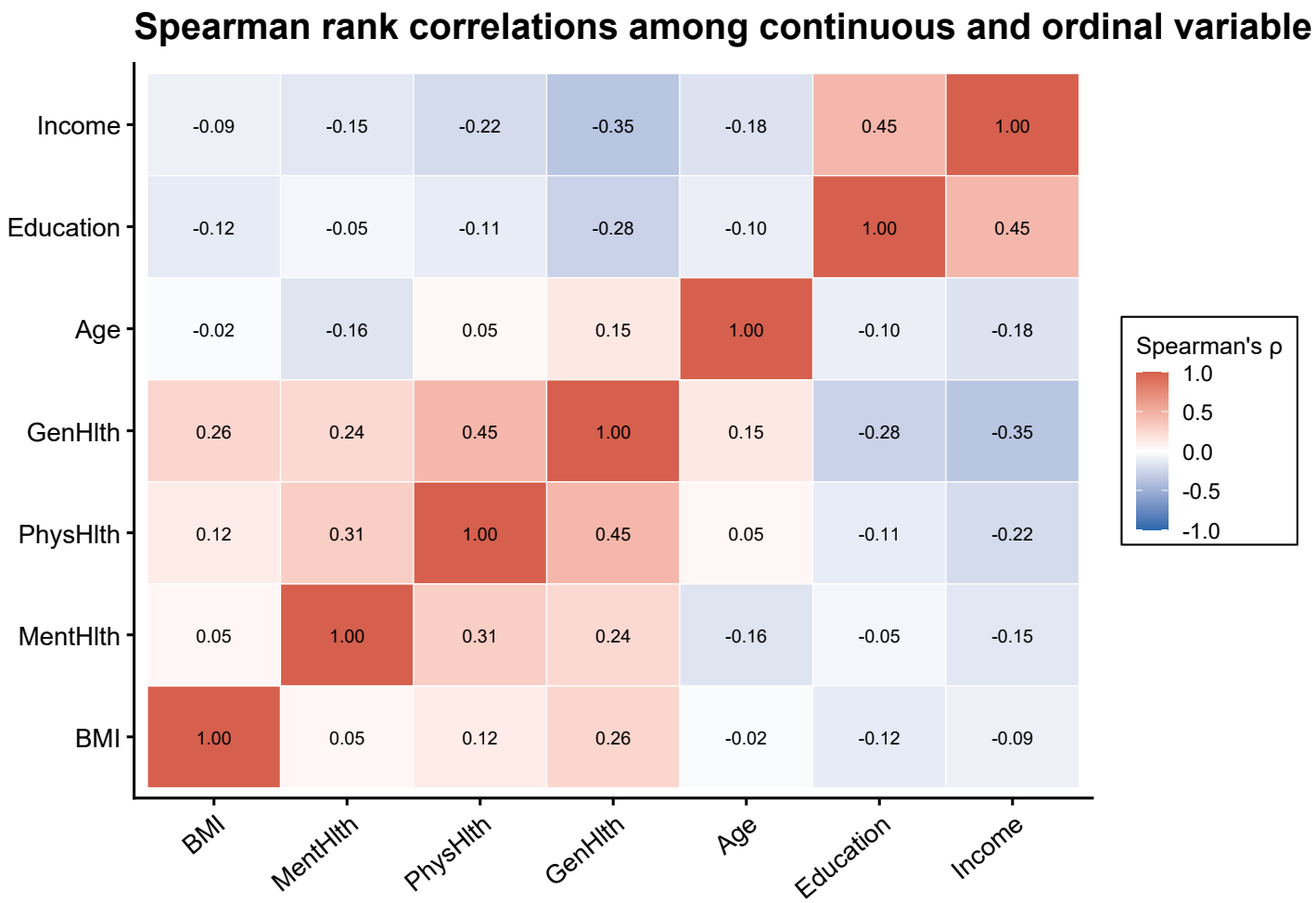
The exploratory analysis provides several insights relevant for model development. The distribution of mentally unhealthy days is highly right-skewed with a substantial concentration at zero, suggesting that standard linear modelling would be inappropriate and motivating the use of count-based models such as Poisson or negative binomial regression, potentially with zero-inflation.

Socioeconomic gradients observed across income and education indicate that demographic and socioeconomic variables should be included as key predictors.

Finally, the strong co-occurrence between mental and physical health indicators suggests that physical health variables such as poor physical health days, difficulty walking, and chronic disease indicators may play an important explanatory role in modelling mental health burden.

3.x.x Spearman Correlation Heatmap: Continuous and Ordinal Variables

```r
ggplot(spear_tidy, aes(x = Var1, y = Var2, fill = rho)) +
  geom_tile(colour = "white") +
  geom_text(
    aes(label = sprintf("%.2f", rho)),
    size   = 3.5,
    colour = "black"
  ) +
  scale_fill_gradient2(
    low      = "#2166AC",
    mid      = "white",
    high     = "#D6604D",
    midpoint = 0,
    limits   = c(-1, 1),
    name     = "Spearman's \u03c1"
  ) +
  labs(
    title = "Spearman rank correlations among continuous and ordinal variables",
    x     = NULL,
    y     = NULL
  ) +
  theme(
    axis.text.x = element_text(angle = 40, hjust = 1),
    axis.text.y = element_text(hjust = 1)
  )
```



The Spearman rank correlations indicate mostly **weak to moderate** monotonic associations among the continuous and ordinal variables.

- The strongest positive correlations are between **GenHlth and PhysHlth** ($\rho \approx 0.45$) and between **Education and Income** ($\rho \approx 0.45$), suggesting moderate monotonic relationships in both cases.
- **GenHlth** is moderately **negatively** correlated with **Income** ($\rho \approx -0.35$) and **Education** ($\rho \approx -0.28$). Because higher GenHlth scores denote poorer health, these negative correlations imply that higher income and education are associated with **better** self-rated health.
- **MentHlth** and **PhysHlth** show a modest positive association ($\rho \approx 0.31$), consistent with some co-occurrence of mental and physical health burden.
- Most other pairwise correlations are small ($|\rho| < 0.3$). In particular, **BMI** and **Age** have only weak associations with the other variables in this set.

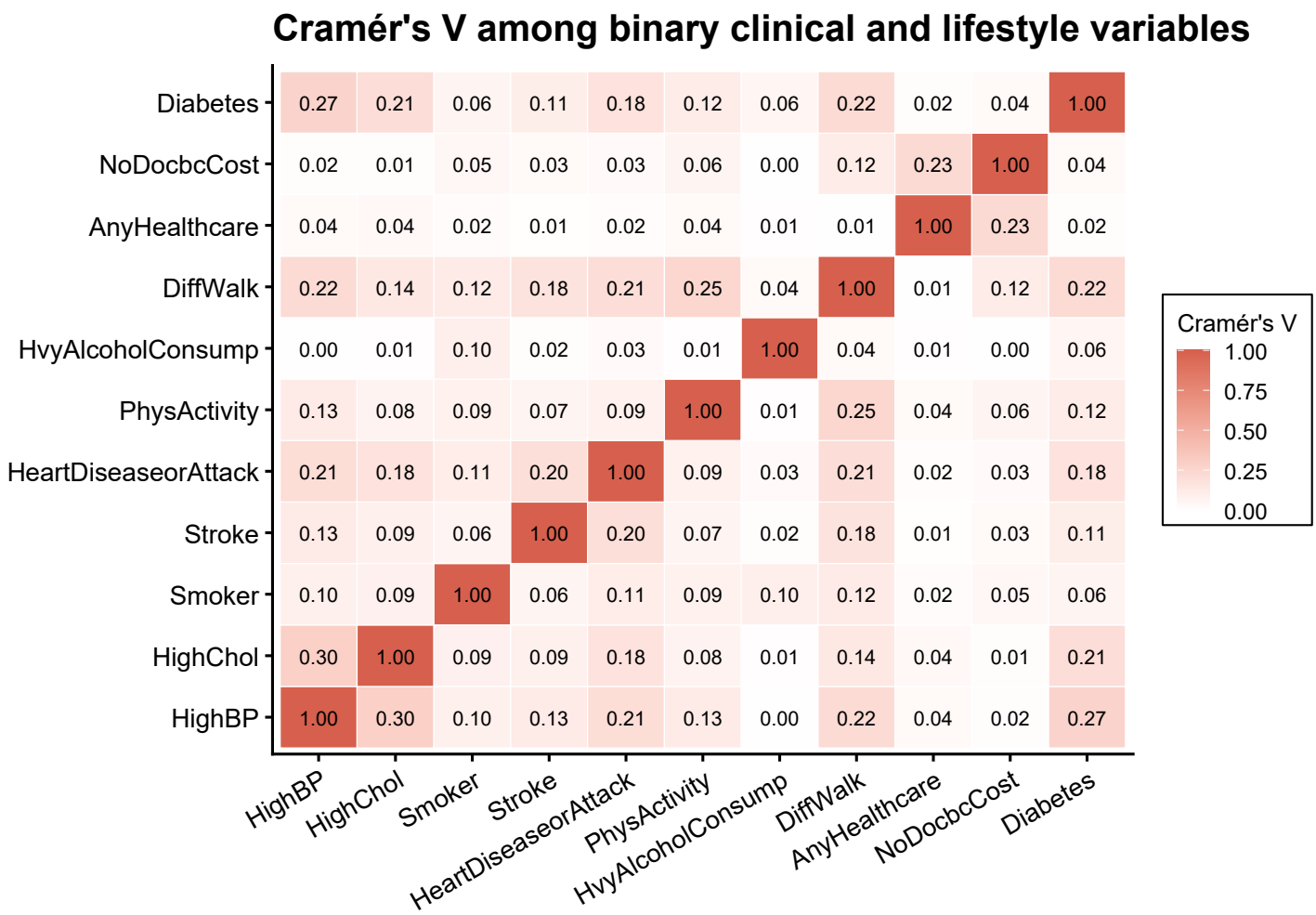
Overall, the correlation matrix does **not** indicate problematic collinearity: apart from a few moderate associations (GenHlth–PhysHlth, Education–Income), most variables are only weakly related. This supports their concurrent use as predictors in multivariable models, subject to routine multicollinearity checks.

3.x.x Cramér's V Association Heatmap: Binary Clinical and Lifestyle Variables

To complement the Spearman correlation heatmap for continuous/ordinal variables, the strength of pairwise associations among all binary clinical and lifestyle indicators is quantified using Cramér's V. This symmetric measure summarises the association in each 2×2 contingency table on a 0–1 scale, without assuming any directionality. Diabetes status is deliberately excluded.

Together, Sections 3.14 and 3.15 establish the correlation structure among all key variables, identifying which predictors are inter-related and justifying the need for multivariable adjustment in all subsequent models.

```r
ggplot(cv_tidy, aes(x = Var1, y = Var2, fill = V)) +
  geom_tile(colour = "white") +
  geom_text(
    aes(label = sprintf("%.2f", V)),
    size   = 3.8,
    colour = "black"
  ) +
  scale_fill_gradient2(
    low       = "white",
    high      = "#D6604D",
    midpoint = 0,
    limits   = c(0, 1),
    name     = "Cramér's V"
  ) +
  labs(
    title = "Cramér's V among binary clinical and lifestyle variables",
    x = NULL,
    y = NULL) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
```



The Cramér's V matrix indicates that most pairwise associations among the binary clinical and lifestyle variables are **weak**, with the majority of values below 0.20.

The strongest association is between **HighBP and HighChol** (V ≈ 0.30), representing a moderate relationship relative to the rest of the matrix. This aligns with known clustering of cardiovascular risk factors.

Other comparatively larger associations include:

- **PhysActivity and DiffWalk** (V ≈ 0.25)
- **HighBP and DiffWalk** (V ≈ 0.22)
- **HighBP and HeartDiseaseorAttack** (V ≈ 0.21)
- **Stroke and HeartDiseaseorAttack** (V ≈ 0.20)

These moderate associations are clinically plausible and reflect shared cardiovascular or functional impairment pathways.

Lifestyle indicators such as **Smoker** and **HvyAlcoholConsump** exhibit generally weak associations with other variables (typically V < 0.15). Healthcare access indicators (**AnyHealthcare**, **NoDocbcCost**) also show minimal association with most clinical variables.

Analytical Implication

The matrix does not suggest substantial redundancy among the binary predictors. Aside from a small number of moderate associations (primarily within cardiovascular-related variables), most relationships are weak.

This supports inclusion of these variables simultaneously in multivariable models, with routine collinearity diagnostics (e.g., VIF or condition indices) as a precaution rather than an expected issue.

4. Refined Research Questions

RQ1: Multidomain Determinants and Prediction of Diabetes Status

Outcome Variable: *Diabetes — Multiclass Outcome*

Research Question

> ***To what extent can diabetes status be explained and predicted using demographic, socioeconomic, behavioural, and cardiometabolic factors, and which predictors contribute most strongly to distinguishing between no diabetes, prediabetes, and diabetes?***

Rationale

Exploratory analysis of the CDC Diabetes Health Indicators dataset indicates that diabetes status is structured by multiple interacting determinants. Section 3.2.1 demonstrated a strong **age gradient**, with prediabetes and diabetes concentrated in older age groups, suggesting that demographic factors must be accounted for when evaluating other risk indicators.

Cardiometabolic clustering was evident across several visualisations. The joint distribution of **high blood pressure and high cholesterol** increased substantially across diabetes categories, while the risk-factor accumulation plot showed a clear gradient in which diabetes prevalence rises with the number of cardiometabolic risk indicators. This pattern aligns with established evidence linking obesity, hypertension, and dyslipidaemia to type 2 diabetes risk (Ismail et al., 2021).

Behavioural factors also exhibited variation across diabetes categories. Physical activity levels tended to decline with worsening diabetes status, while smoking and alcohol consumption showed weaker separation. Because behavioural variables are potentially modifiable, evaluating their contribution to diabetes risk is relevant for prevention strategies (Aune et al., 2015).

Finally, socioeconomic gradients were visible in the education–income distribution, where diabetes prevalence was higher among lower socioeconomic groups. These patterns suggest that diabetes risk is embedded within broader social determinants of health, which influence exposure to risk factors and access to care (Hill-Briggs et al., 2021).

Together, these findings motivate a multivariable modelling approach to estimate adjusted associations and evaluate predictive performance across diabetes categories.

'EDI Discussed in Task1'

RQ2: The Impact of Socioeconomic Moderation on Cardiovascular Risk

Outcome Variables: *HeartDiseaseorAttack* — binary outcome

Research Question

> ***To what extent do socioeconomic determinants, specifically income and education, moderate the impact of clinical risk factors on the prevalence of heart disease, and can the capture of complex feature interactions provide superior predictive insights compared to traditional additive assumptions?***

Rationale

Exploratory analysis identified distinct gradients in cardiovascular health, where the prevalence of heart disease increases substantially across older age groups and among individuals with specific clinical markers. While factors such as high blood pressure, elevated cholesterol, and body mass index are established universal risk indicators, their influence is rarely isolated. In real-world clinical contexts, these physiological markers exist within a broader socioeconomic framework, suggesting that the "weight" of a clinical risk factor may be compounded or mitigated by an individual's access to resources, environmental stressors, and educational background (Roth et al., 2024).

Traditional analytical frameworks often rely on the assumption that risks are simply additive, yet biological and social systems frequently exhibit synergistic effects where the presence of one factor significantly alters the impact of another. For example, the cardiovascular strain associated with high BMI may be intensified in lower-income brackets due to limited access to preventive care or higher cumulative stress (Collins et al., 2023). By investigating these conditional relationships, this inquiry shifts the focus from simple binary classification toward a more nuanced understanding of risk stratification. This approach allows for the identification of specific "risk profiles," which are intersections of clinical and social data that might be overlooked by frameworks that fail to account for higher-order interactions.

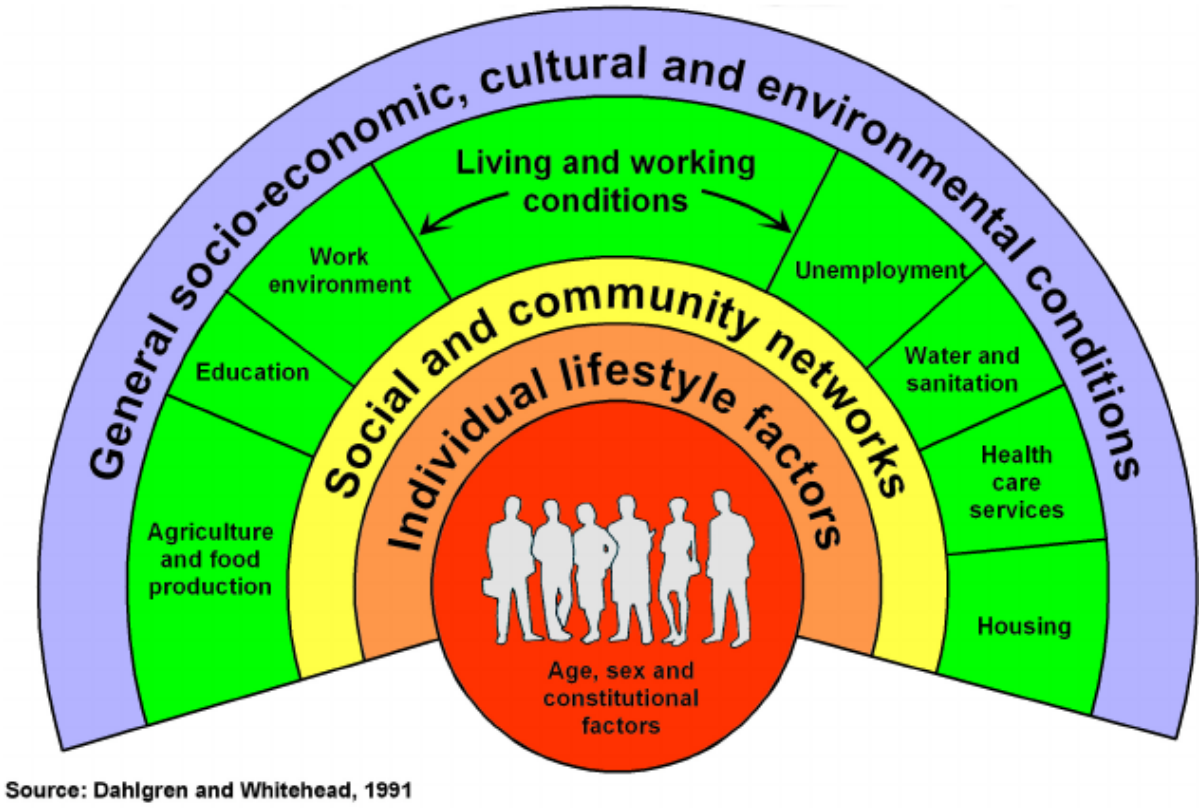


Figure 1: The Dahlgren-Whitehead Rainbow Model illustrating the layers of health determinants, providing the theoretical basis for the moderation analysis of income and education in RQ2.

As visualized in figure, individual clinical risks like blood pressure and BMI are nested within broader socioeconomic environments. This framework justifies the shift from simple predictive modelling to an investigation of how these external layers moderate physiological risk.

EDI and Ethical Considerations

The application of predictive modelling in healthcare necessitates a rigorous focus on transparency and equity. A primary ethical concern is the risk of "algorithmic bias," where models may inadvertently penalise or misrepresent specific subgroups if demographic variables are treated in isolation. By specifically investigating socioeconomic moderators, this research aims to move beyond a "one-size-fits-all" predictive approach, ensuring that the influence of income and education is explicitly understood rather than buried as "noise" within the data.

Furthermore, the interpretability of these complex interactions is a matter of distributive justice, as clinicians must be able to explain risk assessments to patients in a way that is both accurate and contextually relevant. By quantifying how socioeconomic status shifts the predictive significance of clinical markers, this study helps identify whether existing healthcare inequalities are being reinforced or challenged by the data. Ensuring that the model identifies these non-linear patterns is essential for developing more equitable preventative strategies that reflect the lived realities of diverse population groups (Liu et al., 2023).

RQ3: Socioeconomic and Lifestyle Determinants of Mental Health Burden

Outcome Variable: *MentHlth* — number of self-reported mentally unhealthy days in the past 30 days (0–30)

Research Question

> *To what extent do income, education, physical activity, smoking, and heavy alcohol consumption predict the number of mentally unhealthy days reported in the past 30 days, and does the association between physical activity and mental health burden differ across income groups?*

Rationale

The *MentHlth* variable represents a count of mentally unhealthy days within the past month (0–30). Exploratory analysis revealed a strongly right-skewed distribution with a large concentration of zero values, indicating that many respondents report no mentally unhealthy days while a smaller subset experience substantial burden. Such distributions violate assumptions of linear models and often produce overdispersion and excess zeros, making standard Poisson or logistic approaches unsuitable.

A Zero-Inflated Negative Binomial (ZINB) model is therefore appropriate because it separates two processes:

1. the probability that individuals experience no mental health burden, and
2. the frequency of mentally unhealthy days among those who do report burden (Rust et al., 2014; Liu et al., 2023).

The EDA further revealed clear socioeconomic gradients in mental health outcomes. Respondents in the lowest income categories exhibited substantially higher prevalence of high mental health burden compared with those in higher income groups. Physical activity also showed a consistent inverse relationship with mentally unhealthy days, suggesting potential behavioural protection.

These patterns motivate a modelling approach that jointly evaluates socioeconomic position and lifestyle behaviours. In particular, testing an interaction between income and physical activity allows examination of whether the protective association of physical activity differs across socioeconomic strata.

Prior research supports the relevance of these predictors. Socioeconomic disadvantage has been consistently linked with elevated risk of mental health disorders and poorer psychological well-being (Kessler et al., 2010), while analyses of BRFSS data have shown that ZINB regression provides an effective framework for modelling unhealthy-days indicators with excess zeros (Rust et al., 2014; Liu et al., 2023).

EDI / Ethical Considerations

Socioeconomic gradients in mental health outcomes raise important equity considerations. If income remains strongly associated with mental health burden after adjustment for behavioural factors, this suggests that structural determinants, such as economic insecurity and access to resources, play a major role in shaping mental health outcomes. Understanding these relationships can inform more equitable public health strategies that address structural conditions rather than focusing solely on individual behaviour.

6. Multidomain Determinants and Prediction of Diabetes Status

6.1 Modelling Objective

This section addresses the following research question:

> ***To what extent can diabetes status be explained and predicted using demographic, socioeconomic, behavioural, and cardiometabolic factors, and which predictors contribute most strongly to distinguishing between no diabetes, prediabetes, and diabetes?***

The objective is twofold:

1. **Explanatory** – to estimate the adjusted associations between predictor domains and diabetes status.
2. **Predictive** – to compare the classification performance of statistical and machine learning models.

Diabetes status is treated as a **multiclass outcome** (No diabetes / Prediabetes / Diabetes), requiring modelling approaches suitable for nominal categorical outcomes.

6.2. Outcome Variable

The dependent variable is **diabetes status**, categorised into:

- **No diabetes**
- **Prediabetes**
- **Diabetes.**

Exploratory analysis demonstrated class imbalance, with the majority of respondents in the no-diabetes category and substantially smaller proportions in the prediabetes and diabetes groups. This imbalance is accounted for in model training and evaluation.

6.3 Training and Validation

6.3.1 Data Partitioning and Resampling:

The dataset is partitioned into a 75% training set and a 25% testing set using stratified sampling on diabetes status. Within the training set, 5-fold stratified cross-validation is implemented to estimate model performance during development. The held-out test set is reserved for final evaluation to ensure unbiased performance assessment.

```
set.seed(42)

diabetes_split <- initial_split(
  diab,
  prop = 0.75,
  strata = Diabetes
)

diab_train <- training(diabetes_split)
diab_test  <- testing(diabetes_split)


diabetes_folds <- vfold_cv(
  diab_train,
  v = 5,
  strata = Diabetes
)

# Sanity checks
diab_train |>  count(Diabetes) |> mutate(prop = n/sum(n))
```

Diabetes	n	prop
No diabetes	160255	0.8422992
Prediabetes	3485	0.0183171
Diabetes	26519	0.1393837

```
diab_test |> count(Diabetes) |>  mutate(prop = n/sum(n))
```

Diabetes	n	prop
No diabetes	53448	0.8427492
Prediabetes	1146	0.0180697
Diabetes	8827	0.1391810

6.3.2 Recipe Definition and Preprocessing Strategy

- **Outcome specification** – Models `Diabetes` as a three-level categorical outcome (No diabetes, Prediabetes, Diabetes).
- **Near-zero variance removal** – Removes predictors with minimal variability to prevent instability and redundant information.
- **Ordinal scoring** – Converts ordered factors (`Age`, `Education`, `Income`, `GenHlth`) into numeric scores while preserving their natural ranking. This introduces an assumption of approximately equal spacing between adjacent categories; however, the primary objective is to retain monotonic structure in a computationally efficient representation suitable for both parametric and tree-based models.
- **Dummy encoding** – Transforms remaining nominal predictors into binary indicator variables suitable for modelling.
- **Normalization** – Standardizes numeric predictors (mean = 0, sd = 1) to improve optimisation and comparability across models.
- **SMOTE balancing** – Applies synthetic oversampling to minority classes within resampling folds to mitigate severe class imbalance.

```
set.seed(42)

rec_rq1 <- recipe(Diabetes ~ ., data = diab_train) |>
  step_nzv(all_predictors()) |>
  step_ordinalscore(Age, Education, Income, GenHlth) |>
  step_dummy(all_nominal_predictors()) |>
  step_normalize(BMI, MentHlth, PhysHlth) |>
  step_smote(Diabetes, over_ratio = 0.5, seed = 42)

print(rec_rq1)
```

6.3.3 Model Specification

We now define the candidate models for RQ1 using the **tidymodels** framework.

Each model is specified independently of preprocessing and will later be combined with the common recipe using **workflows** and **workflow sets**.

Model	Type	Primary Function	Key Strengths
Multinomial Logistic Regression	Parametric	Linear Baseline	Interpretable log-odds; direct coefficient analysis.

Random Forest	Ensemble (Bagging)	Non-linear Discovery	Captures complex interactions; robust to multicollinearity.
XGBoost	Ensemble (Boosting)	Predictive Performance	High efficiency on tabular data; optimized for accuracy.

Hyperparameters marked `tune()` will be optimized via grid search during cross-validation.

```
# Multinomial logistic regression
mod_multinom <-
  multinom_reg() |>
  set_engine("nnet") |>
  set_mode("classification")

# Random Forest
mod_rf <-
  rand_forest(
    trees = 256,
    mtry  = tune(),
    min_n = tune()
  ) |>
  set_engine("ranger") |>
  set_mode("classification")

# Gradient Boosted Trees
mod_xgb <-
  boost_tree(
    trees = 512,
    tree_depth = tune(),
    learn_rate = tune(),
    loss_reduction = tune(),
    mtry = tune(),
    min_n = tune()
  ) |>
  set_engine("xgboost") |>
  set_mode("classification")
```

6.3.4 Workflowset Construction

We now integrate:

- The **RQ1 preprocessing recipe**
- The **candidate model specifications**

  using `workflow_set()`.

Each workflow represents one **model–preprocessing pair**.
The workflow set enables parallel evaluation of all candidate models under identical resampling conditions.

```
# Workflow set
wf_set_rq1 <-
  workflow_set(
    preproc = list(rq1_recipe = rec_rq1),
    models  = list(
      multinom = mod_multinom,
```

```
      rf        = mod_rf,
      xgb       = mod_xgb))
```

6.3.5 Hyperparameter Tuning

Metrics computed

Model performance is evaluated using multiple complementary metrics to account for class imbalance and provide a robust assessment of predictive performance:

- **Accuracy** – overall classification rate
- **Balanced Accuracy** – mean recall across classes, correcting for class imbalance
- **Cohen's Kappa** – agreement between predictions and observed classes beyond chance
- **F1-score (macro)** – harmonic mean of precision and recall averaged across classes
- **ROC-AUC (multiclass)** – ability of the model to discriminate between diabetes categories

```
metrics_diab <-
  metric_set(
    accuracy,
    bal_accuracy,
    kap,
    f_meas,
    roc_auc
  )
```

Hyperparameter Tuning Strategy

We employ **grid search** to optimize hyperparameters across candidate models:

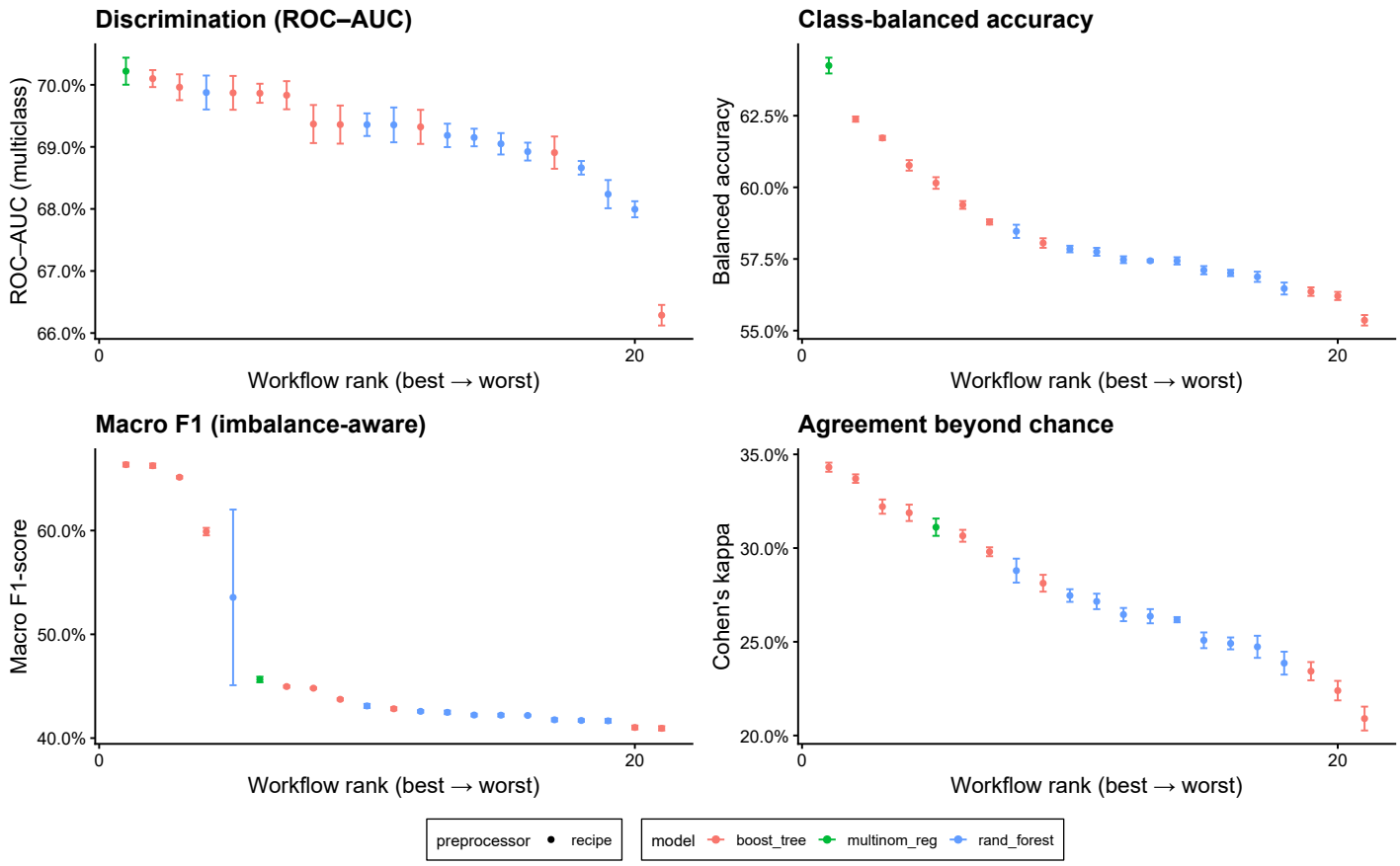
- **Grid size:** 10 configurations per model, balancing exploration of the hyperparameter space with computational feasibility (3 models × 10 grids × 5 CV folds = 150 model fits)
- **Search method:** Grid search over random/Bayesian approaches provides systematic coverage of parameter combinations and is computationally manageable for this dataset size
- **Tuning metric: Macro-averaged F1 score** serves as the primary optimization metric because it gives equal weight to all three diabetes classes (No diabetes, Prediabetes, Diabetes), preventing models from optimizing solely for majority class performance
- **Cross-validation:** 5-fold stratified CV estimates out-of-sample performance while maintaining class proportions in each fold

The best hyperparameter configuration for each model is selected based on highest mean Macro F1 across folds.

```
# set.seed(42)
# ctrl <- control_grid(
#   save_pred = TRUE,
#   save_workflow = FALSE,
#   verbose = TRUE
# )
#
# wf_set_rq1_fit <-
#   wf_set_rq1 |>
#   workflow_map(
#     "tune_grid",
#     resamples = diabetes_folds,
#     grid = 10,
#     metrics = metrics_diab,
#     control = ctrl
#     )
```

```r
wf_set_rq1_fit <- readRDS('wf_set_rq1_fit.rds')
```

```r
m1 <- pretty_autoplot(
  wf_set_rq1_fit,
  "roc_auc",
  "ROC-AUC (multiclass)",
  "Discrimination (ROC-AUC)")
m2 <- pretty_autoplot(
  wf_set_rq1_fit,
  "bal_accuracy",
  "Balanced accuracy",
  "Class-balanced accuracy")
m3 <- pretty_autoplot(
  wf_set_rq1_fit,
  "f_meas",
  "Macro F1-score",
  "Macro F1 (imbalance-aware)")
m4 <- pretty_autoplot(wf_set_rq1_fit, "kap", "Cohen's kappa", "Agreement beyond chance")

(m1 | m2) / (m3 | m4) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```



ROC–AUC.
ROC–AUC measures a model's ability to discriminate between the three diabetes categories across all possible classification thresholds. The highest ROC–AUC value is achieved by the **multinomial logistic regression model**, which appears as the top-ranked workflow in this metric. However, beyond this leading configuration, most of the remaining high-ranking workflows consist of **XGBoost and random forest models**, with values clustered between approximately 0.69 and 0.70.

This indicates that while logistic regression achieves the single best ROC–AUC score, tree-based models provide similarly competitive discrimination across multiple configurations.

Balanced Accuracy.
Balanced accuracy reflects the average recall across classes, giving equal importance to each diabetes category. The **multinomial logistic regression model again appears as the top-ranked workflow**, achieving the highest balanced accuracy among the candidate models. Several **XGBoost configurations follow closely**, while random forest models occupy the middle range of the ranking. Although performance differences are relatively modest, the results suggest that multiple model types are capable of achieving comparable class-balanced recall.

Macro F1 Score.
Macro F1 was selected as the primary tuning metric because it balances precision and recall equally across classes and is therefore appropriate under class imbalance. In this panel, the **top-ranked workflows are clearly dominated by XGBoost models**, which achieve substantially higher macro F1 scores than the other model types. Random forest configurations perform considerably lower, and the multinomial logistic regression model appears further down the ranking. This pattern indicates that boosted trees are better able to balance precision and recall across the three diabetes categories.

Cohen's Kappa.
Cohen's kappa measures agreement between predicted and observed classifications beyond chance. The highest kappa values are achieved by **XGBoost configurations**, which occupy several of the top-ranked positions in this panel. The multinomial logistic regression model appears among the higher-performing workflows but does not exceed the best boosted-tree configurations. Random forest models generally show lower kappa values and appear further down the ranking. Overall, this metric again indicates stronger agreement for boosted tree models compared with the other modelling approaches.

6.3.6 Extract tuning results

```
# Extract tuning results
rf_res  <- extract_workflow_set_result(wf_set_rq1_fit, "rq1_recipe_rf")
xgb_res <- extract_workflow_set_result(wf_set_rq1_fit, "rq1_recipe_xgb")

# Select best hyperparameters using Macro F1
best_rf  <- select_best(rf_res, metric = "f_meas")
best_xgb <- select_best(xgb_res, metric = "f_meas")

# Extract workflows
rf_wf       <- extract_workflow(wf_set_rq1_fit, "rq1_recipe_rf")
xgb_wf      <- extract_workflow(wf_set_rq1_fit, "rq1_recipe_xgb")
multinom_wf <- extract_workflow(wf_set_rq1_fit, "rq1_recipe_multinom")

# Finalise tuned workflows
rf_final_wf  <- finalize_workflow(rf_wf, best_rf)
xgb_final_wf <- finalize_workflow(xgb_wf, best_xgb)

# Multinomial requires no tuning, so it is already final
multinom_final_wf <- multinom_wf
```

```
rf_final_fit       <- fit(rf_final_wf, data = diab_test)
xgb_final_fit      <- fit(xgb_final_wf, data = diab_test)
multinom_final_fit <- fit(multinom_final_wf, data = diab_test)
```

6.4 Model Evaluation

```
pred_rf  <- get_test_preds( rf_final_fit,
                            "Random forest", diab_test, Diabetes)
pred_xgb <- get_test_preds(xgb_final_fit,
                           "XGBoost",       diab_test, Diabetes)
```

```r
pred_mn  <- get_test_preds(multinom_final_fit,
                           "Multinomial",   diab_test, Diabetes)

pred_all <- bind_rows(pred_rf, pred_xgb, pred_mn)

pred_all <- pred_all |>
  rename(.pred_No_diabetes = `.pred_No diabetes`)
```

```r
roc_all <- pred_all |>
  group_by(model) |>
  roc_curve(
    truth = Diabetes,
    .pred_No_diabetes,
    .pred_Prediabetes,
    .pred_Diabetes
  )
```

6.4.1 ROC Curves

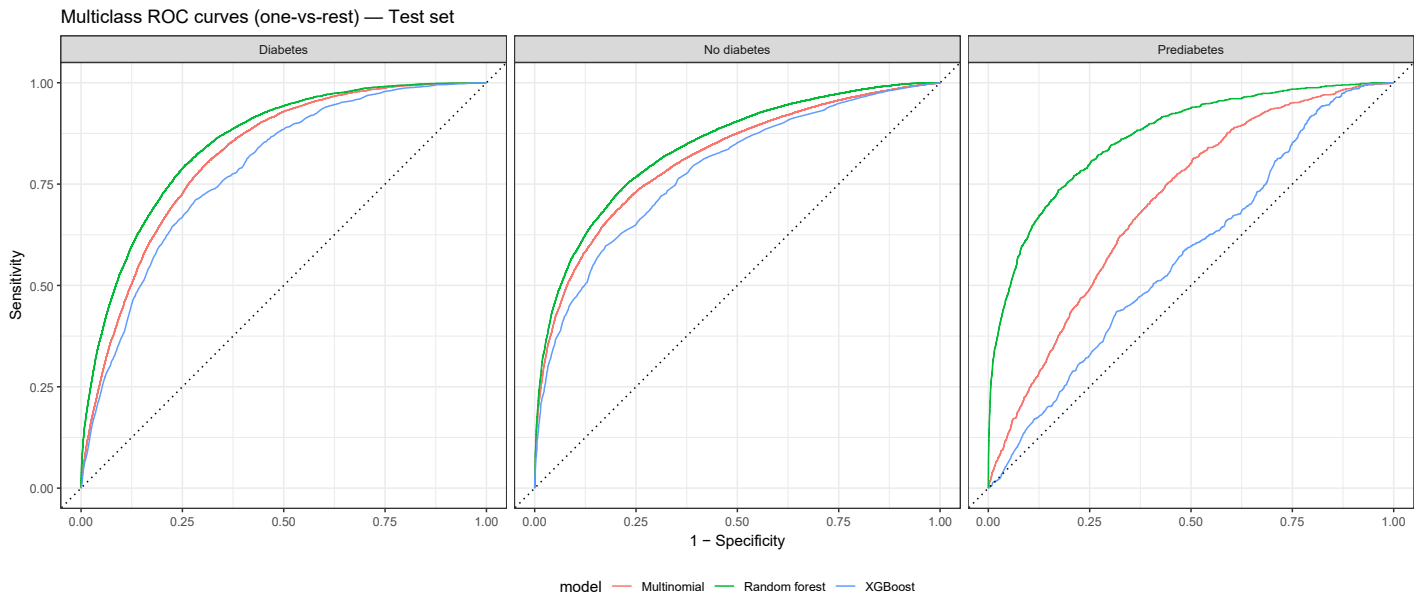
Multiclass ROC-AUC Evaluation

ROC (Receiver Operating Characteristic) curves for multiclass outcomes require extending the binary framework. We employ the **one-vs-rest approach**:

- Each class (No diabetes, Prediabetes, Diabetes) is treated as a separate binary problem against all other classes combined
- For each class, ROC curves plot True Positive Rate vs False Positive Rate across probability thresholds
- **Area Under Curve (AUC)** summarizes discriminative ability: AUC = 0.5 indicates random guessing, AUC = 1.0 indicates perfect separation
- Despite class imbalance, ROC-AUC remains informative for comparing relative model performance, though it may be optimistic for rare classes (Prediabetes, Diabetes)

Higher AUC values indicate better ability to distinguish each diabetes category from alternatives. Models with consistently high AUC across all three classes demonstrate robust multiclass discrimination.

```r
autoplot(roc_all) +
  labs(
    title = "Multiclass ROC curves (one-vs-rest) - Test set",
    x = "1 - Specificity",
    y = "Sensitivity"
  ) +
  theme(
    legend.position = 'bottom'
  )
```



The multiclass ROC curves provide additional insight into discrimination performance across the three classes.

For **diabetes** and **no-diabetes**, all models show strong discriminative ability, with curves well above the diagonal reference line. Differences between models are relatively small, though the tree-based models appear slightly stronger in some regions of the curve.

For the **prediabetes class**, however, the multinomial logistic regression model clearly outperforms both random forest and XGBoost. The ROC curve for the multinomial model remains consistently above the other two models, indicating better sensitivity–specificity trade-offs for detecting this intermediate category.

6.4.2 Confusion Matrices

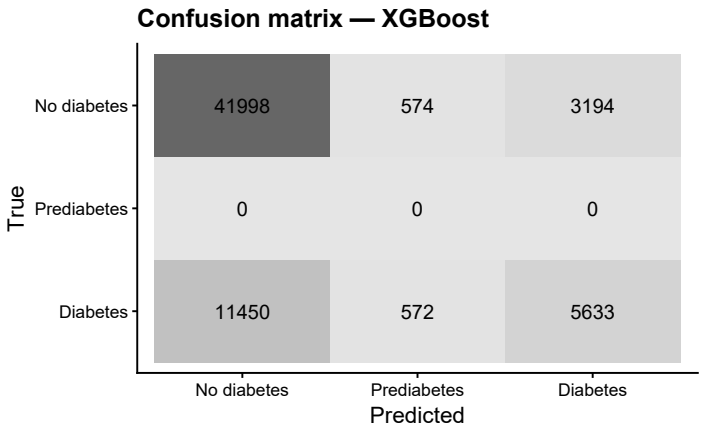
Diagnosing Prediabetes Classification Failure

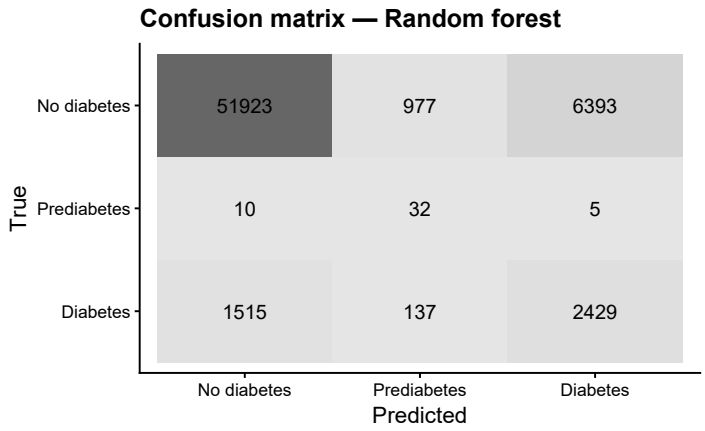
The confusion matrices reveal a critical pattern: **tree-based models (Random Forest, XGBoost) completely fail to predict the Prediabetes class**, collapsing to a binary No diabetes vs Diabetes distinction. This occurs due to several factors:

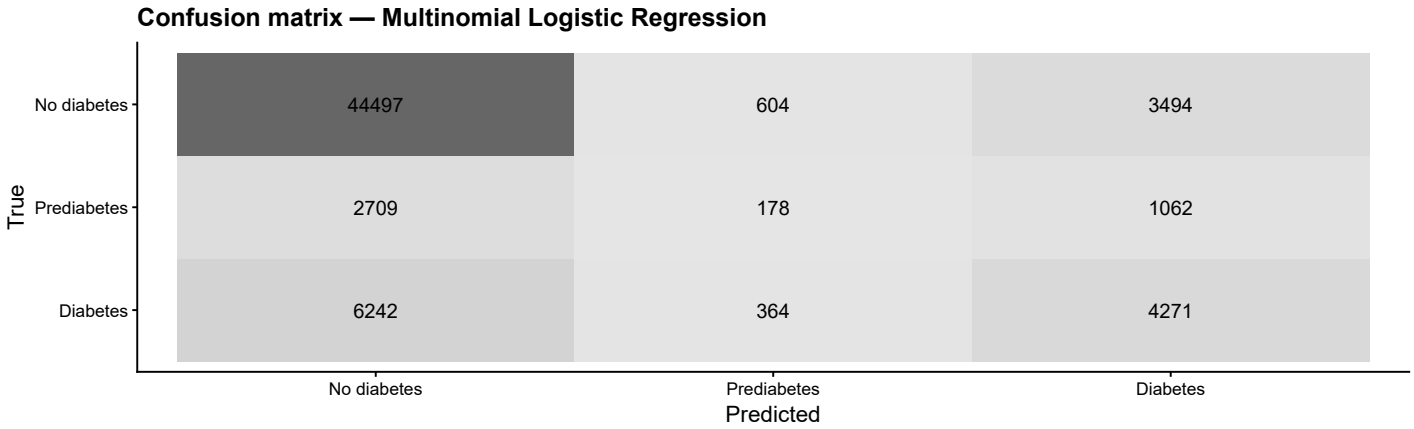
- **Class overlap:** Prediabetes represents an intermediate metabolic state sharing features with both No diabetes and Diabetes, creating ambiguous decision boundaries
- **Minority class optimization:** Despite SMOTE augmentation, tree-based algorithms optimize splits based on impurity reduction. When Prediabetes remains the smallest class, trees preferentially split on features that distinguish the dominant classes (No diabetes vs Diabetes), ignoring subtle Prediabetes-specific patterns
- **Cost function limitations:** Standard multi-class loss functions (e.g., cross-entropy) do not penalize misclassifying Prediabetes into adjacent categories differently than into distant ones, failing to capture ordinal structure

Multinomial logistic regression, by contrast, maintains some Prediabetes prediction by directly modeling class probabilities without hierarchical splitting, though accuracy remains modest. Alternative approaches (e.g., ordinal regression, cost-sensitive learning, focal loss) could better handle this intermediate class in future work.

```
p_cm_xgb <- plot_cm(pred_xgb, "Confusion matrix - XGBoost")
p_cm_rf  <- plot_cm(pred_rf,  "Confusion matrix - Random forest")
p_cm_mn  <- plot_cm(pred_mn,  "Confusion matrix - Multinomial Logistic Regression")

(p_cm_xgb | p_cm_rf) / p_cm_mn +
  plot_layout(guides = "collect")
```







The confusion matrices reveal notable differences in classification behaviour across the three models.

The **multinomial logistic regression model** is the only model that correctly identifies a meaningful number of **prediabetes cases** (150 observations). In contrast, both **XGBoost and random forest completely fail to predict the prediabetes class**, assigning all observations in this category to either the no-diabetes or diabetes classes. This indicates that the tree-based models effectively collapse the problem into a two-class classification task, ignoring the intermediate category.

For the **no-diabetes class**, all three models achieve high numbers of correct predictions due to the dominance of this category in the dataset. However, the multinomial model shows slightly higher correct classifications than the boosted tree model.

For the **diabetes class**, all models correctly identify several thousand cases, though misclassification between diabetes and no-diabetes remains present across all models.

Overall, the multinomial model demonstrates a **more balanced classification behaviour**, particularly by retaining predictive capacity for the minority prediabetes class.

Overall Assessment

Although gradient boosting and random forest performed competitively during cross-validation, the held-out test set evaluation suggests that **multinomial logistic regression** provides the most balanced performance across classes. In particular, it is the only model capable of correctly identifying prediabetes cases while maintaining strong discrimination for the other two classes.

6.5 Model Interpretation

Triangulation Strategy for Model Interpretation

To ensure robust identification of diabetes risk factors, **methodological triangulation** is employed by comparing interpretations across different analytical approaches:

- **Multinomial logistic regression (Log-Odds):** Provides parametric coefficient estimates assuming linear additive effects. Interpretable but limited to linear relationships

- **SHAP values (XGBoost):** Model-agnostic feature attribution capturing non-linear effects and interactions through tree-based partitioning
- **Variable Importance (XGBoost gain):** Measures total loss reduction when features are used for splits, reflecting predictive utility

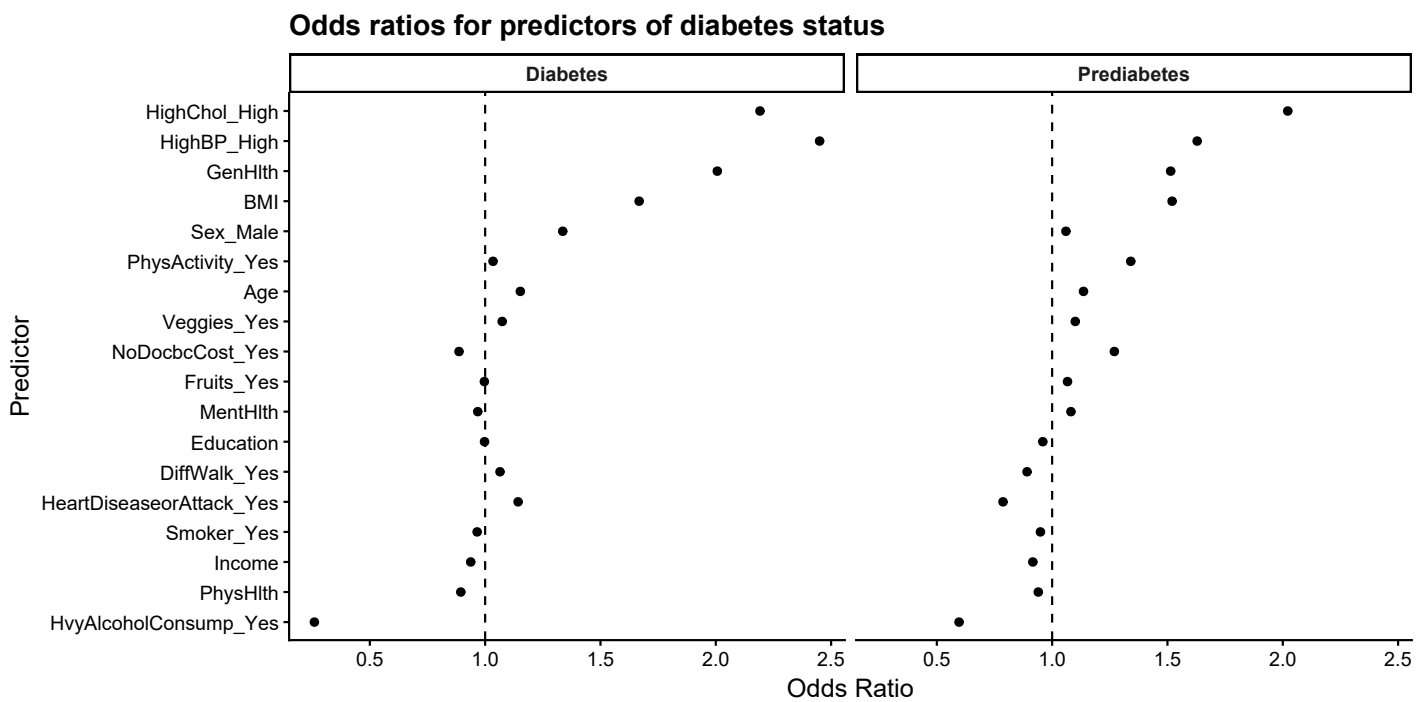
Convergence between methods strengthens confidence that identified risk factors are robust across modelling assumptions. **Divergence** indicates non-linearities, interactions, or violations of parametric assumptions that only flexible models capture. Where findings conflict, consistency patterns are prioritized and discrepancies are noted as areas requiring further investigation.

6.5.1 Log-Odds Ratios (Multinomial Logistic Regression)

Multinomial logistic regression estimates the log-odds of belonging to each diabetes category relative to a reference category. In this analysis, **No diabetes** serves as the reference class.

Positive coefficients indicate **higher odds of belonging to the specified diabetes category**, while negative coefficients indicate **lower odds relative to the reference group**.

```
multinom_coefs_tidy |>
  filter(term != "(Intercept)") |>
  ggplot(aes(x = odds_ratio, y = reorder(term, odds_ratio))) +
  geom_point()  +
  geom_vline(xintercept = 1, linetype = "dashed") +
  facet_wrap(~ y.level) +
  labs(
    x = "Odds Ratio",
    y = "Predictor",
    title = "Odds ratios for predictors of diabetes status"
  )
```



Multinomial Logistic Regression: Odds Ratio (OR) Summary

OR > 1 indicates increased odds; OR < 1 indicates decreased odds.

Synthesis of Findings

- **Primary Drivers:** Clinical indicators (Hypertension, Hypercholesterolemia, and BMI) represent the most significant risk. High blood pressure, in particular, yields a **139% increase** in the odds of diabetes ($OR = 2.39$).

- **The BMI Gradient:** The data shows a clear progression where BMI impacts diabetes ($OR = 1.72$) more severely than prediabetes ($OR = 1.51$), confirming its role in disease advancement.
- **Behavioral Mediation:** The lack of independent effect for physical activity ($OR = 1.00$) suggests its impact is likely **fully mediated** by metabolic variables like BMI and blood pressure.
- **Statistical Anomalies:** The negative association with heavy alcohol ($OR = 0.31$) likely reflects **selection bias** (e.g., "sick quitters" or health-conscious adjustments after diagnosis) rather than a biological protective effect.

Conclusion

Diabetes status in this population is best characterized by a **multifactorial cardiometabolic profile**. While age and sex are relevant, clinical markers of obesity and metabolic syndrome remain the most effective predictors for distinguishing between healthy, prediabetic, and diabetic individuals.

6.5.2 Global SHAP importance

SHAP Interpretation Methodology

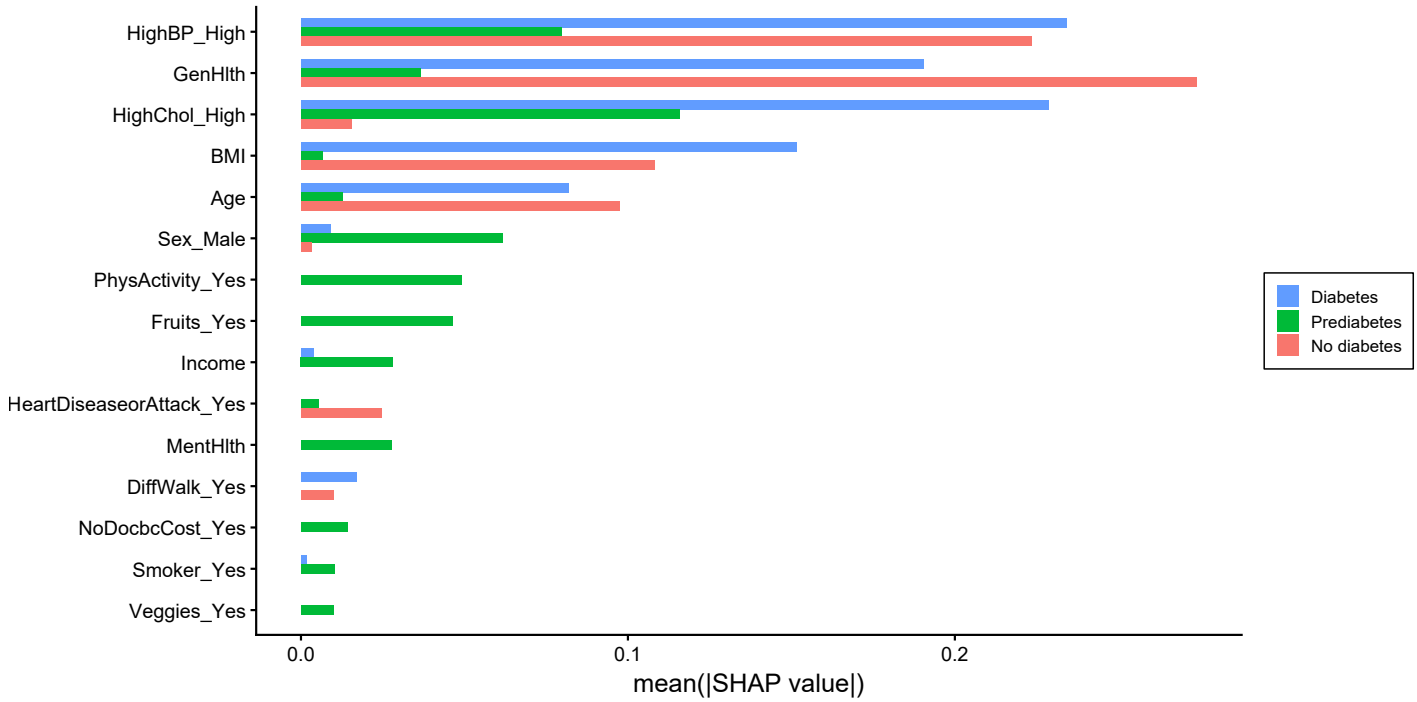
To complement the parametric log-odds interpretation, we employ **SHAP (SHapley Additive exPlanations)** to quantify feature contributions in the XGBoost model. SHAP offers several advantages:

- **Model-agnostic foundation** grounded in game theory, enabling comparison across model types
- **TreeSHAP algorithm** provides exact Shapley values by leveraging XGBoost tree structure (no Monte Carlo approximation needed)
- **Multiclass compatibility** computes separate contributions for each diabetes category (No diabetes, Prediabetes, Diabetes)
- **Mean absolute SHAP values** reveal global feature importance by measuring how much each feature, on average, shifts predicted probabilities from baseline

By comparing SHAP importance with logistic regression odds ratios, we assess method convergence—agreement strengthens confidence in identified risk factors, while discrepancies highlight non-linearities the parametric model cannot capture.

```r
xgb_engine <- extract_fit_engine(xgb_final_fit)

test_processed <- bake(
  prep(extract_preprocessor(xgb_final_wf)),
  new_data = diab_test
)

X <- test_processed |>
  select(-Diabetes) |>
  as.matrix()

set.seed(123)
X_sample <- X[sample(nrow(X), 10000), ]

# shp <- shapviz(xgb_engine, X_pred = X_sample)
# write_rds(shp, "shp_rq1.rds")

# Load precomputed SHAP values to avoid recomputing expensive shapviz calculations
shp <- read_rds("shp_rq1.rds")
```

```r
sv_importance(shp, kind = "bar") +
  scale_fill_discrete(
    labels = c(
      "No diabetes",
      "Prediabetes",
      "Diabetes")
  )
```



The SHAP importance plot indicates that cardiometabolic indicators dominate the predictive structure of the model. High blood pressure, general health status, high cholesterol, and BMI exhibit the largest mean absolute SHAP contributions across classes, indicating they play the strongest role in shifting predicted probabilities of diabetes status. Age also contributes meaningfully, while behavioural variables such as smoking, fruit consumption, and alcohol use show comparatively smaller effects. This pattern closely mirrors the multinomial logistic regression results, reinforcing the conclusion that diabetes status in this dataset is primarily driven by metabolic and health status indicators rather than lifestyle variables alone.

6.5.3 Variable Importance

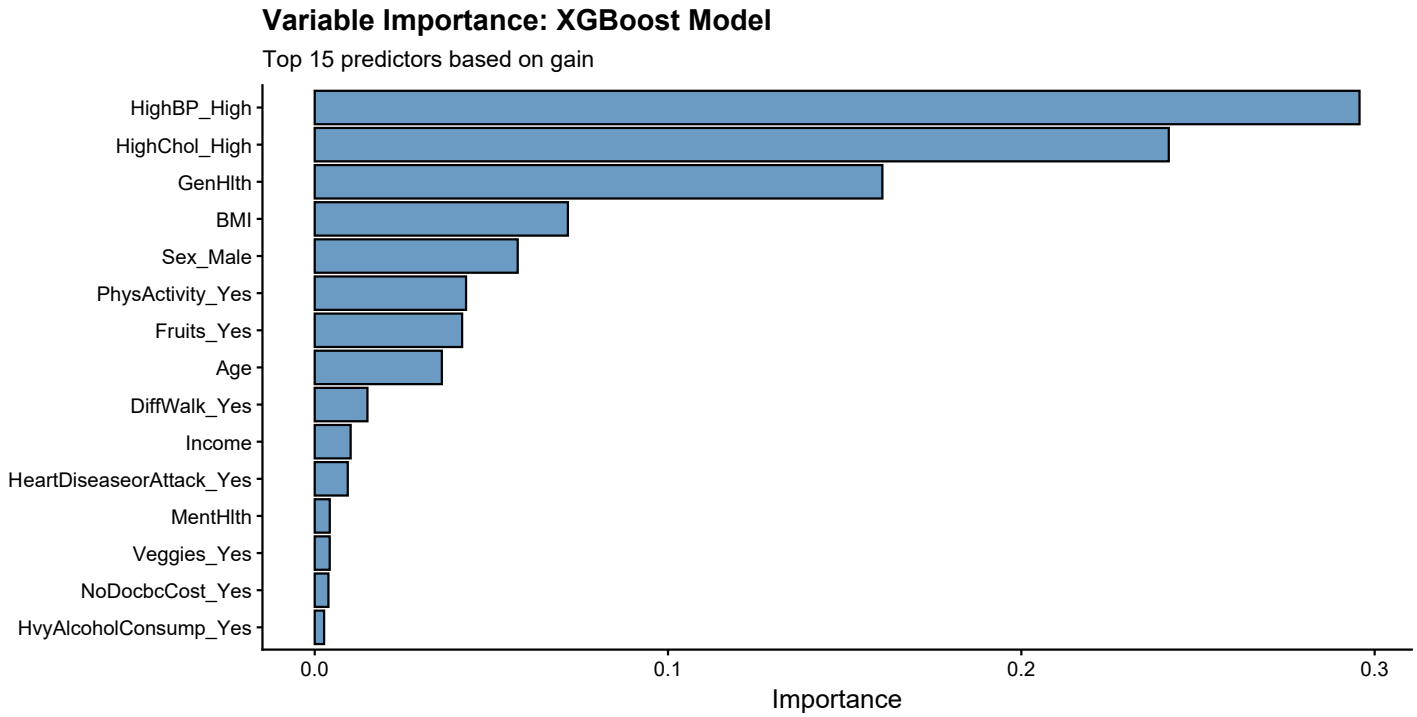
Variable Importance: Gain Metric

XGBoost variable importance is computed using the **gain** metric, which quantifies each feature's contribution to model performance:

- **Gain** measures the total reduction in the loss function (e.g., multi-class log-loss) achieved whenever a feature is used to split nodes across all trees in the ensemble
- Features with high gain contribute substantially to improving predictions by creating splits that better separate diabetes categories
- **Gain vs alternatives:** Unlike split frequency (which counts how often a feature is used) or cover (which weights by number of observations), gain directly captures predictive impact and effect magnitude

Higher gain values indicate features that most effectively reduce prediction error when incorporated into the model's decision rules. This metric complements SHAP importance by focusing on in-model utility rather than marginal contributions to individual predictions.

```
xgb_final_fit |>
  extract_fit_parsnip() |>
  vip(num_features = 15,
      geom = "col",
      aesthetics = list(fill = "steelblue",color = 'black', alpha = 0.8)) +
  labs(title = "Variable Importance: XGBoost Model",
       subtitle = "Top 15 predictors based on gain")
```



- **Consistency with Statistical Findings:** The VIP confirms the results from the multinomial log-odds analysis (Section 6.5.1), with **HighBP**, **BMI**, and **HighChol** consistently appearing as the most influential predictors. This triangulation of methods strengthens the argument that these clinical markers are the primary determinants of diabetes risk in this dataset.
- **The "Prediabetes" Paradox:** Despite the high importance scores for these top features, the model's struggle to classify the "Prediabetes" class suggests a "ceiling effect." While variables like **HighBP** and **GenHlth** (General Health) are excellent at distinguishing between "Healthy" and "Diabetic" individuals, the feature space lacks the specific granularity required to isolate the transitionary prediabetic phase.
- **Lifestyle vs. Clinical Markers:** Interestingly, clinical markers (BMI, Blood Pressure) rank significantly higher than lifestyle factors such as **PhysActivity** or **Fruits**. This implies that while lifestyle choices contribute to the outcome, the physiological manifestations (the "clinical profile") are the more immediate and powerful predictors for the gradient-boosted trees.

6.6 Conclusion: Model Interpretability and Determinants

The comprehensive analysis of diabetes status, conducted through both statistical and machine learning frameworks, demonstrates that diabetes risk in this population is best characterized by a multifactorial cardiometabolic profile. By triangulating the results from multinomial log-odds, XGBoost Variable Importance, and SHAP values, this study provides a robust validation of the primary drivers of the disease.

- **Triangulation of Risk Factors:** There is a clear consensus across all analytical methods regarding the primary determinants of health. The multinomial logistic regression identified **High Blood Pressure** as the strongest categorical predictor, with an associated 139% increase in the odds of a diabetes diagnosis ($OR = 2.39$). This finding is directly mirrored in the **XGBoost Variable Importance Plot (VIP)**, where HighBP ranks as a top-tier feature, and is further clarified by the **SHAP summary plot**, which illustrates how high values for blood pressure, BMI, and cholesterol consistently push the model's prediction toward the diabetic class.
- **Predictive Nonlinearity and Impact:** While the linear multinomial model provides clear odds ratios, the SHAP analysis reveals the nuanced, non-linear impacts of features such as **Age** and **General Health**. The SHAP distribution reveals a significant clustering of high-risk impact for older age groups, suggesting that risk does not simply increase linearly, but rather accelerates as multiple clinical markers, such as BMI and cholesterol, coincide.
- **The "Prediabetes" Gap and Structural Challenges:** A critical finding of this synthesis is the discrepancy in class identification across models. While the machine learning models (XGBoost and Random Forest) performed

competitively in overall discrimination, they effectively collapsed the classification into a binary task, failing to isolate the **Prediabetes** class. The SHAP values show a significant overlap in the middle of the feature distribution, explaining this struggle, the clinical signature of prediabetes in this dataset is too similar to the surrounding classes for the boosting algorithm to partition effectively. Conversely, the multinomial logistic regression demonstrated superior balance, maintaining the ability to distinguish the subtle physiological transitions inherent in the intermediate prediabetic category.

- **Socioeconomic and Actionable Clinical Profiles:** Beyond clinical indicators, general health perception and age remained significant predictors, yet lifestyle factors, such as physical activity and diet, showed weaker direct associations than physiological manifestations. This implies that while lifestyle choices are foundational, they may operate indirectly through their impact on measurable markers like BMI and blood pressure. The synthesis confirms that a high-risk profile is defined by the intersection of these clinical markers, which consistently dominate the top of both the VIP and SHAP rankings, identifying them as the most critical targets for screening and preventative intervention.

Method	Key Strength	Primary Top Predictors
Multinomial Logistic	Identifies the “Prediabetes” gradient	HighBP, BMI, Age
XGBoost VIP (Gain)	Highlights overall model efficiency	HighBP, GenHlth, BMI
SHAP Analysis	Shows direction and impact of features	HighBP, BMI, HighChol

In summary, the triangulation of results from the **Odds Ratios**, **SHAP values**, and **XGBoost Variable Importance** provides a robust validation of the cardiometabolic risk factors. For clinical application, the multinomial model remains the most “fit-for-purpose” tool due to its unique capacity to identify individuals in the critical prediabetic phase.

7. Socioeconomic Moderation of Cardiovascular Risk

7.1. Modelling Objective

This section addresses the following research question:

> *To what extent do socioeconomic determinants, specifically income and education, moderate the impact of clinical risk factors on the prevalence of heart disease, and can the capture of complex feature interactions provide superior predictive insights compared to traditional additive assumptions?*

The objective is:

- **Interaction Analysis** – to investigate whether the predictive weight of clinical markers (e.g., BMI, Blood Pressure) shifts significantly across different socioeconomic strata.
- **Predictive Optimization** – to evaluate if non-linear machine learning frameworks (XGBoost) outperform traditional linear frameworks (Logistic Regression) by capturing these latent moderating effects. **Operationalizing Moderation**

The research question asks whether socioeconomic factors (income, education) moderate the effect of clinical risk factors (high blood pressure, BMI) on heart disease. In this analysis, moderation is operationalized as follows:

- **For Logistic Regression:** Serves as an additive baseline without explicit interaction terms. Any moderation effects would require post-hoc interaction term inclusion, which is beyond the exploratory scope of this model.
- **For XGBoost:** Interactions are captured implicitly through recursive tree partitioning. When the model splits on Income after splitting on HighBP (or vice versa), it effectively models conditional effects where the impact of one variable depends on the value of another.
- **Evidence criterion:** Moderation will be assessed through SHAP interaction analysis and partial dependence plots stratified by SES groups. Significant variation in clinical risk effects across income/education levels would support the moderation hypothesis.

This approach leverages XGBoost’s capacity to detect non-linear interactions without requiring manual interaction term specification.

7.2. Outcome Variable

The dependent variable is **HeartDiseaseorAttack**, a binary outcome indicating whether the respondent has ever been diagnosed with coronary heart disease (CHD) or myocardial infarction (MI).

As identified in the exploratory phase, this variable exhibits a notable class imbalance, with the "No" category significantly outnumbering the "Yes" category. This necessitates the use of stratified sampling and specific evaluation metrics, such as the Area Under the ROC Curve (AUC-ROC) and F1-score, to ensure the model does not become biased toward the majority class.

7.3 Training and Validation

7.3.1 Data Partitioning and Resampling

The dataset is partitioned into a 75% training set and a 25% testing set using stratified sampling on heart disease status. To ensure the models are robust and to prevent overfitting during the hyperparameter tuning phase (particularly for the XGBoost model), 5-fold stratified cross-validation is implemented within the training set. The test set remains strictly held-out for the final comparative evaluation between the additive logistic baseline and the interaction-heavy machine learning model.

```
set.seed(42)

heart_split <- initial_split(
  diab,
  prop = 0.75,
  strata = HeartDiseaseorAttack
)

heart_train <- training(heart_split)
heart_test  <- testing(heart_split)

heart_folds <- vfold_cv(
  heart_train,
  v = 5,
  strata = HeartDiseaseorAttack
)

heart_train |> count(HeartDiseaseorAttack) |> mutate(prop = n/sum(n))
```

HeartDiseaseorAttack	n	prop
No	172351	0.9058709
Yes	17909	0.0941291

```
heart_test  |> count(HeartDiseaseorAttack) |> mutate(prop = n/sum(n))
```

HeartDiseaseorAttack	n	prop
No	57436	0.9056449
Yes	5984	0.0943551

7.3.2 Recipe Definition and Preprocessing Strategy

Class Imbalance Strategy

The outcome exhibits substantial class imbalance (~9% heart disease cases). To prevent models from defaulting to majority-class predictions, we apply **downsampling** with `under_ratio = 2.33`, retaining a 70-30 majority-to-minority ratio

rather than full balancing. This approach balances two objectives: (1) providing sufficient minority-class representation for model learning (~30% of downsampled data), and (2) preserving moderate imbalance to reflect population base rates and avoid miscalibrated probability estimates. Downsampling is applied within cross-validation folds to prevent data leakage.

```
heart_recipe <- recipe(HeartDiseaseorAttack ~ ., data = heart_train) |>
  step_nzv(all_predictors()) |>
  step_ordinalscore(Age, Education, Income, GenHlth) |>
  step_dummy(all_nominal_predictors()) |>
  step_normalize(BMI, MentHlth, PhysHlth) |>
  step_downsample(HeartDiseaseorAttack, under_ratio = 2.33)

print(heart_recipe)
```

7.3.3 Model Specification

We define the candidate models for **RQ2** To address the research question, we compare a traditional parametric baseline against a high-capacity gradient-boosted ensemble. This allows us to determine if the inclusion of non-linear feature interactions (socioeconomic moderation) significantly improves the model's ability to identify cardiovascular risk.

The Models

Model	Type	Primary Function	Key Strengths
Logistic Regression	Parametric	Linear Baseline	Provides interpretable Odds Ratios for clinical and SES factors; assumes additive effects.
XGBoost	Ensemble (Boosting)	Interaction Discovery	Specifically optimized to capture non-linearities and moderation effects between variables.

Note

> The selection of **XGBoost,** instead of Random Forest**,** for this research question is informed by its performance in the preceding analysis of diabetes status (RQ1). While RQ1 dealt with a multi-class outcome, XGBoost is equally well-suited for this binary classification of heart disease. Its gradient boosting framework is particularly effective at preventing the model from becoming biased toward the majority "No" class, ensuring that the clinical and socioeconomic patterns of the minority "Yes" class are accurately captured and weighted.

```
lr_spec <- logistic_reg(
  penalty = tune(),
  mixture = tune()) |>
  set_engine("glmnet") |>
  set_mode("classification")


xgb_spec <- boost_tree(
  trees = 1024,
  tree_depth = tune(),
  learn_rate = tune(),
  loss_reduction = tune(),
  stop_iter = 20) |>
  set_engine("xgboost") |>
  set_mode("classification")
```

7.3.4 Workflow Set Definition

Following the methodology in Section 6, we utilize `workflow_sets` to ensure both models are evaluated using the identical finalized recipe and cross-validation folds.

```
heart_models <- workflow_set(
  preproc = list(base_rec = heart_recipe),
  models = list(logistic = lr_spec, xgboost = xgb_spec),
  cross = TRUE
)
```

7.3.5 Metric Selection and Model Execution

Metrics

To evaluate the models, we define a robust metric set that accounts for the class imbalance inherent in the dataset. While **ROC-AUC** provides a general measure of classifier quality, **PR-AUC** and the **F1-score** are prioritized to ensure the model effectively identifies the "Yes" (Heart Disease) cases without excessive false positives.

```
heart_metrics <- metric_set(
  roc_auc,
  pr_auc,
  sens,
  spec,
  f_meas
)
```

Model Execution and Resampling

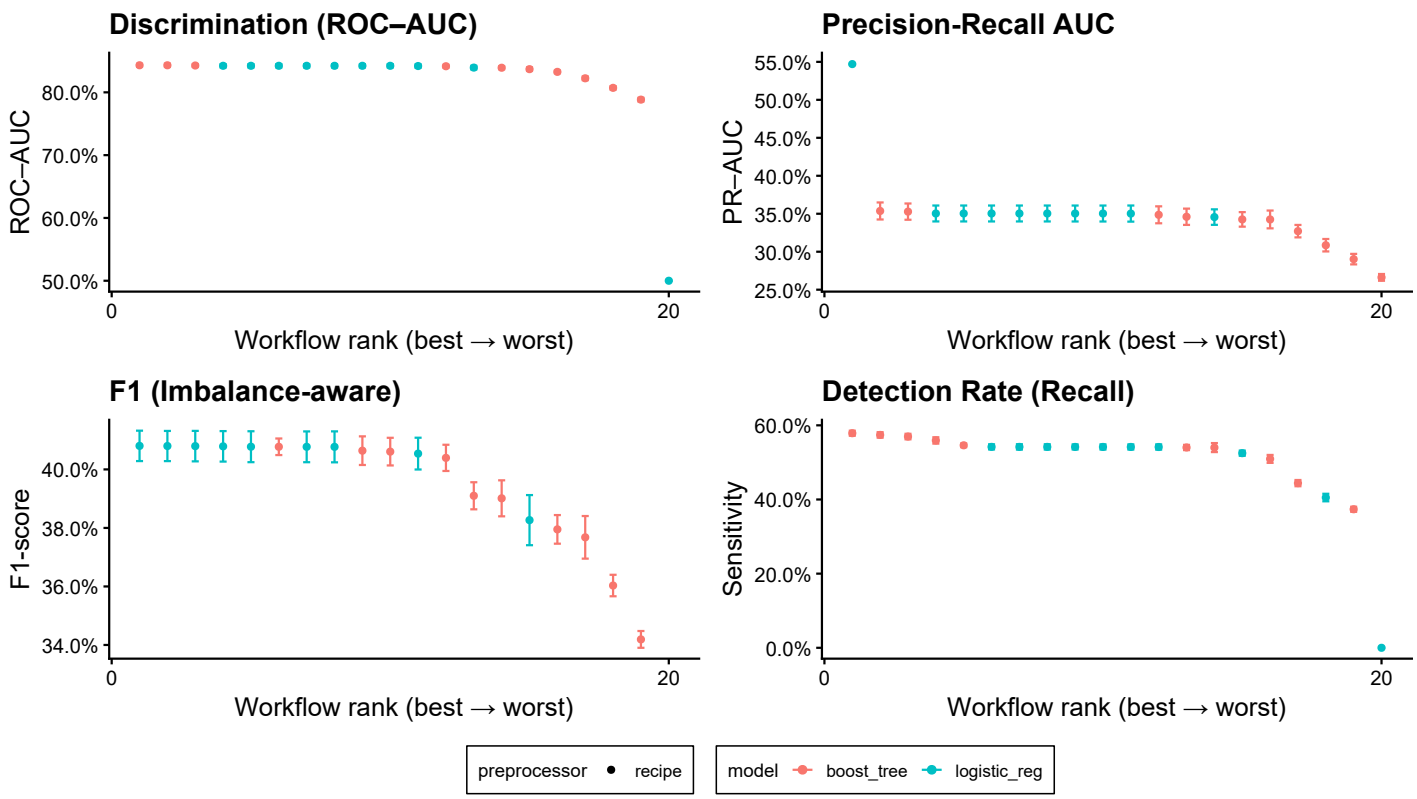
With the metrics defined, we map the `fit_resamples` function across our workflow set. This process uses the 5-fold cross-validation established in Section 7.3 to generate stable performance estimates.

```
# cl <- makePSOCKcluster(parallel::detectCores(logical = FALSE))
# registerDoParallel(cl)
#
# heart_results <- heart_models |>
#   workflow_map(
#     fn = "tune_grid",
#     resamples = heart_folds,
#     grid = 10,
#     metrics = heart_metrics,
#     seed = 42,
#     control = control_grid(
#       save_pred = TRUE,
#       parallel_over = "everything",
#       verbose = TRUE,
#       event_level = "second")
#   )
#
# stopCluster(cl)
#
# write_rds(heart_results, "heart_results.rds")
heart_results <- read_rds('heart_results.rds')
```

7.3.6 Comparative Performance Visualization

```
m1 <- pretty_autoplot(heart_results,
                      "roc_auc", "ROC-AUC", "Discrimination (ROC-AUC)")
m2 <- pretty_autoplot(heart_results,
```

```
                        "pr_auc", "PR-AUC", "Precision-Recall AUC")
m3 <- pretty_autoplot(heart_results,
                        "f_meas", "F1-score", "F1 (Imbalance-aware)")
m4 <- pretty_autoplot(heart_results,
                        "sens", "Sensitivity", "Detection Rate (Recall)")

(m1 | m2) / (m3 | m4) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```



This figure compares the performance of all candidate workflows across four evaluation metrics ranked from best to worst.

For ROC–AUC, most workflows achieve very similar discrimination performance around the mid-0.80 range, indicating that both model families are generally capable of separating the classes effectively. Logistic regression workflows appear more stable across the ranking, while some boosted tree configurations show a noticeable decline toward the lower ranks, suggesting greater sensitivity to preprocessing or tuning choices.

The Precision–Recall AUC plot shows lower absolute values, which is expected in imbalanced classification settings. Both models perform similarly across most workflows, although boosted tree models display slightly greater variability, with a clearer performance drop among the worst-ranked configurations.

The F1-score results show a narrow band among the higher-ranked workflows, indicating that several configurations achieve a balanced trade-off between precision and recall. Logistic regression workflows maintain relatively consistent F1 values, whereas boosted tree models again exhibit more spread, with some configurations performing noticeably worse.

Finally, the Detection Rate (Recall) plot highlights differences in sensitivity. Top-ranked workflows for both models achieve similar recall levels, but logistic regression maintains more consistent detection rates across ranks. Some boosted tree workflows show reduced recall in lower ranks, indicating weaker ability to identify positive cases in those configurations.

Overall, logistic regression workflows demonstrate more stable performance across preprocessing variations, while boosted tree models show slightly greater variability. Despite this, the best-performing configurations of both models achieve broadly comparable predictive performance across all evaluation metrics.

7.4 Final Model Extraction and Test Set Evaluation

This stage identifies the best-performing workflow for each modelling approach based on cross-validated tuning results. The logistic regression and XGBoost workflows with the highest Precision–Recall AUC are selected, as PR–AUC provides a more informative evaluation metric in the presence of class imbalance.

For each model type, the optimal hyperparameter configuration obtained during tuning is applied to the corresponding workflow, producing a finalised model specification. These final workflows therefore represent the best-performing logistic regression and boosted tree models identified during the resampling stage.

The purpose of this step is to prepare the final candidate models for unbiased evaluation on the held-out test dataset, enabling a direct comparison of their predictive performance.

```
best_lr_id <- "base_rec_logistic"
best_xgb_id <- "base_rec_xgboost"

final_lr_wflow <-
  heart_results |>
  extract_workflow(best_lr_id) |>
  finalize_workflow(
    heart_results |>
      extract_workflow_set_result(best_lr_id) |>
      select_best(metric = "pr_auc")
  )

final_xgb_wflow <-
  heart_results |>
  extract_workflow(best_xgb_id) |>
  finalize_workflow(
    heart_results |>
      extract_workflow_set_result(best_xgb_id) |>
      select_best(metric = "pr_auc")
  )
```

The finalised logistic regression and XGBoost models are evaluated on the held-out test set to obtain an unbiased estimate of predictive performance.

This evaluation provides a fair comparison of the two modelling approaches under identical conditions. The resulting performance metrics summarise how well each model generalises to unseen data, allowing the relative effectiveness of logistic regression and the boosted tree model to be assessed for the prediction task.

```
test_results_lr <- last_fit(
  final_lr_wflow, split = heart_split, metrics = heart_metrics)

test_results_xgb <- last_fit(
  final_xgb_wflow, split = heart_split, metrics = heart_metrics)
```

```
final_comparison <- bind_rows(
  test_results_lr |> collect_metrics() |> mutate(model = "Regularized LR"),
  test_results_xgb |> collect_metrics() |> mutate(model = "Tuned XGBoost")
) |>
  select(model, .metric, .estimate) |>
  pivot_wider(names_from = .metric, values_from = .estimate)

print(final_comparison)
```

```
# A tibble: 2 x 6
  model           sens  spec f_meas roc_auc pr_auc
  <chr>          <dbl> <dbl>  <dbl>   <dbl>  <dbl>
1 Regularized LR 1     0      0.950   0.5    0.953
```

```
2 Tuned XGBoost  0.882 0.550  0.915   0.840  0.979
```

The final test-set evaluation compares the predictive performance of the Regularized Logistic Regression model and the Tuned XGBoost model across multiple metrics.

The Regularized Logistic Regression model achieves a sensitivity of 1.00, specificity of 0.00, F1-score of 0.950, ROC–AUC of 0.50, and PR–AUC of 0.953. While the high sensitivity and F1-score may initially appear strong, the specificity of zero indicates that the model predicts all observations as the positive class. This behaviour results in no ability to correctly identify negative cases, which is reflected in the ROC–AUC of 0.50, equivalent to random classification. Consequently, the model fails to provide meaningful class discrimination despite apparently high values on certain metrics.

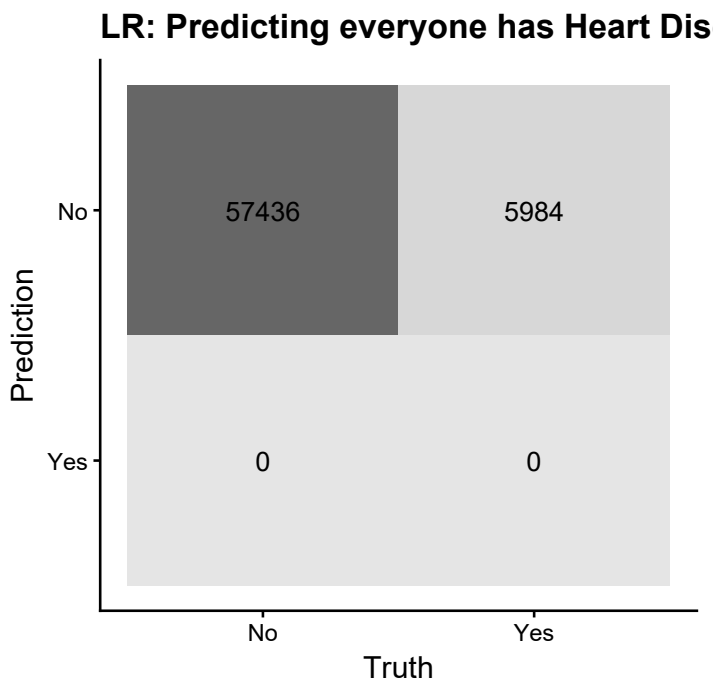
In contrast, the Tuned XGBoost model demonstrates substantially stronger and more balanced predictive performance. It achieves a sensitivity of 0.882, specificity of 0.550, F1-score of 0.915, ROC–AUC of 0.840, and PR–AUC of 0.979. The high ROC–AUC indicates strong discrimination between classes, while the high PR–AUC suggests effective performance under class imbalance. Although the specificity is moderate, the model successfully identifies a meaningful proportion of both positive and negative cases, resulting in a more balanced and practically useful classifier.

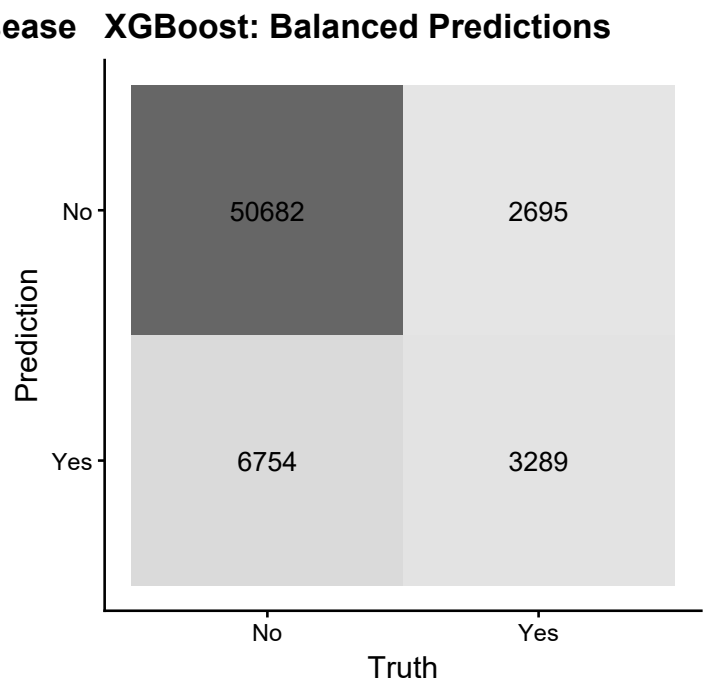
Overall, the results indicate that the Tuned XGBoost model substantially outperforms the Regularized Logistic Regression model. The boosted tree model provides strong discrimination and maintains a reasonable balance between sensitivity and specificity, suggesting that the underlying relationships between predictors and heart disease are better captured by a flexible non-linear model than by a linear logistic framework.

```r
all_preds <- bind_rows(
  test_results_lr |> collect_predictions() |> mutate(model = "Logistic Regression"),
  test_results_xgb |> collect_predictions() |> mutate(model = "XGBoost")
)
```

7.4.3 Confusion Matrix Analysis

```r
cm_lr <- test_results_lr |>
  collect_predictions() |>
  conf_mat(HeartDiseaseorAttack, .pred_class) |>
  autoplot(type = "heatmap") +
  labs(title = "LR: Predicting everyone has Heart Disease")

cm_xgb <- test_results_xgb |>
  collect_predictions() |>
  conf_mat(HeartDiseaseorAttack, .pred_class) |>
  autoplot(type = "heatmap") +
  labs(title = "XGBoost: Balanced Predictions")

cm_lr + cm_xgb
```





The confusion matrices provide a clearer interpretation of how each model performs at the class prediction level.

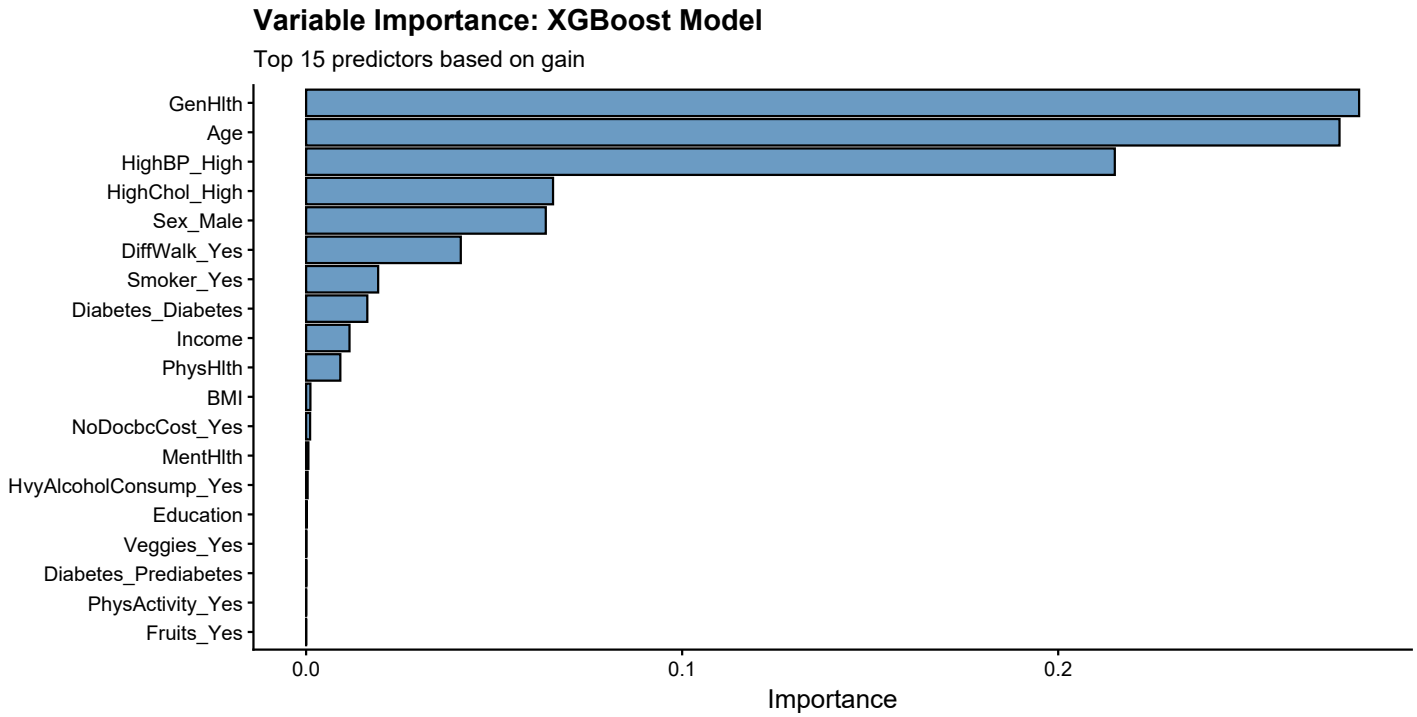
The regularized logistic regression model predicts every observation as the positive class (Heart Disease). As a result, all true negative cases (57,436) are misclassified as positive, while all true positive cases (5,984) are also predicted as positive. This explains the perfect sensitivity (1.00) and zero specificity observed earlier. Although the model detects all positive cases, it completely fails to identify individuals without heart disease, demonstrating that it has collapsed into a trivial classifier that does not meaningfully distinguish between classes.

In contrast, the tuned XGBoost model produces more balanced predictions across both classes. It correctly identifies 50,682 individuals without heart disease and 3,289 individuals with heart disease, while misclassifying 6,754 non-cases as positive and 2,695 true cases as negative. This reflects a more balanced trade-off between sensitivity and specificity, consistent with the stronger ROC–AUC and overall performance observed in the test-set evaluation.

Overall, the confusion matrices reinforce the earlier metric-based comparison. Logistic regression fails to provide meaningful class separation, whereas XGBoost demonstrates effective discrimination by correctly identifying a substantial proportion of both positive and negative cases.

7.4.4 Variable Importance Analysis

```
final_xgb_fit <- extract_fit_parsnip(test_results_xgb)

final_xgb_fit |>
  vip(num_features = 20,
      geom = "col",
      aesthetics = list(fill = "steelblue",color = 'black', alpha = 0.8)) +
  labs(title = "Variable Importance: XGBoost Model",
       subtitle = "Top 15 predictors based on gain")
```



The variable importance plot presents the relative contribution of predictors to the final XGBoost model based on gain, which measures how much each variable improves the model's decision splits.

General health (GenHlth) emerges as the most influential predictor, followed closely by age. This indicates that overall health status and age group are the strongest contributors to distinguishing individuals with and without heart disease in the model. High blood pressure and high cholesterol also rank highly, reinforcing the importance of cardiometabolic risk factors in predicting heart disease outcomes.

Several additional variables contribute moderate predictive value, including sex, difficulty walking, smoking status, and diabetes status. These predictors likely capture functional limitations and behavioural risk factors associated with cardiovascular health.

In contrast, lifestyle variables such as fruit consumption, vegetable consumption, physical activity, and alcohol use show minimal importance within the model. Socioeconomic indicators such as education and income also appear to contribute relatively little compared with clinical and health-status variables.

Overall, the importance ranking suggests that the XGBoost model primarily relies on indicators of general health and cardiometabolic risk, with demographic and behavioural factors playing a secondary role in predicting heart disease status.

7.5 SHAP-Based Model Interpretation

To interpret the predictions of the final XGBoost model, SHAP (SHapley Additive exPlanations) values are computed. SHAP provides a model-agnostic framework for explaining complex machine learning models by decomposing each prediction into additive contributions from individual predictors.

Because the test dataset is large, a random sample of 5,000 observations is drawn to make the SHAP computation tractable while still providing a representative explanation of the model's behaviour. The predictors used for explanation correspond to the processed feature matrix used by the fitted XGBoost engine.

```
# xgb_engine <- extract_fit_engine(final_xgb_fit)
#
# test_processed <- bake(
#   prep(extract_preprocessor(final_xgb_wflow)),
#   new_data = testing(heart_split)
# )
```

```
#
# X <- test_processed |>
#   select(-HeartDiseaseorAttack) |>
#   as.matrix()
#
# set.seed(123)
# X_sample <- X[sample(nrow(X), 10000), ]
# shp <- shapviz(xgb_engine, X_pred = X_sample)
#
# write_rds(shp,"shp_rq2.rds")


# Load precomputed SHAP values to avoid recomputing expensive shapviz calculations
shp <- read_rds("shp_rq2.rds")
```

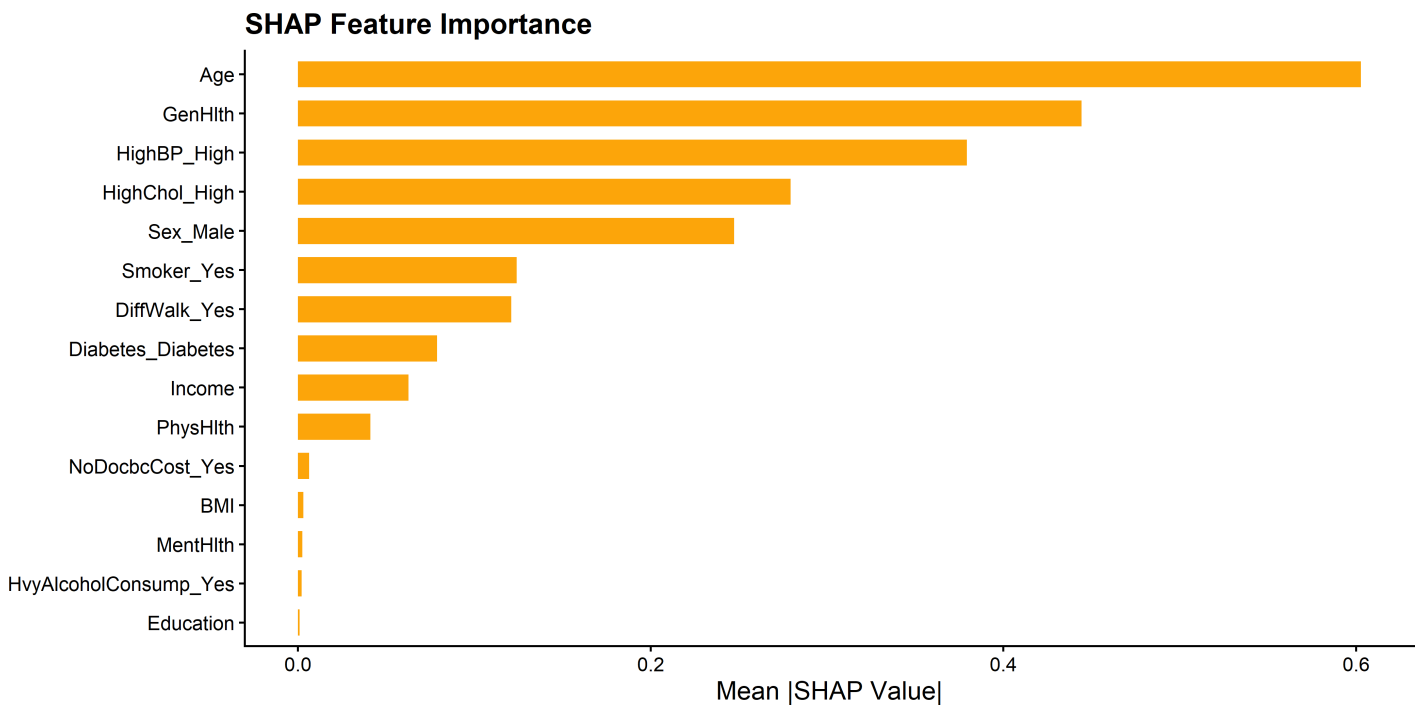
The SHAP values are subsequently sign-adjusted to align with the outcome orientation used in this analysis. In this dataset the positive event is encoded as the second level of the outcome factor (No = 0, Yes = 1), whereas the SHAP implementation assumes that the first level corresponds to the event of interest. As a result, the SHAP values are multiplied by −1 so that positive contributions correctly represent increased predicted risk of heart disease and negative contributions represent decreased risk. This adjustment ensures that subsequent SHAP visualisations are interpreted consistently with the outcome definition used throughout the analysis.

```
shp_flipped <- shapviz(-shp$S, X = shp$X, baseline = -shp$baseline)
```

7.5.1 Global Feature Importance (SHAP)

The SHAP feature importance plot ranks predictors according to their average absolute SHAP value, which represents the overall magnitude of each variable's contribution to the model's predictions across all observations.

```
sv_importance(shp_flipped, kind = "bar", max_display = 15) +
  labs(title = "SHAP Feature Importance", x = "Mean |SHAP Value|")
```



Age emerges as the most influential predictor, indicating that changes in age group have the largest average effect on predicted heart disease risk. General health (GenHlth) is the second most important variable, suggesting that self-reported health status strongly influences the model's assessment of cardiovascular risk.

Cardiometabolic indicators such as high blood pressure (HighBP) and high cholesterol (HighChol) also show substantial importance, reinforcing their established role as key risk factors for cardiovascular disease. These variables contribute strongly to the model's decision structure and appear consistently among the top predictors.

Several demographic and behavioural variables provide moderate contributions, including sex, smoking status, difficulty walking, and diabetes status. These predictors likely capture functional health limitations and behavioural risk factors associated with cardiovascular conditions.

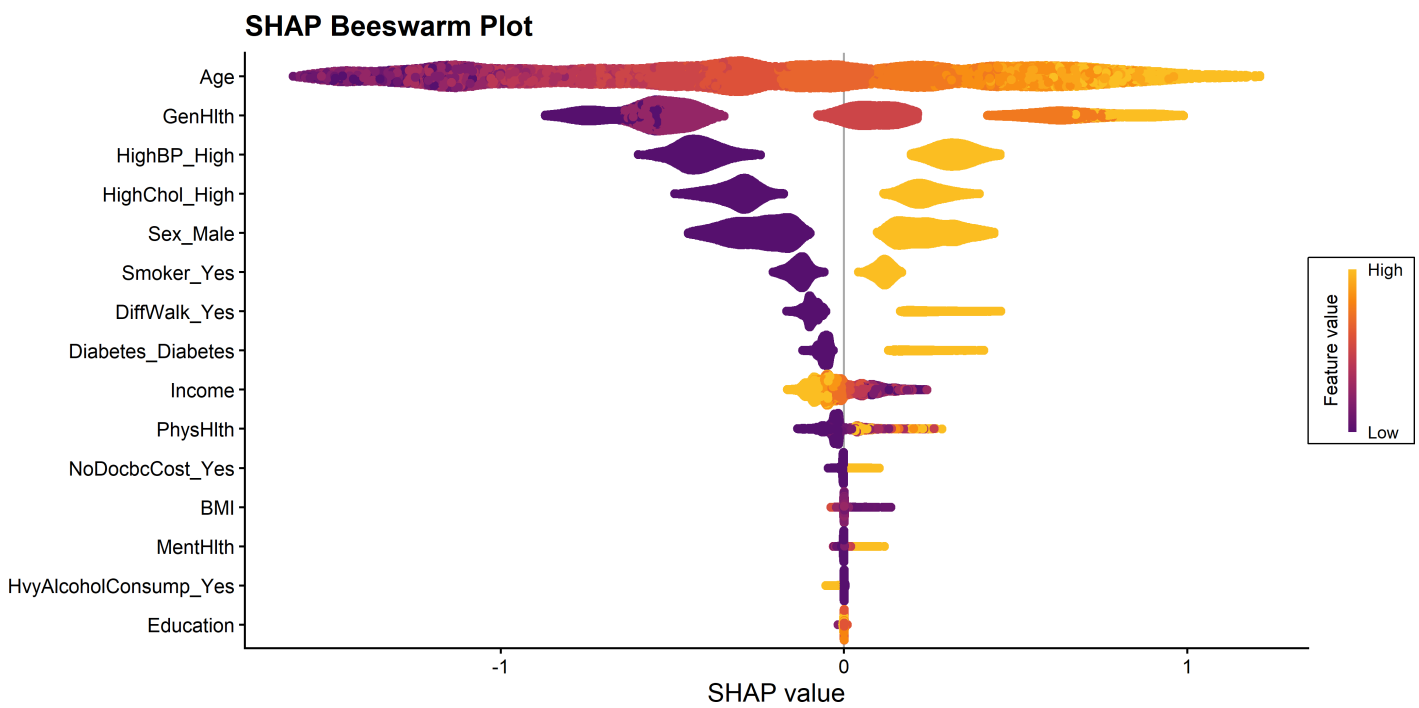
In contrast, variables such as BMI, mental health, alcohol consumption, education, and healthcare cost barriers show relatively small mean SHAP values. This suggests that, within the context of the full model, they contribute comparatively little to the prediction of heart disease status.

Overall, the SHAP importance ranking broadly aligns with the earlier model-based variable importance results, highlighting age, general health, and cardiometabolic risk factors as the dominant predictors driving the XGBoost model's predictions.

7.5.2 Distribution of Feature Effects (SHAP Beeswarm Plot)

The SHAP beeswarm plot provides a detailed view of how individual feature values influence the predicted probability of heart disease across observations. Each point represents a single observation, where the horizontal position indicates the SHAP value (the contribution of the feature to the prediction), and colour represents the magnitude of the feature value.

```
sv_importance(shp_flipped, kind = "beeswarm", max_display = 15) +
  labs(title = "SHAP Beeswarm Plot")
```



Age shows the widest spread of SHAP values, indicating that it has the strongest and most consistent influence on model predictions. Higher age categories are associated with positive SHAP values, meaning they increase the predicted risk of heart disease, whereas lower age values tend to reduce predicted risk.

General health (GenHlth) also demonstrates a strong directional effect. Poorer health levels (higher values) shift predictions toward higher heart disease risk, while better self-reported health contributes negatively to the prediction. Similarly, clinical risk factors such as high blood pressure and high cholesterol consistently push predictions toward increased risk when present.

Demographic and behavioural factors such as sex, smoking, and difficulty walking exhibit moderate contributions, where the presence of these conditions tends to increase predicted risk but with smaller magnitudes compared to primary clinical indicators.

Socioeconomic variables, particularly income and education, display comparatively smaller SHAP magnitudes. Their distributions are centred closer to zero, indicating that while they contribute to predictions, their direct effects are weaker than clinical risk factors. This suggests that socioeconomic determinants may influence predictions more subtly, potentially through interactions with health-related variables rather than through strong independent effects.

Overall, the beeswarm distribution reinforces that the model's predictions are primarily driven by clinical health indicators, while demographic and socioeconomic factors contribute secondary adjustments to predicted heart disease risk.

7.5.3 SHAP Dependence Analysis

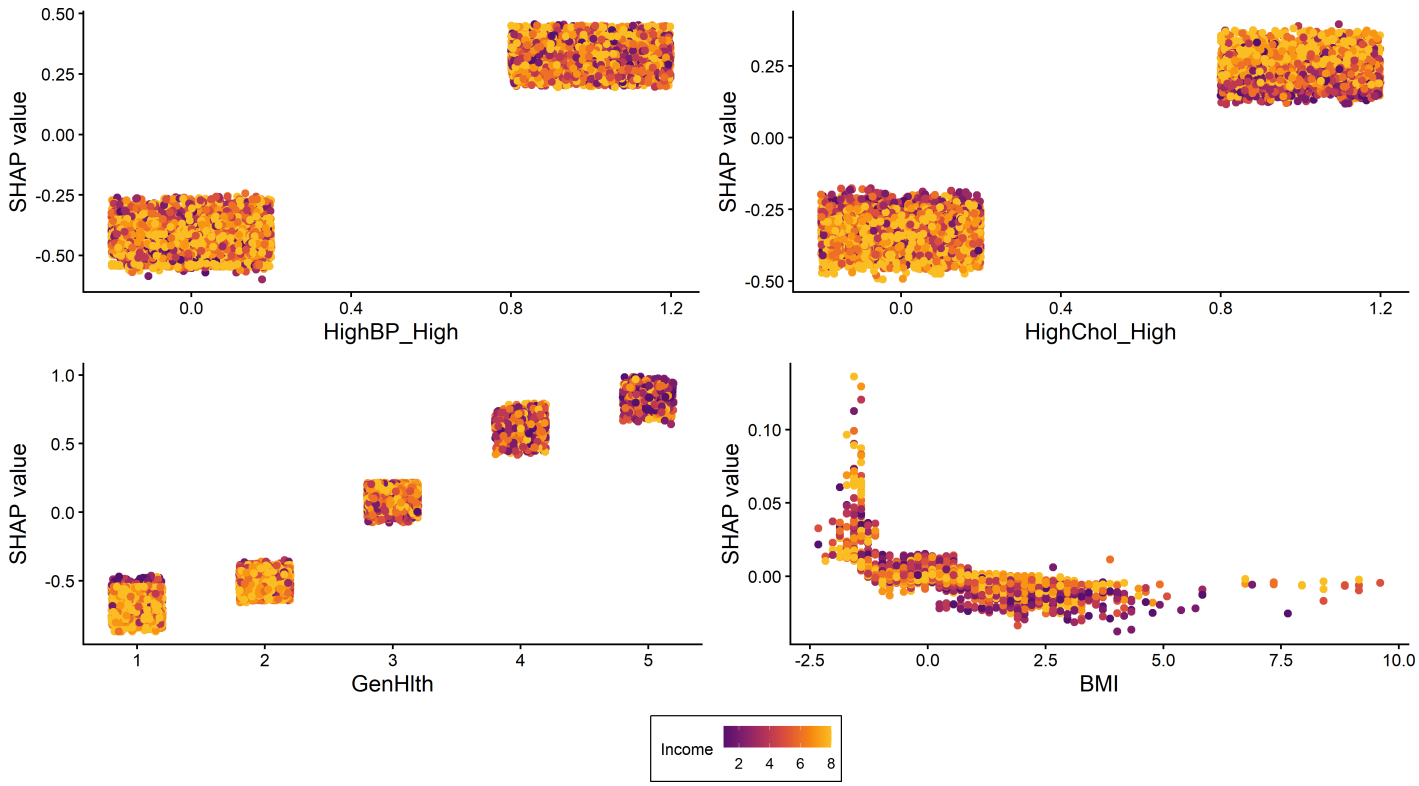
SHAP dependence plots are used to examine how the effect of key clinical risk factors on predicted heart disease risk varies across levels of socioeconomic status. Unlike global SHAP importance measures, dependence plots allow the relationship between a specific predictor and the model output to be visualised while simultaneously colouring observations by a second variable.

In this analysis, the plots are used to explore whether socioeconomic determinants, specifically income and education, moderate the impact of major clinical risk factors. By visualising SHAP values for variables such as high blood pressure, high cholesterol, general health, and BMI while colouring observations by socioeconomic indicators, it becomes possible to assess whether the contribution of these clinical predictors differs across socioeconomic groups.

This approach directly addresses the research question by investigating whether complex interaction effects between socioeconomic determinants and clinical risk factors are captured by the XGBoost model, and whether these interactions provide insights beyond traditional additive modelling assumptions.

Clinical Risk Factors by Income

```
p1 <- sv_dependence(shp_flipped, v = "HighBP_High", color_var = "Income")
p2 <- sv_dependence(shp_flipped, v = "HighChol_High", color_var = "Income")
p3 <- sv_dependence(shp_flipped, v = "GenHlth", color_var = "Income")
p4 <- sv_dependence(shp_flipped, v = "BMI", color_var = "Income")

(p1 + p2) / (p3 + p4) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```



The dependence plots illustrate how the contribution of key clinical risk factors to heart disease prediction varies across different income levels.

For high blood pressure, the SHAP values shift clearly from negative to positive when the condition is present, indicating that individuals with high blood pressure are predicted to have a substantially higher risk of heart disease. The colour distribution across the two clusters shows little systematic separation by income level, suggesting that the effect of high blood pressure on predicted risk is relatively consistent across socioeconomic groups.

A similar pattern is observed for high cholesterol. When high cholesterol is present, SHAP values increase markedly, indicating an increased contribution to predicted heart disease risk. As with high blood pressure, the colour gradient representing income does not display strong vertical separation within the clusters, implying that income does not strongly moderate the impact of high cholesterol on model predictions.

General health demonstrates a strong monotonic relationship with SHAP values. As self-reported health worsens (moving from excellent toward poor health), the SHAP contribution increases steadily, indicating a progressively higher predicted risk of heart disease. While observations from different income levels are present within each health category, the dominant pattern remains the effect of worsening health rather than a clear socioeconomic interaction.

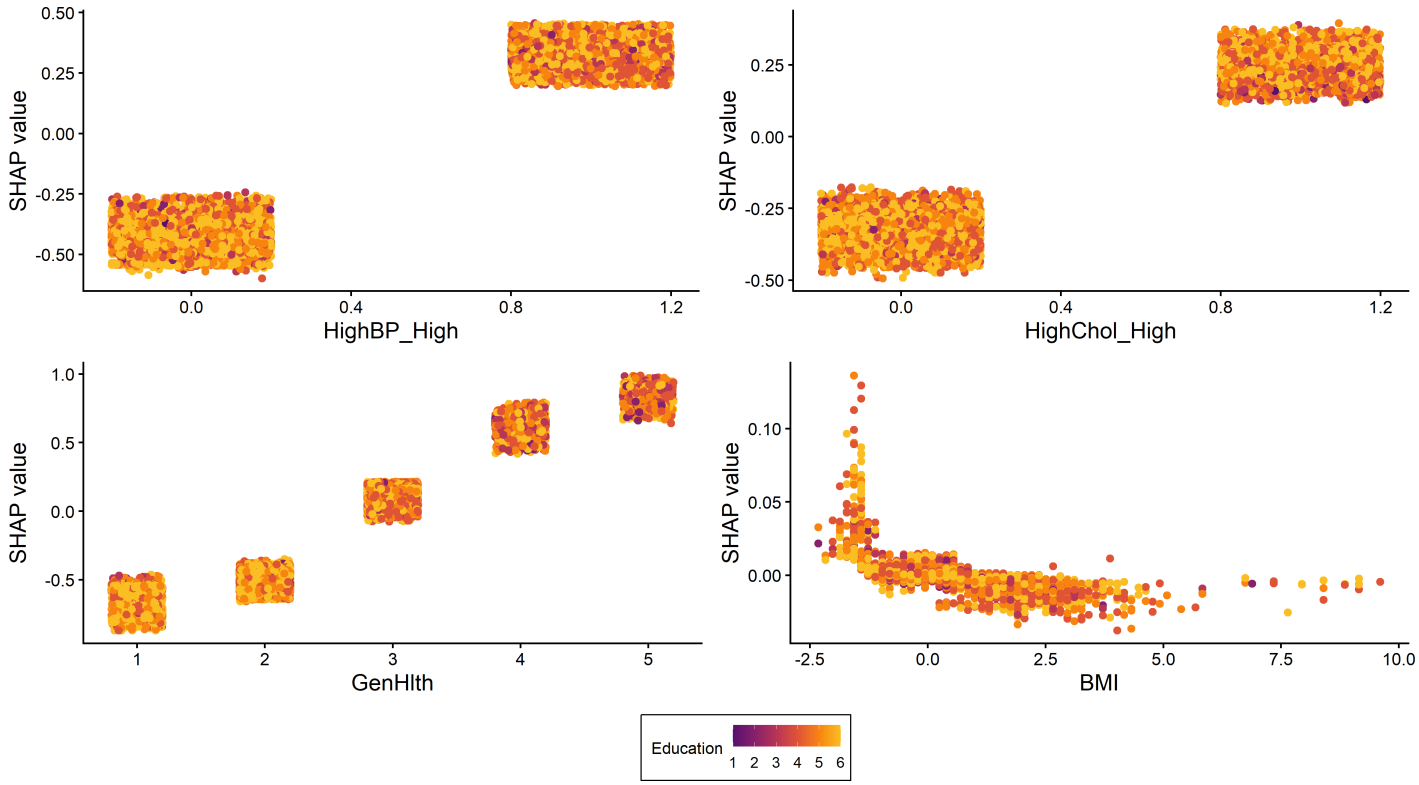
The BMI dependence plot shows a weaker and more diffuse pattern. SHAP values remain close to zero across most BMI values, with only modest variation in predicted risk. Income levels are mixed throughout the distribution, again indicating limited evidence that income substantially modifies the influence of BMI on predicted heart disease risk.

Overall, these results suggest that the primary drivers of heart disease prediction in the model are the clinical risk factors themselves. Although the XGBoost model is capable of capturing complex feature interactions, the dependence plots provide limited visual evidence that income strongly moderates the impact of these clinical predictors on predicted heart disease prevalence.

Clinical Risk Factors by Education

```
p5 <- sv_dependence(shp_flipped, v = "HighBP_High", color_var = "Education")
p6 <- sv_dependence(shp_flipped, v = "HighChol_High", color_var = "Education")
```

```r
p7 <- sv_dependence(shp_flipped, v = "GenHlth", color_var = "Education")
p8 <- sv_dependence(shp_flipped, v = "BMI", color_var = "Education")

(p5 + p6) / (p7 + p8) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```



When coloured by education rather than income, the SHAP dependence plots show a very similar overall structure. High blood pressure and high cholesterol both produce clear upward shifts in SHAP values when present, indicating that these conditions contribute positively to predicted heart disease risk. However, the colour distribution is broadly mixed within each cluster, providing little visual evidence that educational attainment substantially changes the effect of either condition.

General health again shows the clearest monotonic relationship with model output. As self-reported health worsens, SHAP values increase steadily from negative to strongly positive, indicating progressively greater predicted cardiovascular risk. Although observations from different education levels are present within each category, the dominant pattern is the strong main effect of general health rather than a clear education-based separation.

BMI continues to display a comparatively weak and diffuse pattern. Most SHAP contributions remain close to zero, with only modest positive contributions at the lower end and slight negative contributions across much of the mid-range. Educational levels are intermingled throughout the plot, suggesting limited moderation of the BMI effect by education.

Overall, the education-coloured dependence plots provide little evidence that educational attainment strongly moderates the relationship between these clinical risk factors and predicted heart disease risk. As with the income-based plots, the dominant signal comes from the direct contribution of the clinical variables themselves. This suggests that, within the XGBoost model, socioeconomic moderation by education is limited for these particular predictors.

7.5.4 Local SHAP Explanations: Individual-Level Predictions

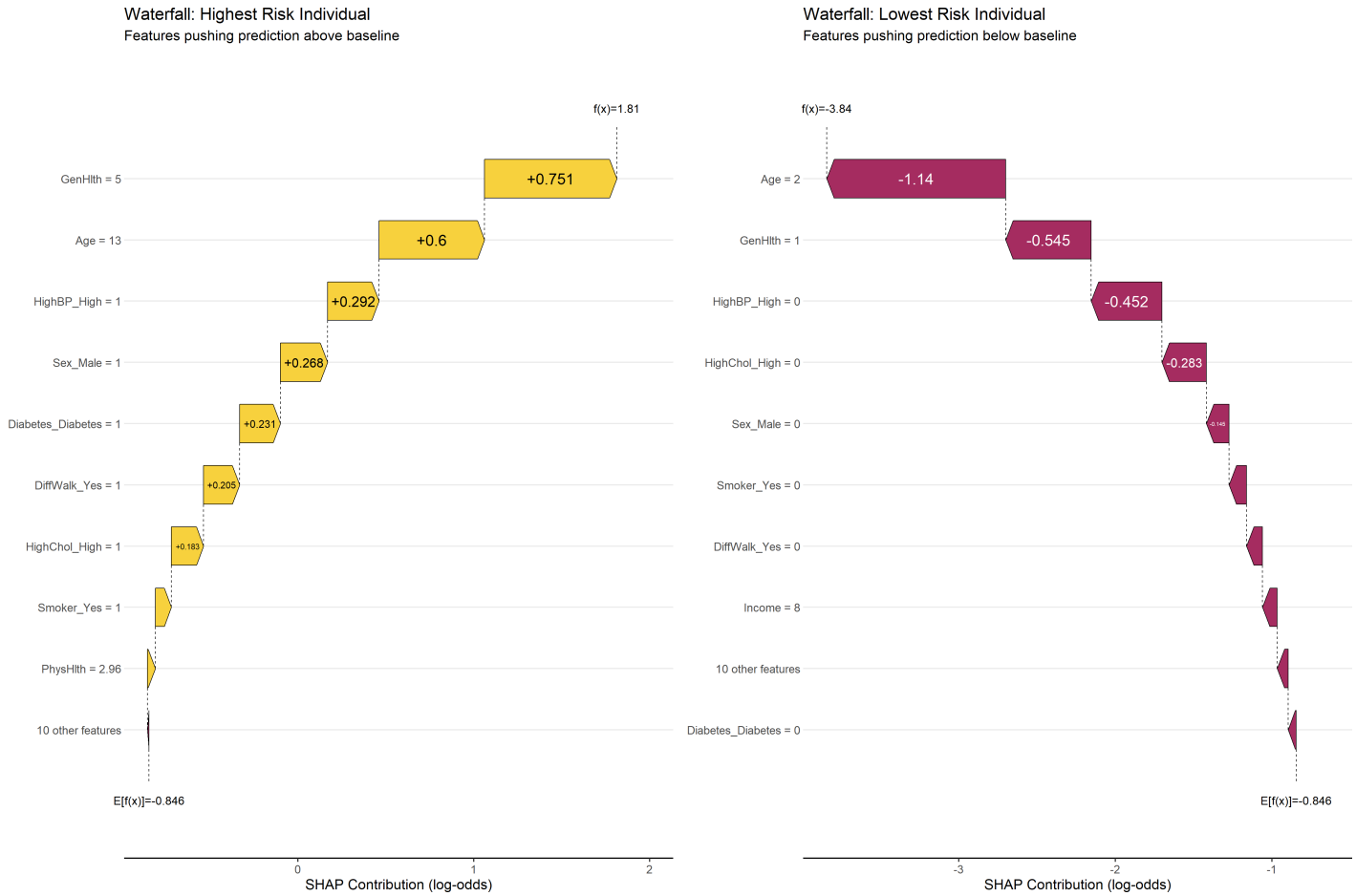
To complement the global interpretation of the model, SHAP local explanation plots are used to examine how individual predictions are formed. Two observations are selected: the individual with the highest predicted risk of heart disease and the individual with the lowest predicted risk. These cases are identified by selecting the rows with the largest and smallest total SHAP contributions across all features.

The waterfall plots show how each predictor shifts the model output away from the baseline prediction for a specific individual. Features with positive SHAP values increase the predicted probability of heart disease, while negative values decrease it. For the highest-risk individual, several clinical variables contribute positively to the prediction, cumulatively pushing the model output well above the baseline. In contrast, for the lowest-risk individual, protective factors or the absence of risk conditions contribute negatively, pulling the prediction below the baseline.

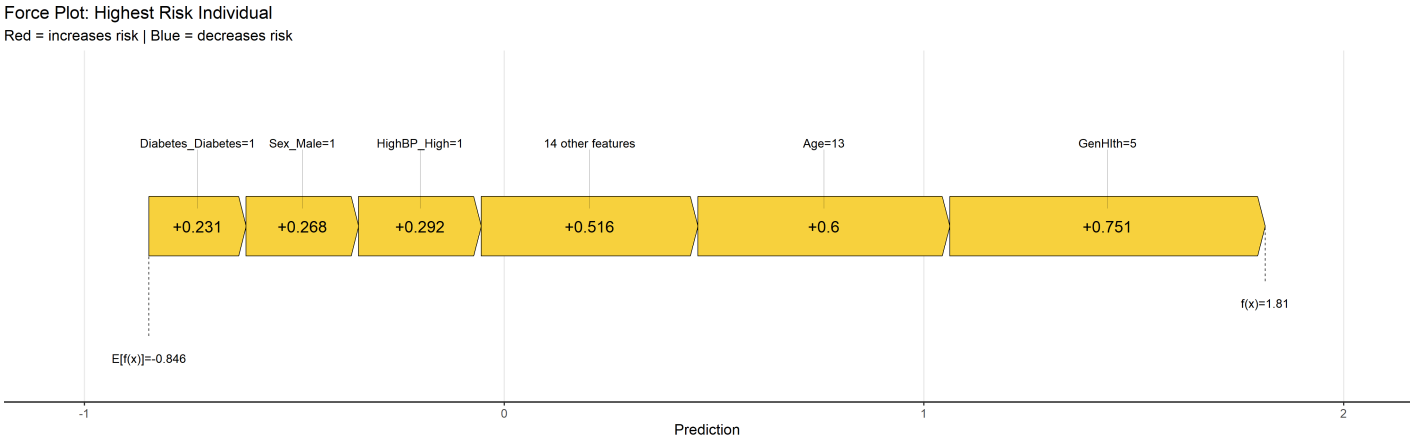
The force plots provide a complementary visualisation of the same explanations by displaying how features push the prediction toward higher or lower risk simultaneously. Red features indicate variables increasing predicted heart disease risk, while blue features indicate variables decreasing it.

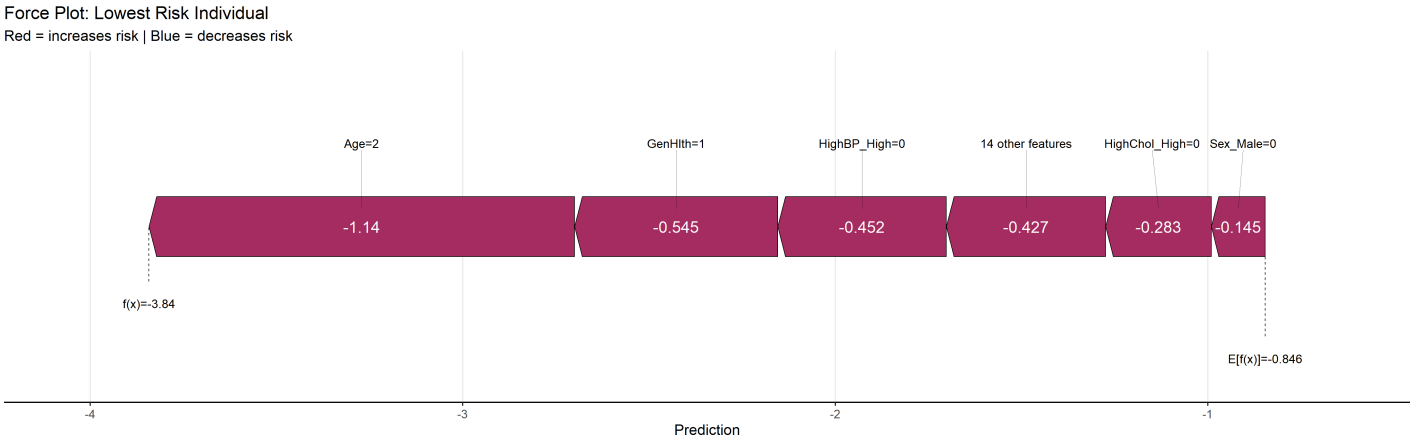
These local explanations demonstrate how the XGBoost model combines multiple demographic, clinical, and socioeconomic predictors to generate individual-level predictions. This illustrates the model's ability to capture complex interactions and heterogeneous effects across individuals, providing interpretable insights beyond aggregate feature importance measures.

```
low_risk_idx <- which.min(rowSums(as.matrix(shp_flipped$S)))
high_risk_idx <- which.max(rowSums(as.matrix(shp_flipped$S)))

w1 <- sv_waterfall(shp_flipped, row_id = high_risk_idx) +
  labs(title = "Waterfall: Highest Risk Individual",
       subtitle = "Features pushing prediction above baseline",
       x = "SHAP Contribution (log-odds)")

w2 <- sv_waterfall(shp_flipped, row_id = low_risk_idx) +
  labs(title = "Waterfall: Lowest Risk Individual",
       subtitle = "Features pushing prediction below baseline",
       x = "SHAP Contribution (log-odds)")

f1 <- sv_force(shp_flipped, row_id = high_risk_idx) +
  labs(title = "Force Plot: Highest Risk Individual",
       subtitle = "Red = increases risk | Blue = decreases risk")

f2 <- sv_force(shp_flipped, row_id = low_risk_idx) +
  labs(title = "Force Plot: Lowest Risk Individual",
       subtitle = "Red = increases risk | Blue = decreases risk")

(w1 | w2)
```


Waterfall: Highest Risk Individual
Features pushing prediction above baseline
f(x)=1.81
GenHlth = 5
+0.751
Age = 13
+0.6
HighBP_High = 1
+0.292
Sex_Male = 1
+0.268
Diabetes_Diabetes = 1
+0.231
DiffWalk_Yes = 1
+0.205
HighChol_High = 1
+0.183
Smoker_Yes = 1
PhysHlth = 2.96
10 other features
E[f(x)]=-0.846
0
1
2
SHAP Contribution (log-odds)
Waterfall: Lowest Risk Individual
Features pushing prediction below baseline
f(x)=-3.84
Age = 2
-1.14
GenHlth = 1
-0.545
HighBP_High = 0
-0.452
HighChol_High = 0
-0.283
Sex_Male = 0
Smoker_Yes = 0
DiffWalk_Yes = 0
Income = 8
10 other features
Diabetes_Diabetes = 0
E[f(x)]=-0.846
-3
-2
-1
SHAP Contribution (log-odds)

```
(f1 / f2)
```





The local SHAP explanations illustrate how the model combines individual features to generate predictions for specific observations. Two contrasting cases are shown: the individual with the highest predicted risk of heart disease and the individual with the lowest predicted risk.

For the highest-risk individual, the waterfall and force plots show that several clinical and demographic risk factors push the prediction substantially above the baseline. Poor general health (GenHlth = 5) provides the largest positive contribution to the prediction, followed by older age (Age = 13). Additional factors such as the presence of high blood pressure, male sex, diabetes, difficulty walking, and high cholesterol all contribute positively to the predicted risk. Together, these factors accumulate to shift the model output well above the baseline log-odds, indicating a strong predicted likelihood of heart disease.

In contrast, the lowest-risk individual shows a sequence of negative SHAP contributions that push the prediction far below the baseline. Younger age (Age = 2) provides the strongest protective effect, followed by excellent self-reported health (GenHlth = 1). The absence of major clinical risk factors, including high blood pressure, high cholesterol, smoking, and diabetes, further reduces the predicted risk. Female sex and higher income also contribute modestly toward lowering the prediction.

These local explanations demonstrate how the XGBoost model integrates multiple predictors to generate individual-level predictions. High predicted risk emerges when several clinical risk factors accumulate simultaneously, whereas low predicted risk arises from the combination of younger age, good health status, and the absence of cardiometabolic conditions.

7.5.5 SHAP Interaction Analysis

To further investigate whether the model captures complex relationships between predictors, SHAP interaction values are computed for the XGBoost model. While standard SHAP values quantify the independent contribution of each feature to the prediction, SHAP interaction values decompose these effects into pairwise feature interactions.

This allows the model output to be examined in terms of how two variables jointly influence predictions rather than assuming purely additive effects. Such interactions are particularly relevant for the research question, which investigates whether

socioeconomic determinants such as income and education modify the impact of clinical risk factors on heart disease prevalence.

By computing SHAP interaction values, it becomes possible to identify which pairs of variables most strongly interact within the model and to determine whether socioeconomic variables meaningfully alter the contribution of key clinical predictors. This provides a more detailed understanding of whether the non-linear modelling approach captures interaction effects beyond traditional additive assumptions.

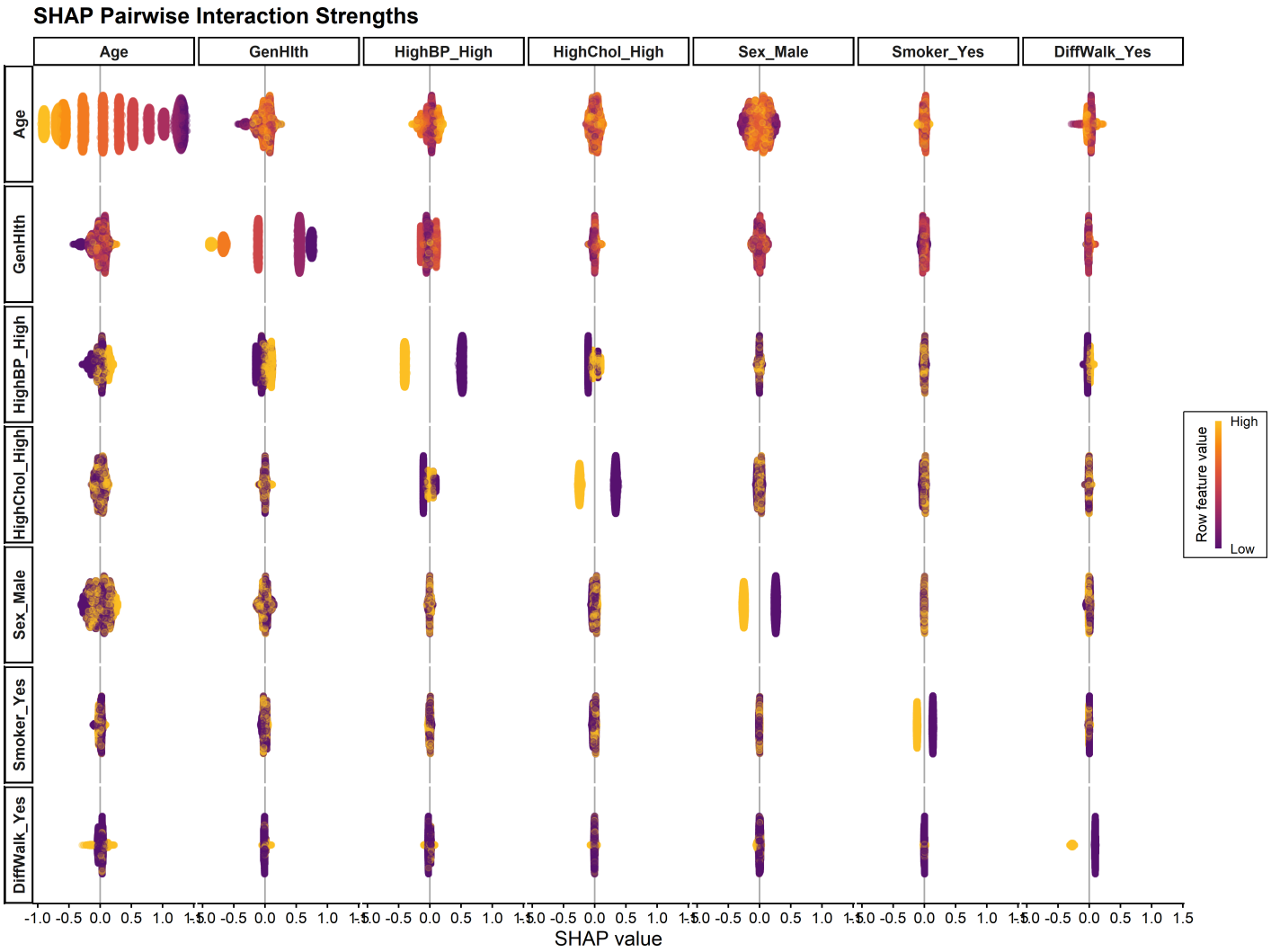
```
if (!file.exists("shp_i.rds")) {
  shp_i <- shapviz(xgb_engine, X_pred = X_sample, interactions = TRUE)
  write_rds(shp_i, "shp_i.rds")
} else {
  shp_i <- read_rds("shp_i.rds")
}
```

SHAP Pairwise Interactions

The SHAP interaction plot summarises the strength of pairwise interactions between predictors in the XGBoost model. Interaction strength reflects how much two variables jointly influence predictions beyond their individual contributions.

This visualisation helps identify whether important predictors operate largely independently or whether meaningful interaction effects are present. In the context of this research question, the interaction matrix is used to assess whether socioeconomic variables such as income and education meaningfully interact with clinical risk factors in shaping predicted heart disease risk.

```
sv_interaction(shp_i) +
  labs(title = "SHAP Pairwise Interaction Strengths")
```



The SHAP interaction plot highlights the strength and direction of pairwise interactions between key predictors in the XGBoost model. Each panel shows how the contribution of one feature varies depending on the value of another feature, allowing interaction effects to be examined beyond simple additive relationships.

The strongest patterns are observed for age and general health, which display wide spreads of SHAP values across several panels. This indicates that these variables not only have strong independent effects but also interact with multiple other predictors when determining heart disease risk. In particular, the interaction between age and general health shows substantial variation, suggesting that the effect of worsening health status becomes more pronounced among older individuals.

Interactions involving clinical risk factors such as high blood pressure and high cholesterol are present but appear more limited in magnitude. Their effects are largely consistent across other predictors, indicating that these variables contribute relatively stable increases in predicted risk rather than strongly interacting with other features.

Demographic and behavioural variables, including sex, smoking, and difficulty walking, show smaller interaction spreads. These predictors appear to contribute modest adjustments to risk but do not display strong pairwise interaction patterns with the primary clinical variables.

Overall, the interaction matrix suggests that the XGBoost model captures some non-additive relationships, particularly involving age and general health. However, most other predictors exhibit relatively weak interaction effects, indicating that much of the model's predictive signal is still driven by the direct contributions of key clinical and demographic variables rather than strong pairwise interactions.

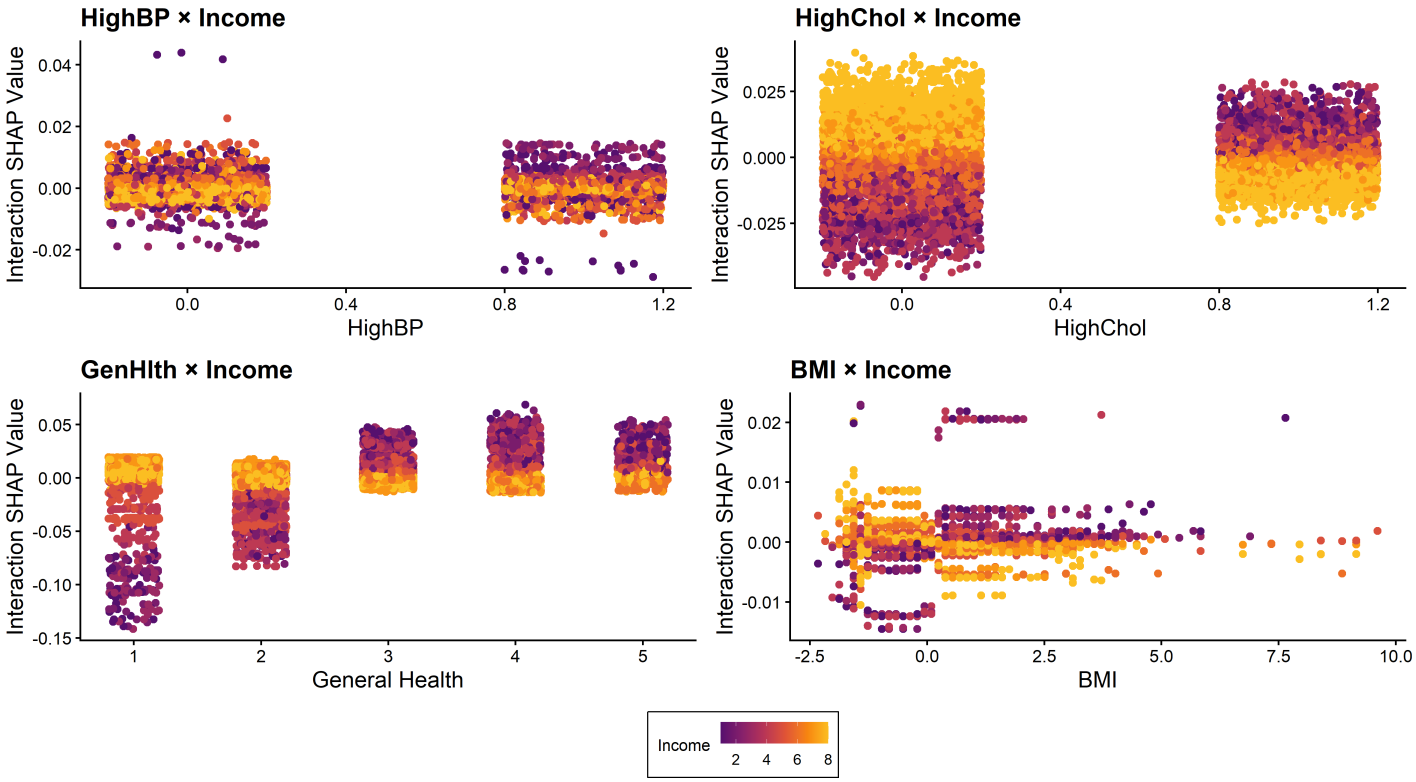
7.5.6 SHAP Interaction Dependence

These SHAP interaction dependence plots are used to explore whether the contribution of key clinical risk factors to predicted heart disease risk varies across socioeconomic groups. By visualising interaction SHAP values while colouring observations by socioeconomic variables, the plots allow potential moderation effects to be examined.

Unlike standard dependence plots, interaction SHAP values isolate the portion of a feature's effect that arises specifically from interactions with other predictors. This enables the identification of non-additive relationships within the model and helps assess whether socioeconomic determinants modify the influence of clinical risk factors on predicted heart disease risk.

Income Moderation

```
p1 <- sv_dependence(
  shp_i,
  v = "HighBP_High",
  color_var = "Income",
  interactions = TRUE) +
  labs(
    title = "HighBP × Income",
    x = "HighBP",
    y = "Interaction SHAP Value")

p2 <- sv_dependence(
  shp_i,
  v = "HighChol_High",
  color_var = "Income",
  interactions = TRUE) +
  labs(
    title = "HighChol × Income",
    x = "HighChol",
    y = "Interaction SHAP Value")

p3 <- sv_dependence(
  shp_i,
  v = "GenHlth",
  color_var = "Income",
  interactions = TRUE) +
  labs(
    title = "GenHlth × Income",
    x = "General Health",
    y = "Interaction SHAP Value")

p4 <- sv_dependence(
  shp_i,
  v = "BMI",
  color_var = "Income",
  interactions = TRUE) +
  labs(
    title = "BMI × Income",
    x = "BMI",
    y = "Interaction SHAP Value")

(p1 + p2) / (p3 + p4) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```



The interaction dependence plots illustrate how the contribution of key clinical risk factors to predicted heart disease risk varies across income levels when interaction effects are explicitly isolated.

For high blood pressure, interaction SHAP values remain tightly clustered around zero for both categories of the variable. Although the presence of high blood pressure increases predicted risk overall, the interaction component with income appears minimal, indicating that income does not substantially alter the effect of this condition in the model.

A similar pattern is observed for high cholesterol. The interaction values show limited spread and remain close to zero across income levels, suggesting that the influence of high cholesterol on heart disease risk is largely independent of income.

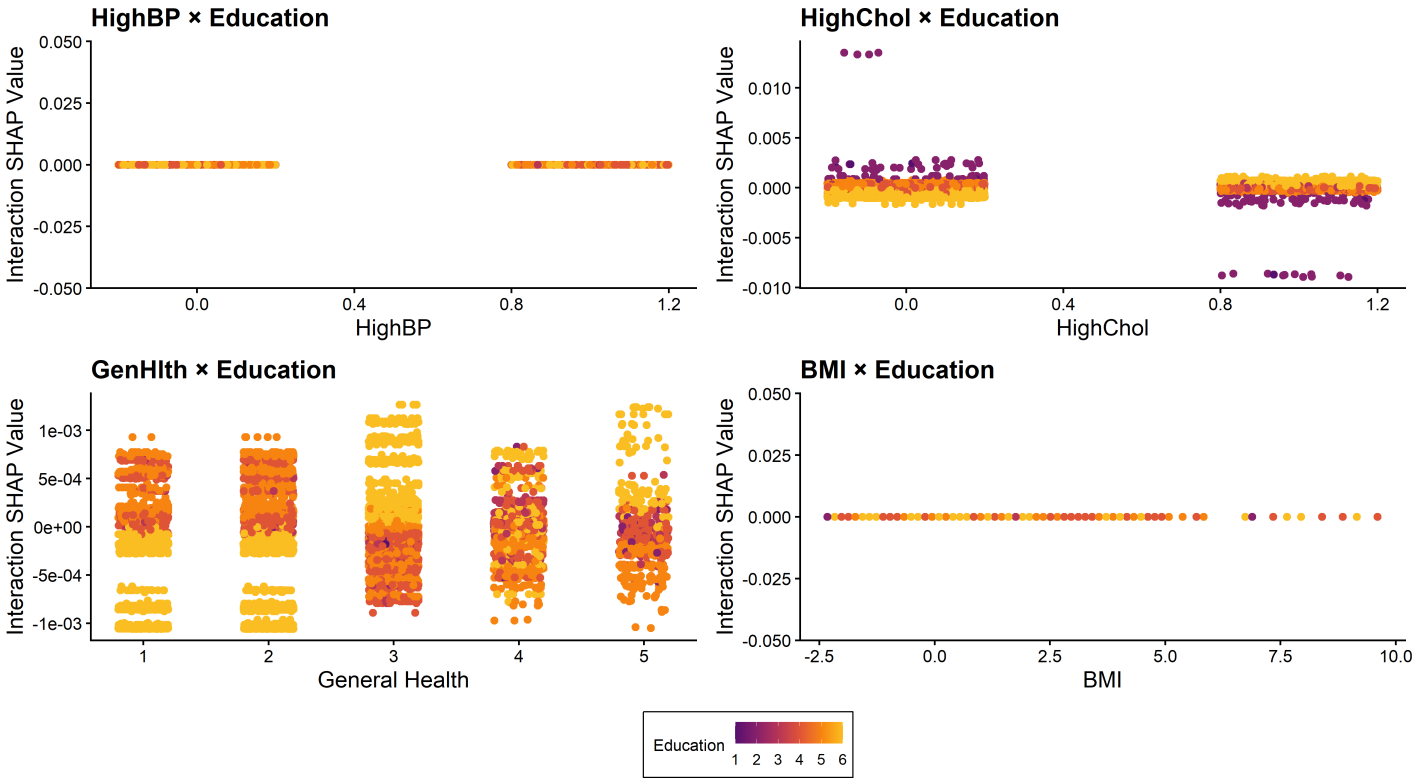
General health displays slightly greater variation in interaction SHAP values, particularly for poorer health categories. However, while income levels are distributed across the observations, there is no strong or systematic separation by income group. This suggests that worsening health increases predicted risk consistently across income levels rather than being strongly moderated by socioeconomic status.

The BMI interaction plot shows very small interaction values across the majority of observations, indicating that BMI contributes little through interaction effects with income. Most points remain close to zero, suggesting that any influence of BMI on predicted risk operates largely through its direct effect rather than through socioeconomic moderation.

Overall, the interaction SHAP values indicate that while the XGBoost model captures complex non-linear relationships, the interaction between income and these clinical risk factors is relatively weak. The primary contribution to predicted heart disease risk therefore appears to come from the direct effects of the clinical variables themselves rather than strong socioeconomic interaction effects.

```
p5 <- sv_dependence(
  shp_i,
  v = "HighBP_High",
  color_var = "Education",
  interactions = TRUE) +
  labs(
    title = "HighBP × Education",
    x = "HighBP",
    y = "Interaction SHAP Value")
```

```r
p6 <- sv_dependence(
  shp_i,
  v = "HighChol_High",
  color_var = "Education",
  interactions = TRUE) +
  labs(
    title = "HighChol × Education",
    x = "HighChol",
    y = "Interaction SHAP Value")

p7 <- sv_dependence(
  shp_i,
  v = "GenHlth",
  color_var = "Education",
  interactions = TRUE) +
  labs(
    title = "GenHlth × Education",
    x = "General Health",
    y = "Interaction SHAP Value"
  )

p8 <- sv_dependence(
  shp_i,
  v = "BMI",
  color_var = "Education",
  interactions = TRUE) +
  labs(
    title = "BMI × Education",
    x = "BMI",
    y = "Interaction SHAP Value")

(p5 + p6) / (p7 + p8) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```



The education-based interaction plots show patterns very similar to those observed for income. For high blood pressure and high cholesterol, interaction SHAP values remain tightly clustered around zero regardless of educational level. This indicates that although these clinical conditions strongly influence predicted heart disease risk, their interaction with education contributes very little additional effect within the model.

General health again displays slightly more variation in interaction SHAP values across categories. However, the spread remains relatively small and educational levels are distributed across each health category without clear separation. This suggests that worsening self-reported health increases predicted risk consistently across educational groups rather than being strongly moderated by education.

For BMI, interaction SHAP values are effectively zero for nearly all observations. This indicates that BMI contributes almost no interaction effect with education in the model, and its influence on predictions is primarily through its direct effect rather than through socioeconomic modification.

Overall, the interaction dependence plots provide little evidence that educational attainment strongly moderates the relationship between these clinical risk factors and predicted heart disease risk. Consistent with the income-based analysis, the dominant signal in the model arises from the direct effects of the clinical predictors rather than strong interaction effects with socioeconomic variables.

7.5.7 Summary of SHAP Analysis

The SHAP analysis provides a comprehensive interpretation of how the XGBoost model generates predictions at both global and local levels. Global feature importance and beeswarm visualisations show that the model's predictions are primarily driven by clinical and health-related variables, particularly age, general health status, high blood pressure, and high cholesterol. These predictors consistently exert the largest influence on predicted heart disease risk across the dataset.

Dependence plots further illustrate the direction and structure of these relationships, demonstrating clear increases in predicted risk for worsening health indicators and the presence of cardiometabolic conditions. Socioeconomic variables such as income and education contribute comparatively smaller direct effects, suggesting that clinical risk factors remain the dominant predictors of heart disease prevalence in the model.

Interaction analyses were conducted to assess whether socioeconomic determinants moderate the influence of clinical predictors. Both pairwise interaction strength visualisations and interaction dependence plots indicate that most interactions between socioeconomic variables and clinical risk factors are relatively weak. While the XGBoost model is capable of

capturing non-linear and interaction effects, the results suggest that the primary predictive signal arises from the independent contributions of the clinical variables rather than strong socioeconomic moderation effects.

Finally, local SHAP explanations demonstrate how individual predictions are formed through the accumulation of multiple risk factors. High predicted risk emerges when several adverse health conditions occur simultaneously, whereas low predicted risk is associated with younger age, better general health, and the absence of key cardiometabolic conditions.

With respect to the research question, the SHAP interaction analyses were used to investigate whether socioeconomic determinants, specifically income and education, moderate the impact of these clinical risk factors. Both the interaction strength visualisation and the interaction dependence plots show that most interactions between socioeconomic variables and clinical predictors are relatively small. The contribution of major risk factors such as high blood pressure, high cholesterol, and general health appears largely consistent across different socioeconomic groups.

This suggests that although the XGBoost model is capable of capturing complex feature interactions, the predictive signal is dominated by the direct effects of clinical risk factors rather than strong moderation by income or education. Socioeconomic variables still contribute to predictions, but their role appears to be secondary compared with physiological and health-status indicators.

Overall, the results indicate that complex machine learning models can detect interaction structures within the data, but in this case the moderation effect of socioeconomic determinants on the relationship between clinical risk factors and heart disease prevalence appears limited. Consequently, the predictive insights provided by the model primarily reinforce the importance of established clinical risk factors rather than revealing strong socioeconomic interaction effects.

7.6 Conclusions: Discussion and Limitations

This section addressed the second research question by investigating whether complex machine learning models can capture interactions between socioeconomic determinants and clinical risk factors when predicting heart disease prevalence. To do this, a modelling framework was implemented using the tidymodels ecosystem, comparing a regularized logistic regression model with a tuned XGBoost model. Logistic regression was included as a baseline representing traditional additive modelling assumptions, while XGBoost was used to capture potential non-linear relationships and interaction effects within the data.

Models were trained using cross-validated tuning and evaluated on a held-out test set using metrics appropriate for imbalanced classification, including ROC–AUC, PR–AUC, F1-score, sensitivity, and specificity. The final evaluation demonstrated that the tuned XGBoost model substantially outperformed logistic regression in terms of discrimination and overall predictive performance. Logistic regression effectively collapsed to predicting the majority class, highlighting the limitations of a simple additive framework for this prediction task.

To interpret the behaviour of the XGBoost model, SHAP-based explainability techniques were applied. Global importance and beeswarm plots showed that predictions were primarily driven by clinical and health-related predictors such as age, general health status, high blood pressure, and high cholesterol. Dependence plots confirmed clear monotonic relationships between worsening health indicators and increasing predicted heart disease risk.

A central aim of this section was to evaluate whether socioeconomic determinants, specifically income and education, moderate the impact of these clinical risk factors. SHAP interaction analyses were therefore conducted to isolate pairwise feature interactions. The results indicated that although the XGBoost model is capable of capturing non-linear and interaction effects, the moderation of clinical risk factors by income and education appears relatively limited. Interaction SHAP values for combinations such as high blood pressure × income, high cholesterol × income, and general health × education remained small compared with the direct effects of the clinical variables themselves.

These findings suggest that, within this dataset, the prevalence of heart disease is primarily associated with established clinical risk factors rather than strong socioeconomic moderation of those factors. While socioeconomic variables contribute to the predictive model, their influence appears secondary compared with physiological and health-status indicators. Consequently, the use of a flexible machine learning model provided improved predictive performance and interpretability, but did not reveal strong interaction structures between socioeconomic and clinical predictors.

Several limitations should be considered when interpreting these results. First, the analysis relies on self-reported survey data, which may contain measurement error or reporting bias. Second, the variables used to represent socioeconomic status are relatively coarse indicators, which may limit the ability to detect nuanced socioeconomic interactions. Third, the dataset is cross-sectional, meaning causal relationships cannot be inferred from the observed associations. Finally,

although SHAP provides powerful interpretability tools, the explanations remain dependent on the structure of the fitted model and should be interpreted as model-based insights rather than definitive causal effects.

Overall, this section demonstrates that machine learning models such as XGBoost can provide improved predictive performance and detailed interpretability through explainable AI techniques. However, in the context of this dataset, the dominant determinants of predicted heart disease risk remain clinical health indicators, with socioeconomic moderation effects appearing relatively modest.